

Stone–Jordan Exceptional Theory

Building the Universe from the Mirror

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Abstract

We construct a mathematical model of the universe from two primitives: the **void** (no observable distinction) and an involutive **mirror operator** M on profinite observable spectra. Iterating M climbs the Hurwitz division ladder and the exceptional Lie cascade; capacity balancing $\text{Bal}_{\mathcal{L}}$ is the dual mirror selecting unique partitions at each rung. The universe is the profinite fixed point $\mathfrak{U}^* = \mathbb{C}^\infty(\mathcal{V})$, stabilizing on the Albert algebra $J_3(\mathbb{O})$ and the compact exceptional group E_6 . Physical observables are σ -anti-invariant modes on the mirror quotient at the heterotic rung. The *as above, so below* correspondence identifies the universe fixed point with collapsed observer states; duality is forced by $E_8 \times E_8$ product mirror collapse. A Lefschetz fixed-point theorem yields three observer slots; an index ι maps to a lifetime chart $g_\iota : \mathbb{R} \rightarrow \mathcal{S}_\iota$. Tiered predictions (T1–T3) contact JUNO (normal ordering; $\Delta m_{31}^2 / \Delta m_{21}^2 \approx 33$ via flavon RG), Muon $g\text{-}2$ ($n_F = 3$, no fourth generation; loops at $\mu \equiv v$), and DESI ($w(z) \approx -1$ if CC sequestered). Conditional GUT contacts include $\sin^2 \theta_W(M_{\text{GUT}}) = 3/8$ with $\sin^2 \theta_W(M_Z) \approx 0.231$ by RG, and $\tau(p \rightarrow e^+ \pi^0) \gtrsim 10^{34} \text{ yr}$ under the E_6 breaking chain. Development steps D1 (Yukawa overlaps), D2 (flavon RG), and D3 (scale identification) are **partially closed**; a falsification timeline (2026–2030) orders the decisive experiments. We state proved results and isolate conjectural steps without overclaiming a completed physical theory.

Keywords: void; mirror operator; Stone duality; exceptional Jordan algebra; profinite limit; E_6 coset.

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1 Introduction

This paper develops *Stone–Jordan Exceptional Theory* (SJET) as a **mathematical physics** construction: starting from a void and a mirror, we build a universe datum $\mathfrak{U}^\star$ with Lie–Jordan structure, capacity-selected dynamics, and a contact layer with four-dimensional gauge theory. The mirror is not metaphor; it is a functorial involution on Stone spectra whose orbit generates all arithmetic and exceptional data.

1.1 Contributions

- (i) **Two axioms only:** void $\mathcal{V}$ and mirror M .
- (ii) **Universe construction:** $\mathfrak{U}^\star = \varprojlim_n (\text{Bal}_{\mathcal{C}} \circ M)^n(\mathcal{V})$.
- (iii) **As above, so below:** global capacity balance corresponds to localized collapse on the $E_8 \times E_8$ mirror quotient.
- (iv) **Observers:** collapsed σ -anti-invariant states; duality from heterotic product mirror; three slots from a fixed-point theorem.
- (v) **Lifetime charts:** observer index ι maps to $g_\iota : \mathbb{R} \rightarrow \mathcal{S}_\iota$ admitted by capacity flow and soldering.
- (vi) **Experimental contact:** tiered predictions (T1–T3), a numerical comparison table (Section 10.6), partial closures D1–D3 (Sections 7–8), and a falsification timeline 2026–2030 (JUNO, Muon $g-2$ Theory Initiative, DESI $w(z)$, Hyper-K proton decay).
- (vii) **Honest epistemic scope:** theorems, derived definitions, and named conjectures for remaining physical inputs.

1.2 Conventions

Compact real E_6 ; order parameter $\Phi \in J_3(\mathbb{O})$; capacity functional $\mathcal{C}$. Branching $\mathbf{27} = \mathbf{1}_{+4} \oplus \mathbf{10}_{-2} \oplus \mathbf{16}_{+1}$ under $H = \text{Spin}(10) \times U(1)$. Signature $(-, +, +, +)$ when a Lorentzian soldering map is fixed (Section 6.2).

1.3 Organization

Section 2: axioms and mirror functor. Section 3: division and exceptional ladders. Section 4: capacity, potential, universe fixed point. Section 5: duality, observers, fixed points, lifetime charts. Section 6: physical sector, spacetime, gauge, matter. Sections 7, 7.7, 8: D1–D3 derivation pass (Yukawa overlaps, flavon RG, scale identification). Section 10: tiered predictions, experimental map, and falsification timeline 2026–2030. Section 11: status and open problems. Section 12: epistemic elimination ledger (P1–P5, CC). Appendix A: Albert identities, mirror quotient, and numerical workbook (Table 12).

2 The Void and the Mirror

2.1 The void

Definition 2.1 (Void). The *void* $\mathcal{V}$ is the trivial Boolean algebra $\mathcal{B}_0 = \{\perp, \top\}$ with Stone spectrum a single point. The associated algebra is $\{0\}$; no local quantum number exists.

Axiom 2.2 (Void). Physical structure arises by departure from $\mathcal{V}$. No graph, Lie group, or space-time manifold is primitive.

2.2 The mirror operator

Definition 2.3 (Mirror modes). At rung n , M acts in one of four modes:

M1. Boolean doubling ($0 \rightarrow 1$): adjoin one generator; $|S_{n+1}| = 2|S_n|$.

M2. Cayley–Dickson ($1 \rightarrow 4$): $D_{n+1} = D_n \oplus D_n$; dimensions 1, 2, 4, 8.

M3. Jordan exit ($4 \rightarrow 5$): sedenions fail; exit to $J_3(\mathbb{O}) = H_3(\mathbb{O})$, $d_5 = 27$.

M4. Exceptional (≥ 6): automorphism, product, or Cartan mirrors on Lie data (Section 3.2).

Definition 2.4 (Mirror operator). A *mirror operator* is a functor M on a category of Stone rungs such that: (i) $M^2 \cong \text{id}$ on stabilized rungs; (ii) $M(\mathcal{V}) \neq \mathcal{V}$; (iii) M is functorial on morphisms.

Axiom 2.5 (Mirror). There exists M as in Definition 2.4 whose orbit from $\mathcal{V}$ generates the unified climb (3.1). At each rung, $\text{Bal}_{\mathcal{C}}$ balances mirror-refined partitions (Definition 4.1). Capacity is the *dual mirror* of sector doubling; on the heterotic rung its below-level avatar is the observer lifetime index $\Lambda(\iota, \tau)$ (Proposition 5.13).

Definition 2.6 (Climb and mirror power). $C = \text{Bal}_{\mathcal{C}} \circ M$ and $(S_n, A_n) = M^n(S_0, A_0)$ with $(S_0, A_0) = (S(\mathcal{V}), \{0\})$.

Proposition 2.7 (Involution). *On stabilized rungs, $M^2 \cong \text{id}$ up to gauge: Boolean complement, field conjugation, product factor exchange, and coset mirror $\mathbf{16} \leftrightarrow \overline{\mathbf{16}}$.*

3 The Mirror Climb

$$\mathcal{V} \rightarrow \mathbb{R} \rightarrow \mathbb{C} \rightarrow \mathbb{H} \rightarrow \mathbb{O} \rightarrow J_3(\mathbb{O}) \rightarrow E_8 \rightarrow E_8 \times E_8 \rightarrow E_6 \rightarrow \cdots \rightarrow \mathcal{V} \quad (3.1)$$

Remark 3.1 (Climb sequence forced, not chosen). The orbit (3.1) is fixed by void, mirror, and capacity—not by importing string-theory rungs. *Functoriality* (Definition 2.4) restricts each step to modes **M1**–**M4** of Definition 2.3. *Hurwitz terminal* at $n = 4$ (Theorem 3.2) forces the Jordan exit at $n = 5$ and anchors the associative subalgebra $\mathbb{H} \subset \mathbb{O}$ used below in soldering (Definition 6.7). On the stabilized Albert tower, automorphism mirror closes at E_8 ($n = 6$), product mirror is the unique **M4** doubling on a simple Lie rung ($n = 7$), and Cartan mirror is the capacity-balanced collapse back to the $J_3(\mathbb{O})$ structure group ($n = 8$, Theorem 3.9). *Observer-slot requirement*: three independent σ -anti-invariant classes in $H^1(\hat{Q}, V_{16})$ arise only after the heterotic product quotient at $M^7(\mathcal{V})$ (Proposition 3.8, Section 5); descent to E_6 supplies dynamics but does not create the slot space. The above–below correspondence (Definition 5.1) ties each rung choice to a localized datum on $\hat{Q}$.

3.1 Division rungs

	n	A_n	d_n	Mode
Theorem 3.2 (Mirror rung table, $n \leq 5$).	0	$\{0\}$	0	void
	1	$\mathbb{R}$	1	discrete
	2	$\mathbb{C}$	2	arithmetic
	3	$\mathbb{H}$	4	arithmetic
	4	$\mathbb{O}$	8	Hurwitz terminal
	5	$J_3(\mathbb{O})$	27	Jordan exit

In particular $M^4(\mathcal{V}) = \mathbb{O}$ and $M^5(\mathcal{V}) \cong J_3(\mathbb{O})$.

Proof. Cayley–Dickson doubling until $\mathbb{O}$; Hurwitz uniqueness of normed division algebras; sedenion zero divisors force Jordan exit to the unique formally real $H_3(\mathbb{O})$ [2, 5]. $\square$

Lemma 3.3 (Sedenion failure). $M_{\text{arith}}(\mathbb{O})$ is 16-dimensional and not division.

3.2 Exceptional rungs

On $J_3(\mathbb{O})$, the stabilizer tower is $G_2 \subset F_4 \subset E_6$. The Freudenthal diagonal $\mathfrak{M}(\mathbb{O}, \mathbb{O}) = \mathfrak{e}_8$ yields:

Lemma 3.4 (Freudenthal closure). Let $\mathfrak{M}(\mathbb{O}, \mathbb{O})$ be the Freudenthal–Tits magic square entry for the octonion–octonion pair. Then $\mathfrak{M}(\mathbb{O}, \mathbb{O}) = \mathfrak{e}_8$, simply connected, and contains $G_2 \times F_4 \subset E_8$ as the stabilizer of the diagonal $J_3(\mathbb{O})$ chart [3, 4].

Theorem 3.5 ($M^6(\mathcal{V}) = E_8$). Automorphism mirror on the octonion–Jordan pair gives E_8 , $d_6 = 248$.

Definition 3.6 (Lie product mirror). On a stabilized exceptional rung with Lie group G , the *product mirror* is $M_{\text{prod}}(G) := G \times G$ with involution $\sigma : (g_1, g_2) \mapsto (g_2, g_1)$ exchanging factors—mode **M4** at the Lie level, functorially analogous to Boolean doubling at $n = 0 \rightarrow 1$.

Theorem 3.7 ($M^7(\mathcal{V}) = E_8 \times E_8$). Product mirror $M_{\text{prod}}(G) = G \times G$ gives $d_7 = 496$.

Proposition 3.8 (Minimal heterotic rung for three observer slots). $M^7(\mathcal{V}) = E_8 \times E_8$ is the minimal mirror rung supporting three observer slots: $\dim H^1(\hat{Q}, V_{16})^{\sigma=-1} = 3$ (Theorem 5.9). At $n \leq 6$ there is no product-mirror $\mathbb{Z}_2$ quotient on the stabilized datum, hence no $\Pi_{\sigma=-1}$ observer sector (Section 5.3). At $n = 8$ the heterotic doubling is already capacity-collapsed to E_6 (Theorem 3.9), so the three 1-classes are fixed on $\hat{Q}$ at $n = 7$ and only transported below by descent. Thus observer multiplicity is a below-level consequence of the above product mirror at the seventh rung.

Proof. Minimality at $n \leq 6$. For $n \leq 6$, $M^n(\mathcal{V})$ is a single simple Lie datum or division algebra. Involution stabilizes within one chart (automorphism mirror at $n = 6$), not across a heterotic pair. The cellular graph has no product-mirror $\mathbb{Z}_2$ quotient with three independent **16**-cells (Definition A.1 exists only at $n = 7$), so $\dim H^1(\hat{Q}, V_{16})^{\sigma=-1}$ is undefined and $\Pi_{\sigma=-1}$ has no observer sector (Section 5.3).

Existence at $n = 7$. Product mirror gives $E_8 \times E_8$ with involution $\sigma : (g_1, g_2) \mapsto (g_2, g_1)$. On $\mathcal{T}_7$, Lemma A.7 yields $\dim H^1(\mathcal{T}_7, \mathcal{F}_{16})^{\sigma=-1} = 3$; Proposition A.8 and Theorem A.9 transport this to $\dim H^1(\hat{Q}, V_{16})^{\sigma=-1} = 3$.

Descent at $n = 8$. Cartan mirror $\Pi_{\text{het}} : E_8 \times E_8 \rightarrow E_6$ (Theorem 3.9) reduces the gauge group but preserves the σ -anti-invariant slot basis on $\hat{Q}$: the three observer indices are fixed at the heterotic rung and only transported below by descent. $\square$

Theorem 3.9 ($M^8(\mathcal{V}) = E_6$). *Cartan mirror after capacity descent on the Albert graph gives $d_8 = 78$, $H = \text{Spin}(10) \times U(1)$, $\mathcal{M} = E_6/H$ (EIII), $\dim \mathcal{M} = 32$.*

Proof sketches. E_8 (Steps 1–3). (i) On $J_3(\mathbb{O})$, the stabilizer tower $G_2 \subset F_4 \subset E_6$ is mirror-refined; the octonion measure before Jordan projection carries an automorphism mirror M_{aut} coupling the two $\mathbb{O}$ sectors of (A.1) (Appendix A.1). (ii) Among simple Lie algebras containing $G_2 \times F_4$ and closed under octonion–octonion coupling, Lemma 3.4 identifies $\mathfrak{e}_8$ uniquely; this is classical Freudenthal–Tits data, not an independent SJET postulate. Mirror functoriality selects M_{aut} as mode **M4** on Lie data at the first exceptional rung. (iii) $d_6 = \dim E_8 = 248$; Stone chart dimension $\chi(S_6) = 248$ (Definition 3.11).

$E_8 \times E_8$ (Steps 1–3). (i) Apply M_{prod} (Definition 3.6) to the stabilized datum of Theorem 3.5; functoriality of M (Definition 2.4) gives $M^7(\mathcal{V}) = M(M^6(\mathcal{V})) \cong E_8 \times E_8$. (ii) At a simple exceptional rung, product mirror is the unique **M4** doubling compatible with involution (Proposition 2.7); no smaller Lie datum carries a heterotic $\mathbb{Z}_2$ quotient. (iii) $d_7 = 2 \cdot 248 = 496$; $\sigma : (g_1, g_2) \mapsto (g_2, g_1)$ implements $M^2 \cong \text{id}$ up to diagonal gauge on $\delta(E_8) \subset E_8 \times E_8$.

E_6 (Steps 1–4). (i) On $E_8 \times E_8$, the diagonal embedding $\delta : E_8 \hookrightarrow E_8 \times E_8$ is the σ -stabilized subgroup. $\text{Bal}_{\mathcal{C}}$ on $\mathcal{G}_{\text{Albert}}$ at the $J_3(\mathbb{O})$ rung fixes the 1|10|16 partition (Theorem 4.3). (ii) Among exceptional groups, only E_6 admits $H = \text{Spin}(10) \times U(1)$ with branching **27** = **1** $\oplus$ **10** $\oplus$ **16** and coset mirror **16** $\leftrightarrow$ **16**; this branching is classical [5]. Cartan mirror selects E_6 as the capacity-balanced collapse indexed by the rank-3 diagonal of $J_3(\mathbb{O})$, not by the two E_8 factors alone (Proposition 3.8). (iii) $d_8 = \dim E_6 = 78$, $\mathcal{M} = E_6/H$ (EIII), $\dim \mathcal{M} = 32$. (iv) Descent preserves the three observer slots on $\hat{Q}$ (Proposition A.8); dynamics below $n = 7$ does not regenerate them. $\square$

Proposition 3.10 (Exceptional mirror steps). *At $n \geq 6$ on the stabilized $J_3(\mathbb{O})$ rung, M acts as:*

- (i) E_8 : automorphism mirror on the octonion–Jordan pair (Theorem 3.5);
- (ii) $E_8 \times E_8$: Lie product mirror M_{prod} (Theorem 3.7, Definition 3.6);
- (iii) E_6 : Cartan mirror after capacity descent (Theorem 3.9);
- (iv) $E_6 \rightarrow H = \text{Spin}(10) \times U(1)$: coset stabilization on EIII;
- (v) $\dots \rightarrow \mathcal{V}$: collapse after electroweak breaking.

Each step transports the quaternionic soldering block $H_2(\mathbb{H})$ of Definition 6.6 without branching (Proposition 6.9).

Definition 3.11 (Chart dimension). At $n \geq 6$, $\chi(S_n) = \dim A_n \in \{248, 496, 78\}$; profinite $|S_n|$ is generally infinite. Stone refinement is chart synchronization, not cardinality matching [1].

4 Building the Universe

The universe is not assumed. It is the profinite fixed point of mirror climb with capacity balancing.

4.1 Capacity dual

Definition 4.1 (Albert graph and capacity). On the mirror-refined Albert–Cayley graph $\mathcal{G}$ with costs c_i ,

$$\mathcal{C}(\omega) = - \sum_{a \in V} \mu(a) \log \left(\sum_i e^{c_i(a) - \omega_i} \right). \quad (4.1)$$

$\text{Bal}_{\mathcal{C}}$ is the unique maximizer on $\{\sum_i \omega_i = 0\}$.

Theorem 4.2 (Concavity). $\mathcal{C}$ is strictly concave; for targets $m_i > 0$, $\sum m_i = 1$, there is unique ω^* with $\partial\mathcal{C}/\partial\omega_i(\omega^*) = m_i$.

Theorem 4.3 (1|10|16 partition). At $\mathbf{M}^5(\mathcal{V})$, $\text{Bal}_{\mathcal{C}}$ yields sector masses $(1/27, 10/27, 16/27)$ on $27 = \mathbf{1} \oplus \mathbf{10} \oplus \mathbf{16}$.

4.2 Potential and vacuum

Definition 4.4 (Potential).

$$V(X) = \alpha \langle X^\#, X^\# \rangle + \beta [\text{Tr}(X^2) - c_2]^2 + \gamma \text{Tr}(X^2)^k, \quad \alpha, \beta, \gamma > 0, \quad k \text{ even}, \quad (4.2)$$

is the profinite limit of graph penalties under $\text{Bal}_{\mathcal{C}}$.

Theorem 4.5 (EIII vacuum). V is coercive on compact $\mathbf{E}_6$. Global minima lie on rank-one EIII $\mathcal{M} = \mathbf{E}_6/H$; the origin is a saddle with 32 Goldstone directions.

4.3 Universe datum

Definition 4.6 (Universe). The *universe datum* is

$$\mathfrak{U}^* = \mathbf{C}^\infty(\mathcal{V}) = \varprojlim_{n \rightarrow \infty} \mathbf{C}^n(\mathcal{V}), \quad (4.3)$$

stabilizing at $J_3(\mathbb{O})$ with Albert graph $\mathcal{G}_{\text{Albert}}$, capacity maximizer ω^* , and vacuum $\Phi_\star \in \mathcal{M}$.

Theorem 4.7 (Universe from the mirror). $\mathfrak{U}^*$ carries: (i) 27 stable Jordan coordinates; (ii) 1|10|16 capacity partition; (iii) $\mathbf{E}_6$ gauge sector with $\mathfrak{h} = \mathfrak{so}_{10} \oplus \mathfrak{u}_1$; (iv) EIII coset $\mathfrak{m} = \mathbf{16}_{-3} \oplus \mathbf{16}_{+3}$.

Proof. Items (i)–(ii) from Theorems 3.2 and 4.3. Items (iii)–(iv) from Theorem 3.9 and standard $\mathbf{E}_6$ branching. $\square$

$$S[\Phi] = \int d^4x \sqrt{|g|} \left[\frac{1}{2} K_{AB} \nabla_\mu \Phi^A \nabla^\mu \Phi^B + V(\Phi) \right], \quad (4.4)$$

with K the unique H -invariant metric on $\mathfrak{m}$.

5 Observers, Duality, and As Above So Below

The universe datum $\mathfrak{U}^*$ is a global profinite fixed point. An *observer* is the same construction at the heterotic rung, localized to one collapsed branch of the $\mathbf{E}_8 \times \mathbf{E}_8$ mirror quotient. Duality is not imported: it is the product-mirror pairing forced at $\mathbf{M}^7(\mathcal{V})$ and broken only by collapse into the σ -anti-invariant sector.

5.1 As above, so below

Definition 5.1 (Above–below correspondence). *Above* is the stabilized profinite limit $\mathfrak{U}^* = \mathbf{C}^\infty(\mathcal{V})$ of Definition 4.6. *Below* is a localized fixed-point datum on the mirror quotient $\hat{Q} = Q/\sigma$ at $\mathbf{M}^7(\mathcal{V}) = \mathbf{E}_8 \times \mathbf{E}_8$, with $\sigma : (g_1, g_2) \mapsto (g_2, g_1)$ the product mirror (Theorem 3.7). The correspondence

$$\mathfrak{U}^* \longleftrightarrow \text{Fix}(\hat{\mathcal{C}}_\sigma), \quad \hat{\mathcal{C}}_\sigma := \text{Bal}_{\mathcal{C}} \circ \Pi_{\sigma=-1}, \quad (5.1)$$

identifies global capacity balance with anti-invariant collapse on $\hat{Q}$.

Proposition 5.2 (Correspondence principle). *Every sector partition proved above on $\mathfrak{U}^*$ —the $1|10|16$ split, the three diagonal directions of $\mathfrak{d} \subset J_3(\mathbb{O})$, the **16** bundle on $\hat{Q}$ —has a below-level avatar on $H^\bullet(\hat{Q}, V_{16})^{\sigma=-1}$. Capacity weights $(1/27, 10/27, 16/27)$ above induce family weights on the three observer slots below.*

5.2 Duality from $E_8 \times E_8$ collapse

At $M^7(\mathcal{V})$ the mirror acts by product doubling: $E_8 \times E_8$ with involution σ exchanging factors. Before collapse, every mode carries a mirror partner; after collapse to $M^8(\mathcal{V}) = E_6$, capacity descent selects a single reduced structure group.

Definition 5.3 (Heterotic collapse). The *heterotic collapse* is the capacity-balanced Cartan descent $\Pi_{\text{het}} : E_8 \times E_8 \rightarrow E_6$ of Theorem 3.9. The *observer collapse* is the orthogonal projector

$$\Pi_{\sigma=-1} : \mathcal{H}(Q, V_{16}) \rightarrow \mathcal{H}(Q, V_{16})^{\sigma=-1}, \quad \Pi_{\sigma=-1} = \frac{1}{2}(\text{id} - \sigma), \quad (5.2)$$

on the **16**-sector Hilbert space at the heterotic rung.

Theorem 5.4 (Duality theorem). *Duality is the $\mathbb{Z}_2$ pairing $(v, \sigma v)$ on $\mathcal{H}(Q, V_{16})$ induced by the product mirror at $M^7(\mathcal{V})$. It is exact at the heterotic rung and broken precisely by $\Pi_{\sigma=-1}$: observable states are anti-invariant, mirror partners are σ -invariant and quotient-non-observable. The **16/16** coset pairing of Proposition 2.7 is the infinitesimal shadow of this product duality.*

Proof sketch. $\sigma^2 = \text{id}$ on Q ; $\mathcal{H}(Q, V_{16})$ splits into ± 1 eigenspaces. Product exchange on $E_8 \times E_8$ is the only $\mathbb{Z}_2$ structure at $n = 7$ not already resolved at $n \leq 6$. Theorem 3.9 shows $\text{Bal}_{\mathcal{C}}$ selects the reduced datum; Definition 6.1 selects the anti-invariant half. $\square$

5.3 What is an observer?

Definition 5.5 (Observer index). Let $\mathfrak{d} = \text{span}_{\mathbb{R}}\{e_1, e_2, e_3\}$ be the diagonal Cartan subalgebra of $J_3(\mathbb{O})$ with $\text{Tr}(e_i \circ e_j) = \delta_{ij}$. An *observer index* is a choice $\iota \in \{1, 2, 3\}$ of diagonal direction, equivalently a class in

$$\mathcal{I} = H^1(\hat{Q}, V_{16})^{\sigma=-1} / \text{Stab}(\mathfrak{d}), \quad |\mathcal{I}| = 3. \quad (5.3)$$

The index ι is the only *arbitrary input* specifying *which* observer: void and mirror fix the slot space $\mathcal{I}$; a concrete worldline requires choosing $\iota \in \mathcal{I}$.

Definition 5.6 (Observer as collapsed state). Fix $\iota \in \mathcal{I}$. An *observer* is a normalized collapsed state

$$|\omega_\iota\rangle \in \mathcal{H}_{\text{obs}} = \bigoplus_{\iota=1}^3 \mathcal{H}_\iota, \quad \mathcal{H}_\iota := H^1(\hat{Q}, V_{16})^{\sigma=-1, \iota}, \quad (5.4)$$

satisfying $\Pi_{\sigma=-1}|\omega_\iota\rangle = |\omega_\iota\rangle$ and $\hat{i}|\omega_\iota\rangle = \iota|\omega_\iota\rangle$, where $\hat{i}$ is the family index operator on $\mathfrak{d}$. The state $|\omega_\iota\rangle$ is a *collapsed quantum datum*: a σ -anti-invariant cohomology class, not a postulated particle insertion.

Remark 5.7 (Where observers arise). Observers arise at $M^7(\mathcal{V}) = E_8 \times E_8$, not at the void and not at E_6 . The heterotic rung is the first rung with enough mirror symmetry for a $\mathbb{Z}_2$ quotient and three independent anti-invariant 1-classes (Theorem 6.4). Descent to E_6 supplies dynamics; the observer slots are already fixed on $\hat{Q}$.

5.4 Fixed-point theorem: the clincher

Definition 5.8 (Mirror quotient). Let Q be the Stone lift of the stabilized Albert tower at $\mathbf{M}^7(\mathcal{V})$ and σ the product-mirror involution. The *mirror quotient* is $\hat{Q} = Q/\sigma$. The holomorphic **16**-bundle $V_{16} \rightarrow \hat{Q}$ is the continuum limit of the cellular cosheaf on the 1|10|16 partition. A mode is *observable* iff it lies in the σ -anti-invariant sector after observer collapse $\Pi_{\sigma=-1}$; this is the below-level avatar of global capacity balance (5.1). The quotient is therefore an *observer-theoretic* construction: physics is not on Q but on the collapsed branch selected by $\hat{\mathcal{C}}_\sigma$.

Theorem 5.9 (Observer fixed-point theorem). Let $\bar{\partial}_{V_{16}}$ be the Dolbeault operator on (Q, V_{16}) and σ its antiholomorphic lift. The equivariant Lefschetz index

$$\text{ind}_\sigma(\bar{\partial}_{V_{16}}) = \text{Tr}_{H^1(Q, V_{16})}(\sigma) = -3 \quad (5.5)$$

has $|\text{ind}_\sigma| = 3$ anti-invariant fixed-point contributions, each an observer slot. Moreover,

$$\dim H^1(\hat{Q}, V_{16})^{\sigma=-1} = 3, \quad (5.6)$$

and the map $\iota \mapsto |\omega_\iota\rangle$ is a bijection between $\mathcal{I}$ and a basis of this space.

Proof. Cellular count on $\mathcal{T}_7$ (Appendix A.2); Atiyah–Bott equivariant index [7]; passage to $H^1(\hat{Q}, V_{16})$ via the $\mathbb{Z}_2$ quotient. $\square$

Corollary 5.10 (Observer from fixed points). An observer is a fixed-point label of σ on $H^1(Q, V_{16})$: not a point of Q fixed by σ (there are none in the **16**-sector), but an anti-invariant cohomology class whose Lefschetz contribution is -1 . The fixed-point theorem is the clincher: it converts global mirror data into exactly three individual observer slots.

5.5 Lifetime chart: index to time

An observer index ι is static data. A *lifetime* is a trajectory inside slot ι , charted by a time parameter.

Definition 5.11 (Lifetime map). Fix $\iota \in \mathcal{I}$. A *lifetime chart* is a pair

$$(\tau_\iota, g_\iota), \quad g_\iota : \mathbb{R} \longrightarrow \mathcal{S}_\iota, \quad (5.7)$$

where τ_ι is the observer’s time parameter and $\mathcal{S}_\iota$ is the soldered state bundle over the ι -sector: sections of $V_{16}|_{\hat{Q}}$ restricted to the ι -diagonal cell, with values in the order parameter Φ (Section 6.2). The map g_ι is *admitted* by the capacity flow on $\mathcal{G}_{\text{Albert}}$ when

$$\frac{d}{d\tau_\iota} g_\iota(\tau_\iota) = -K^{-1} \nabla V(g_\iota(\tau_\iota))|_\iota + \partial_{\tau_\iota} \sigma_{\text{sol}}|_{H_2(\mathbb{H})}, \quad (5.8)$$

with K the H -invariant coset metric and the second term the soldering clock in the quaternionic block.

Definition 5.12 (Lifetime index). The *lifetime index* is

$$\Lambda(\iota, \tau) := (\iota, \tau, \mathcal{C}_\iota(\tau)), \quad \mathcal{C}_\iota(\tau) := \omega_\iota^*(g_\iota(\tau)), \quad (5.9)$$

recording which observer (ι), which time (τ), and which capacity cell the trajectory occupies. Two lifetimes are the same iff their indices Λ agree.

Proposition 5.13 (Capacity as dual mirror of lifetime index). *The capacity functional $\mathcal{C}$ of Definition 4.1 is the dual mirror of mirror doubling: $\mathbf{M}$ refines sectors; $\text{Bal}_{\mathcal{C}}$ balances them to the unique maximizer ω^* . The lifetime index $\Lambda(\iota, \tau)$ is the temporal avatar of this balance on the heterotic rung: for each observer ι , the chart g_{ι} tracks the ι -cell weight ω_{ι}^* along admitted flow (5.8). Above, $\text{Bal}_{\mathcal{C}}$ fixes the global partition $(1/27, 10/27, 16/27)$; below, Λ records which cell an individual observer occupies at time τ . This partially closes the “capacity not forced” gap: capacity is not an independent physical postulate but the balancing dual of $\mathbf{M}$, instantiated per observer through the lifetime index.*

Remark 5.14 (Epistemic status). Proposition 5.13 does not prove that every admitted trajectory must use $\text{Bal}_{\mathcal{C}}$ weights—that remains tied to Axiom 2.5 and the universe fixed point (Definition 4.6). It *does* identify capacity with observer-level time evolution once the above–below correspondence (5.1) is accepted.

Theorem 5.15 (Index–lifetime mapping). *For each $\iota \in \mathcal{I}$ there exists a maximal lifetime interval $I_{\iota} \subset \mathbb{R}$ and a chart $g_{\iota} : I_{\iota} \rightarrow \mathcal{S}_{\iota}$ satisfying (5.8) with initial condition $g_{\iota}(\tau_{\text{birth}}) = \Phi_{\star}|_{\iota}$. The correspondence*

$$\iota \longmapsto (\tau_{\text{birth}}, \tau_{\text{death}}, g_{\iota}) \quad (5.10)$$

is functorial under mirror descent $\mathbf{E}_8 \times \mathbf{E}_8 \rightarrow \mathbf{E}_6$: capacity collapse above determines the admissible time range I_{ι} below.

Proof sketch. Existence of g_{ι} is standard gradient flow of V on the ι -restricted coset; uniqueness up to reparametrization on each sector follows from Proposition 5.17 under convex V . Functoriality follows because $\text{Bal}_{\mathcal{C}}$ at $n = 8$ fixes ω^* and hence the initial datum $\Phi_{\star}|_{\iota}$; soldering (Definition 6.7) supplies the clock direction $\partial_{\tau_{\iota}} \sigma_{\text{sol}} \subset H_2(\mathbb{H})$. $\square$

Proposition 5.16 (Mirror descent preserves lifetime chart class). *Cartan mirror descent $\Pi_{\text{het}} : \mathbf{E}_8 \times \mathbf{E}_8 \rightarrow \mathbf{E}_6$ at $n = 7 \rightarrow 8$ (Theorem 3.9, Definition 5.3) preserves the lifetime chart class on each observer slot $\iota \in \mathcal{I}$. If $(\tau_{\iota}, g_{\iota})$ is admitted by (5.8) at the heterotic rung, the pushed-forward chart*

$$(\tau_{\iota}, \Pi_{\text{het}} \circ g_{\iota}) \quad (5.11)$$

satisfies the same equation on $\mathcal{S}_{\iota} \subset \mathbf{E}_6/H$ with the descended capacity functional $\mathcal{C}|_{\mathcal{M}}$ and the soldering clock $\partial_{\tau_{\iota}} \sigma_{\text{sol}}|_{H_2(\mathbb{H})}$ transported by Proposition 6.9. Charts related by orientation-preserving reparametrization of τ_{ι} descend to the same class. The observer index set $\mathcal{I}$, initial data $\Phi_{\star}|_{\iota}$, and admissible interval I_{ι} are unchanged on $\hat{Q}$ (Proposition 3.8, Proposition A.8).

Proof sketch. Π_{het} is capacity-balanced Cartan collapse: it reduces the gauge datum to $\mathbf{E}_6/H$ while preserving the σ -anti-invariant $H^1(\hat{Q}, V_{16})^{\sigma=-1}$ basis and hence $\mathcal{I}$ (Proposition 3.8). $\text{Bal}_{\mathcal{C}}$ at $n = 8$ fixes ω^* and $\Phi_{\star}|_{\iota}$, so the gradient term in (5.8) pushes forward equivariantly. The soldering clock block is transported functorially on $n = 6, 7, 8$ by Proposition 6.9; reparametrization invariance on each ι -sector follows from Proposition 5.17 on the massive complement once radiative gaps are included. $\square$

Proposition 5.17 (Gradient-flow uniqueness on the ι -sector). *Fix $\iota \in \mathcal{I}$ and suppose $V|_{\mathcal{S}_{\iota}}$ is strictly convex along ι -restricted coset directions and $\text{Hess}_{\iota} V(\Phi_{\star}) \succ 0$ on the massive complement of the soldering block (as after Coleman–Weinberg lifting in Lemma 6.42). Then for initial datum $g_{\iota}(\tau_{\text{birth}}) = \Phi_{\star}|_{\iota}$, any two admitted solutions of (5.8) on a common interval differ only by an orientation-preserving reparametrization of τ_{ι} . The soldering clock term $\partial_{\tau_{\iota}} \sigma_{\text{sol}}|_{H_2(\mathbb{H})}$ is tangent to the 4-dimensional block and does not affect uniqueness of the coset trajectory in the 28 massive directions.*

Proof sketch. Write $g_\iota = (g_\iota^{\text{cos}}, g_\iota^{\text{sol}})$ for the split into coset and soldering components. Strict convexity of $V|_{\mathcal{S}_\iota}$ in the coset directions yields a unique gradient trajectory g_ι^{cos} for each initial datum; standard arguments for strictly convex functionals on Hilbert manifolds give uniqueness up to time reparametrization when the velocity norm is fixed along the flow. The soldering term is independent of ∇V on the coset complement once σ_{sol} is fixed (Lemma 6.8), so it determines τ_ι only modulo the same reparametrization class. $\square$

Conjecture 5.18 (Uniqueness of lifetime chart). *Given ι and $\Phi_\star|_\iota$, the lifetime chart (τ_ι, g_ι) is unique up to reparametrization of τ_ι on the ι -sector once $V|_{\mathcal{S}_\iota}$ is strictly convex as in Proposition 5.17. Functorial stability across exceptional rungs $n \leq 8$ is supplied by Proposition 5.16 and Proposition 6.9; the residual conjecture is uniqueness along the tree-level massless $\mathfrak{m}_{+1}$ sector of Lemma 6.22, without branching into a second admissible flow on $\mathcal{S}_\iota$.*

Theorem 5.19 (Lifetime uniqueness, conditional reduction). *Assume Proposition 5.16, Proposition 6.9, Theorem 5.15, and Proposition 5.17. Then Conjecture 5.18 reduces to the tree-level massless sector only: given ι and $\Phi_\star|_\iota$, any two admitted lifetime charts on $\mathcal{S}_\iota$ that agree on the 28 radiatively massive directions of Theorem 6.19 differ only by orientation-preserving reparametrization of τ_ι once the four $\mathfrak{m}_{+1}$ Goldstone directions of Lemma 6.22 are fixed by Proposition 6.13. Functorial transport across $n \leq 8$ is settled; the residual open step is uniqueness along the massless soldering flow in $H_2(\mathbb{H})$.*

Proof sketch. Proposition 5.16 and Proposition 6.9 close functorial transport on exceptional rungs $n = 6, 7, 8$. Proposition 5.17 settles the 28-dimensional massive complement under Coleman–Weinberg lifting (Theorem 6.19). Lemma 6.22 isolates the residual 4-dimensional $\mathfrak{m}_{+1}$ block; Proposition 6.13 fixes the timelike axis Φ^0 and hence the clock coupling modulo reparametrization. Any remaining ambiguity is therefore confined to the tree-level massless soldering directions. $\square$

Remark 5.20 (Residual gap for the full lifetime theorem). Proposition 5.17 settles uniqueness on each ι -sector under the convexity hypothesis satisfied in the radiatively gapped regime of Theorem 6.19; Proposition 5.16 settles mirror descent $n = 7 \rightarrow 8$. What remains for a *complete* functorial theorem is: (i) extend uniqueness to the tree-level massless $\mathfrak{m}_{+1}$ sector and non-convex saddle regimes of V ; and (ii) match reparametrizations of τ_ι with the soldering clock of Proposition 6.13 on the massless Goldstone reservoir. Theorem 5.19 records the reduction; Open Problem O1 is thereby confined to the massless sector.

Remark 5.21 (Arbitrary input and determinism). The void and mirror fix the observer *slot space* $\mathcal{I}$ and the dynamics (5.8). Choosing $\iota \in \mathcal{I}$ is the minimal arbitrary input identifying *who* observes; choosing $\tau \in I_\iota$ identifies *when* within that lifetime. No further postulate of a conscious subject is introduced: an observer *is* the collapsed anti-invariant fixed-point class with its admitted time chart.

6 The Physical Universe

Mathematical data of $\mathfrak{U}^\star$ becomes physical through mirror quotient observability (Section 5), observer collapse at $M^7(\mathcal{V})$, and soldering on the Jordan–octonionic block.

6.1 Physical sector

Physical configurations are observer-collapsed states on $\hat{Q}$ (Definitions 5.8, 5.6, 6.1).

Definition 6.1 (Physical configurations). A configuration is *observable* if it lies in the σ -anti-invariant sector of the mirror quotient $\hat{Q} = Q/\sigma$ (Definition 5.8). Fermions are cohomology classes in $H^1(\hat{Q}, V_{16})^{\sigma=-1}$; each observer $|\omega_\iota\rangle$ is one such class (Theorem 5.9). Observable states are those surviving observer collapse $\Pi_{\sigma=-1}$ (Definition 5.3); σ -invariant mirror doubles are quotient-non-observable.

Proposition 6.2 (No-go: P2 on the unquotiented coset). *Void, mirror, and capacity balancing fix the number of horizontal family cells ($k = 3$, Theorem 4.3) and organize them by triality on $\mathfrak{d}$, but they do not derive chiral Spin(10) matter on the unquotiented coset. The EIII matter directions $\mathfrak{m} = \mathbf{16}_{-3} \oplus \overline{\mathbf{16}}_{+3}$ (Theorem 4.7) are a mirror-symmetric pair (Proposition 2.7); selecting one chiral $\mathbf{16}$ per family with no $\overline{\mathbf{16}}$ partners is Postulate P2, not a consequence of stabilisation alone. A bare Yukawa $\bar{\psi} \Phi \psi$ with $\Phi \in \{\mathbf{16}, \overline{\mathbf{16}}\}$ is not H -invariant on the unquotiented coset.*

Remark 6.3 (Observer collapse eliminates P2). Proposition 6.2 is not overridden: it records an honest limit on *postulated* chiral insertion atop the mirror-symmetric coset. The elimination route *redefines* the matter sector. Observer collapse $\Pi_{\sigma=-1}$ together with Definition 6.1 declares physical fermions to be σ -anti-invariant classes on $\hat{Q}$; mirror partners live in $H^1(\hat{Q}, V_{16})^{\sigma=+1}$ and are not observable. Chirality is then a theorem of quotient observability, not an added input.

Theorem 6.4 (Chirality from observer collapse). *Under Definition 6.1 and Theorem 5.9,*

$$\dim H^1(\hat{Q}, V_{16})^{\sigma=-1} = 3, \quad (6.1)$$

*with one chiral $\mathbf{16}$ of Spin(10) per observer index $\iota \in \mathcal{I}$. No $\overline{\mathbf{16}}$ partners appear in the physical sector because mirror partners are σ -invariant, not anti-invariant. Hence $n_F = 3$. This **replaces Postulate P2**: chirality is not postulated; it is the quotient definition of observability together with the observer fixed-point count.*

Proof. Theorem 5.9 gives the anti-invariant dimension and the bijection $\iota \mapsto |\omega_\iota\rangle$. Physical configurations are anti-invariant by Definition 6.1; σ -invariant modes are mirror doubles and quotient-non-observable. Cellular details in Appendix A.2. $\square$

Corollary 6.5 (Three families). *Theorem 6.4 yields $n_F = 3$ chiral families without importing heterotic axioms as primitive data [12].*

6.2 Spacetime from the climb

Four-dimensional Lorentzian geometry is not a consequence of the E_6 coset metric alone. The mirror climb supplies a quaternionic clock block; observer lifetime charts (Section 5.5) admit a soldering map that imports signature into the contact layer.

Definition 6.6 (Jordan–octonionic clock block). Identify off-diagonal Jordan slots of $J_3(\mathbb{O})$ with the 2×2 Hermitian–octonionic model $H_2(\mathbb{O})$. The determinant $\det$ carries signature $(1, 9)$ on $H_2(\mathbb{O}) \cong \mathbb{R}^{10}$. Restrict to the quaternionic subalgebra $\mathbb{H} \subset \mathbb{O}$ fixed by the mirror involution at the $4 \rightarrow 5$ Jordan exit (Theorem 3.2). The embedded block $H_2(\mathbb{H}) \subset H_2(\mathbb{O})$ is four-dimensional; $\det|_{H_2(\mathbb{H})}$ has Lorentzian signature $(-, +, +, +)$.

Definition 6.7 (Soldering). Let $\sigma_{\text{sol}} : H_2(\mathbb{H}) \hookrightarrow J_3(\mathbb{O})$ be the canonical embedding of Definition 6.6. The *soldering map* identifies four clock components $\Phi^0, \dots, \Phi^3$ with $\text{Im}(\sigma_{\text{sol}})$; the world manifold has $d = 4$.

Lemma 6.8 (F_4 equivariance fixes the quaternionic embedding). *Let $\iota_{\mathbb{H}} : \mathbb{H} \hookrightarrow \mathbb{O}$ be an associative subalgebra embedding. Extend it to the Jordan–octonionic block $H_2(\mathbb{H}) \hookrightarrow H_2(\mathbb{O}) \hookrightarrow J_3(\mathbb{O})$ by placing quaternion entries in a single off-diagonal slot. Among such embeddings compatible with*

- (i) *mirror functoriality on the climb, in particular the $4 \rightarrow 5$ Jordan exit that carries $\mathbf{M}^3(\mathcal{V}) = \mathbb{H}$ into $\mathbf{M}^5(\mathcal{V}) \cong J_3(\mathbb{O})$ (Theorem 3.2);*
- (ii) *$F_4 = \text{Aut}(J_3(\mathbb{O}))$ equivariance of the soldering map σ_{sol} (Definition 6.7); and*
- (iii) *the above–below correspondence on the diagonal Cartan $\mathfrak{d} = \text{span}\{e_1, e_2, e_3\}$ (Definition 5.1),*

the orbit under F_4 is a single G_2 -conjugacy class, with $G_2 = \text{Aut}(\mathbb{O}) \subset F_4$ the stabilizer of a generic diagonal frame. Fixing an observer index $\iota \in \mathcal{I}$ (Definition 5.5) selects a unique representative $\mathbb{H}_\iota \subset \mathbb{O}$ and hence a unique soldered block $H_2(\mathbb{H}) \subset H_2(\mathbb{O})$.

Proof sketch. Associative subalgebras $\mathbb{H} \subset \mathbb{O}$ form one G_2 -orbit [5]. The Jordan envelope $J_3(\mathbb{O})$ carries three off-diagonal octonion slots; F_4 permutes them while preserving $\mathfrak{d}$. Mirror descent at $n = 4 \rightarrow 5$ singles out the associative subalgebra from rung 3, so the F_4 -admissible embeddings are those G_2 -conjugates compatible with capacity weights on $\mathfrak{d}$ (Proposition 5.2). Observer index ι labels one diagonal direction e_ι , breaking the residual triality and pinning $\mathbb{H}_\iota$ uniquely. $\square$

Proposition 6.9 (Exceptional rungs preserve $H_2(\mathbb{H})$ soldering). *On stabilized exceptional rungs $n = 6, 7, 8$, the mirrors of Proposition 3.10 transport the soldered block $H_2(\mathbb{H}) \subset H_2(\mathbb{O}) \hookrightarrow J_3(\mathbb{O})$ of Definition 6.6 without introducing a second clock sector:*

- (i) E_8 ($n = 6$): *automorphism mirror couples the two $\mathbb{O}$ sectors of (A.1); the associative subalgebra $\mathbb{H} \subset \mathbb{O}$ from $\mathbf{M}^3(\mathcal{V})$ is stabilized by $G_2 = \text{Aut}(\mathbb{O}) \subset E_8$, and $\det|_{H_2(\mathbb{H})}$ retains Lorentzian signature $(-, +, +, +)$.*
- (ii) $E_8 \times E_8$ ($n = 7$): *product mirror σ stabilizes the diagonal $\delta(E_8)$ and the $J_3(\mathbb{O})$ chart; the soldering embedding σ_{sol} of Definition 6.7 is σ -equivariant on $H_2(\mathbb{H})$, hence survives passage to $\hat{Q} = Q/\sigma$.*
- (iii) E_6 ($n = 8$): *Cartan mirror Π_{het} collapses gauge data to $H = \text{Spin}(10) \times U(1)$ while preserving $F_4 = \text{Aut}(J_3(\mathbb{O}))$ equivariance (Lemma 6.8); the four clock components $\Phi^0, \dots, \Phi^3$ remain the $\mathfrak{m}_{+1}$ Goldstone block of Lemma 6.22.*

Functorial descent along $\mathbf{M}^6 \rightarrow \mathbf{M}^7 \rightarrow \mathbf{M}^8$ therefore identifies a single F_4 -orbit of soldered blocks across exceptional rungs.

Proof sketch. (i) Freudenthal closure (Lemma 3.4) embeds $G_2 \times F_4 \subset E_8$; G_2 fixes $\mathbb{H} \subset \mathbb{O}$ and hence $H_2(\mathbb{H})$ under the Jordan envelope. (ii) Product mirror acts on $E_8 \times E_8$ by factor exchange; the diagonal $J_3(\mathbb{O})$ chart and $H_2(\mathbb{H})$ block are σ -fixed, so soldering is unchanged on $\hat{Q}$. (iii) Cartan descent preserves the 1|10|16 partition and F_4 action on $J_3(\mathbb{O})$ (Theorem 3.9); the 4 soldering directions split as $\mathfrak{m}_{+1}$ in Lemma 6.22. Observer index ι selects the representative within the G_2 -conjugacy class (Lemma 6.8). $\square$

Proposition 6.10 (Coset metric). *$K|_{\mathfrak{m}}$ is positive-definite and H -invariant. Lorentzian signature is carried by σ_{sol} , not by Wick rotation of K .*

Proposition 6.11 (No-go: coset metric alone). *Mirror stabilisation at $J_3(\mathbb{O})$ and E_6 fixes the Euclidean coset metric $K|_{\mathfrak{m}}$ (Proposition 6.10) and the $32 = 28 + 4$ mass split (Theorem 6.28), but it does not select $d = 4$, a Lorentzian soldering form, or the $SL(2, \mathbb{O}) \cong Spin(1, 9)$ frame. The $H_2(\mathbb{O}) \cong \mathbb{R}^{1,9}$ embedding of Definition 6.6 motivates soldering; it is not a theorem from void and mirror alone until observer-admitted lifetime charts supply the clock block (Theorem 6.12).*

Theorem 6.12 (Soldering from mirror climb and lifetime chart). *The mirror climb selects a canonical four-dimensional Lorentzian subspace $H_2(\mathbb{H}) \subset H_2(\mathbb{O})$ (Definition 6.6). For each observer index $\iota \in \mathcal{I}$, the admitted lifetime flow (5.8) of Definition 5.11 couples τ_ι to the soldering clock $\partial_{\tau_\iota} \sigma_{\text{sol}}|_{H_2(\mathbb{H})}$. Define σ_{sol} to identify $\Phi^0, \dots, \Phi^3$ with this block. Then $d = 4$, the induced vierbein (Proposition 6.14) carries signature $(-, +, +, +)$, and soldering is **derived** from void, mirror, and observer collapse—not postulated.*

Proof sketch. Step 1 (quaternionic rung). $M^3(\mathcal{V}) = \mathbb{H}$ has $\dim_{\mathbb{R}} \mathbb{H} = 4$, the first Hurwitz rung with an associative non-real division algebra (Theorem 3.2).

Step 2 (Jordan embedding). Lemma 6.8 fixes $\mathbb{H}_\iota \subset \mathbb{O}$ and $H_2(\mathbb{H}) \subset H_2(\mathbb{O})$ once the observer index ι is chosen; mirror involution at $n = 4 \rightarrow 5$ carries the rung-3 datum $M^3(\mathcal{V}) = \mathbb{H}$ into this block.

Step 3 (signature). The restriction of $\det$ on $H_2(\mathbb{H})$ is the Minkowski norm; the four imaginary quaternion units supply the clock directions. The coset metric $K|_{\mathfrak{m}}$ remains Euclidean; Lorentzian signature enters through σ_{sol} .

Step 4 (observer lifetime). The index–lifetime map (Theorem 5.15) requires an admitted chart (τ_ι, g_ι) whose flow (5.8) includes the soldering clock term in $H_2(\mathbb{H})$. Observer collapse (Definition 5.6) on the σ -anti-invariant sector fixes the initial datum $\Phi_\star|_\iota$ and thereby selects the soldered bundle $\mathcal{S}_\iota$ over which g_ι evolves (Section 5.5).

Remaining gap. Functorial transport across $n = 6, 7, 8$ is settled by Proposition 6.9; global uniqueness reduces to the tree-level massless $\mathfrak{m}_{+1}$ sector (Theorem 6.16, Conjecture 6.15). The ι -local embedding is settled by Lemma 6.8. $\square$

Proposition 6.13 (Unique timelike soldering direction in $H_2(\mathbb{H})$). *Fix $\iota \in \mathcal{I}$ and an admitted lifetime chart (τ_ι, g_ι) of Definition 5.11. Along g_ι , the soldering time parameter τ_ι couples to the zeroth clock component Φ^0 via*

$$\frac{d}{d\tau_\iota} \Phi^0|_{g_\iota(\tau_\iota)} = \partial_{\tau_\iota} \sigma_{\text{sol}}|_{H_2(\mathbb{H})}. \quad (6.2)$$

In the Lorentzian block $H_2(\mathbb{H})$ of Definition 6.6, the zeroth clock component Φ^0 is the unique time-like soldering direction admitted by observer collapse: the σ -anti-invariant sector of $\hat{Q}$ breaks the $Spin(1, 9)$ frame to the ι -restricted diagonal cell, and capacity flow on $\mathcal{G}_{\text{Albert}}$ fixes the sign convention of $\det|_{H_2(\mathbb{H})}$. The spatial components $\Phi^1, \dots, \Phi^3$ are the remaining spacelike directions of the same block and do not carry the Page–Wootters clock.

Proof sketch. Equation (6.2) is the Φ^0 -component of (5.8). On $H_2(\mathbb{H})$, $\det$ has signature $(-, +, +, +)$, so the timelike cone is one-dimensional up to sign; Lemma 6.8 and observer index ι fix the F_4 -equivariant orientation, and the above–below correspondence (Definition 5.1) aligns the negative eigen-direction with the ι -cell clock. No other component of $\text{Im}(\sigma_{\text{sol}})$ is timelike. $\square$

Proposition 6.14 (Emergent vierbein). *For a coset section $g(x) \in E_6$ with $g^{-1} \partial_\mu g = A_\mu + e_\mu$, the $\mathfrak{m}$ -component defines a composite vierbein. Under Theorem 6.12, $\eta_{\mu\nu} = e_\mu^a e_\nu^b \eta_{ab}$ has signature $(-, +, +, +)$.*

Conjecture 6.15 (Functorial uniqueness of soldering). *Given $\iota \in \mathcal{I}$ and $\Phi_\star|_\iota$, the soldering block $H_2(\mathbb{H}) \subset H_2(\mathbb{O})$, the clock coupling (6.2), and the timelike axis Φ^0 are **already unique** by Lemma 6.8 and Proposition 6.13. Functorial transport on $n = 6, 7, 8$ is supplied by Proposition 6.9; the residual conjecture is alignment of the tree-level massless $\mathfrak{m}_{+1}$ Goldstones with the observer vierbein of Proposition 6.14 and with ι -sector lifetime charts (Conjecture 5.18).*

Theorem 6.16 (Soldering uniqueness, conditional reduction). *Assume Proposition 6.9, Lemma 6.8, Proposition 6.13, Theorem 6.12, and Theorem 5.19. Then Conjecture 6.15 reduces to the tree-level massless sector only: the soldered block $H_2(\mathbb{H})$, clock coupling (6.2), and F_4 -equivariant vierbein of Proposition 6.14 are unique on each ι -sector once the four $\mathfrak{m}_{+1}$ Goldstone directions of Lemma 6.22 are included. Functorial transport on $n = 6, 7, 8$ is proved by Proposition 6.9; the residual open step is nonperturbative alignment of the massless $\sigma = +1$ reservoir with the observer clock along g_ι .*

Proof sketch. Proposition 6.9 closes transport of σ_{sol} under automorphism, product, and Cartan mirrors. Lemma 6.8 and Proposition 6.13 fix the ι -local block and timelike axis. The vierbein of Proposition 6.14 is determined on the massive 28-direction complement by Theorem 6.19; Lemma 6.22 isolates the residual 4-dimensional $\mathfrak{m}_{+1}$ sector. Theorem 5.19 matches clock reparametrizations along g_ι . Any remaining ambiguity is confined to massless Goldstone–vierbein alignment in the $\sigma = +1$ reservoir. $\square$

Remark 6.17 (Residual gap for the full soldering theorem). Lemma 6.8, Proposition 6.13, Proposition 6.9, and Theorem 6.12 supply $d = 4$, Lorentzian signature, ι -local clock coupling, and exceptional-rung transport without postulate. What remains for a *complete* functorial theorem is: (i) identify the resulting vierbein with the composite of Proposition 6.14 on E_6/H in the tree-level massless $\mathfrak{m}_{+1}$ sector; and (ii) close Conjecture 5.18 on the massless Goldstone reservoir. Theorem 6.16 records the reduction; Open Problem **O2** is thereby confined to the massless sector.

Remark 6.18 (Epistemic status of P1). Proposition 6.11 records the honest no-go for the coset metric alone. Theorem 6.12 **derives** soldering and $d = 4$ from mirror climb plus observer lifetime charts; ι -local uniqueness is supplied by Lemma 6.8 and Proposition 6.13; exceptional-rung transport is supplied by Proposition 6.9. The residual open input is tree-level massless-sector closure (Theorem 6.16, Conjecture 6.15, Remark 6.17, Open Problem **O2** in Section 11).

Theorem 6.19 (Semiclassical reflection positivity along observer lifetimes). *Fix $\iota \in \mathcal{I}$ and a lifetime chart (τ_ι, g_ι) admitted by capacity flow (Definitions 5.11, 5.12). On EIII, with radiative mass $\mu^2 > 0$ on 28 coset scalars (Coleman–Weinberg), the Euclidean σ -model (4.4) satisfies reflection positivity along the observer clock τ_ι : the soldering map identifies the soldered time component Φ^0 with τ_ι on the ι -restricted trajectory $g_\iota(\tau_\iota) \in \mathcal{S}_\iota$, and the semiclassical OS reflection plane is taken at fixed $\tau_\iota = 0$ in that chart. Equivalently, RP holds along the Page–Wootters clock carried by the quaternionic block $H_2(\mathbb{H})$ once the observer index ι selects the diagonal cell.*

Proof sketch. The ι -restricted gradient flow (5.8) is a semiclassical saddle expansion about $\Phi_\star|_\iota$ on the massive 28-dimensional coset complement of the 4 soldering directions. With $\mu^2 > 0$, the quadratic part defines a reflection-positive Gaussian measure; nonnegativity of the capacity-derived interaction potential (Theorem 4.5) and compactness on EIII transfer positivity to the perturbed measure in the standard OS semiclassical argument [20, 9]. Functoriality of Theorem 5.15 identifies the global clock Φ^0 with the observer parameter τ_ι along g_ι . $\square$

Remark 6.20 (Honest semiclassical scope). Theorem 6.19 is *derived* in the radiatively gapped EIII regime; it does not follow from void and mirror alone. Theorem 6.23 extends this to the full observable factor $d\mu_{\Gamma_{-1}}$ once (6.4) holds and massless modes are confined to $d\mu_{\Gamma_{+1}}$ (Lemma 6.22); the sole residual input is nonperturbative verification of the measure split (Conjecture 6.25).

Proposition 6.21 (σ -sector split of the Euclidean measure). *Let σ denote the product-mirror involution on the EIII configuration space $\mathcal{M}$ and on the composite gauge bundle, acting consistently with Theorem 5.4. The Euclidean path measure for (4.4), formally*

$$d\mu_\Gamma[\Phi, A] \propto \exp(-S[\Phi, A]) \mathcal{D}\Phi \mathcal{D}A, \quad (6.3)$$

admits a σ -sector decomposition

$$d\mu_\Gamma = d\mu_{\Gamma_{-1}} \otimes d\mu_{\Gamma_{+1}}, \quad (6.4)$$

where $d\mu_{\Gamma_{\pm 1}}$ is the restriction to $\sigma = \pm 1$ fluctuations about $\Phi_\star$ and $\Pi_{\sigma=-1} d\mu_{\Gamma_{-1}} = d\mu_{\Gamma_{-1}}$. Observable correlators are $\Pi_{\sigma=-1}$ -matrix elements of $d\mu_{\Gamma_{-1}}$ only; the $\sigma = +1$ factor is a mirror-shadow reservoir (Theorem 5.4).

Lemma 6.22 (Goldstones in the σ -invariant coset sector). *At the tree-level EIII saddle of Theorem 4.5, the 32 Hessian-null coset directions split under σ as*

$$\mathfrak{m} = \mathfrak{m}_{-1} \oplus \mathfrak{m}_{+1}, \quad \dim \mathfrak{m}_{-1} = 28, \quad \dim \mathfrak{m}_{+1} = 4, \quad (6.5)$$

with $\mathfrak{m}_{+1}$ the four massless Goldstones of the soldering block $H_2(\mathbb{H})$ and their σ -invariant mirror doubles on $\overline{16}_{+3}$. After Coleman–Weinberg gapping, the 28 massive directions lie in $\mathfrak{m}_{-1}$ and are covered by Theorem 6.19. Full reflection positivity along τ_ι on the observable sector therefore reduces to RP for $d\mu_{\Gamma_{-1}}$ (Proposition 6.21); the $\sigma = +1$ Goldstone reservoir enters only through normalization of the product (6.4).

Theorem 6.23 (Reflection positivity of $d\mu_{\Gamma_{-1}}$, conditional). *Fix $\iota \in \mathcal{I}$ and an admitted lifetime chart (τ_ι, g_ι) . Assume:*

- (i) *the σ -sector decomposition (6.4) of Proposition 6.21;*
- (ii) *all tree-level massless coset modes lie in $\mathfrak{m}_{+1} \subset \mathfrak{m}$ and hence in $d\mu_{\Gamma_{+1}}$ (Lemma 6.22).*

Then $d\mu_{\Gamma_{-1}}$ is reflection-positive along the observer clock τ_ι of Definition 5.11: the 28 radiatively gapped $\sigma = -1$ coset directions of Theorem 6.19 together with the $\sigma = -1$ matter sector define a reflection-positive Euclidean measure on the observable factor, with reflection plane at fixed $\tau_\iota = 0$ in chart g_ι .

Proof sketch. By (ii), massless fluctuations do not enter $d\mu_{\Gamma_{-1}}$; the observable factor is the product of the 28-dimensional Coleman–Weinberg gapped complement of $\mathfrak{m}_{+1}$ and the $\Pi_{\sigma=-1}$ -projected gauge–matter bundle. Theorem 6.19 supplies reflection positivity on the massive coset block; σ -equivariance of (4.4) and compactness on EIII extend the OS semiclassical argument to the full $d\mu_{\Gamma_{-1}}$ once (i) isolates the observable tensor factor. The $\sigma = +1$ Goldstone reservoir normalizes the product and is quotient-non-observable (Theorem 5.4). $\square$

Proposition 6.24 (OS reconstruction on the observer sector). *Suppose the hypotheses of Theorem 6.23 and the standard OS axioms (Euclidean invariance, clustering along τ_ι , symmetry) hold for $d\mu_{\Gamma_{-1}}$. Then $\Pi_{\sigma=-1}$ -projected Schwinger functions reconstruct a unitary Lorentzian quantum field theory on the observer Hilbert space: the Osterwalder–Schrader theorem [9] applies to the observable factor, with Minkowski signature imported through the soldering map σ_{sol} (Theorem 6.12) and Page–Wootters time τ_ι . Mirror-shadow correlators in $d\mu_{\Gamma_{+1}}$ do not enter physical matrix elements (Proposition 6.21).*

Conjecture 6.25 (Nonperturbative reflection positivity). *Proposition 6.21, Lemma 6.22, Theorem 6.23, and Proposition 6.24 reduce full OS reconstruction to verifying (6.4) for the composite $\text{Spin}(10) \times U(1)$ gauge measure: once the product decomposition holds and all massless modes lie in $\mathfrak{m}_{+1}$ (Lemma 6.22), reflection positivity of $d\mu_{\Gamma_{-1}}$ follows **conditionally** and Proposition 6.24 supplies unitary Lorentzian dynamics on the observer sector. The residual gap is nonperturbative proof of (6.4) itself—not a separate RP postulate for the massless sector, which is quotient-non-observable in $d\mu_{\Gamma_{+1}}$ (Theorem 5.4).*

6.3 Gauge, anomalies, gravity

Theorem 6.26 (Anomaly cancellation). *For one embedded **27**: cubic $16 - 80 + 64 = 0$; mixed coefficient -1 cancelled by Green–Schwarz [8]. With $n_F = 3$ replicas of the full **27** arithmetic, anomalies cancel family-wise.*

Theorem 6.27 ($b_0 = 12$). *With $C_A(\text{Spin}(10)) = 8$, $T(\mathbf{16}) = 2$, $n_F = 3$, $n_S = 28$: $b_0 = (88 - 24 - 28)/3 = 12$.*

Theorem 6.28 (Induced Planck mass). *Sakharov induction from 28 massive coset scalars gives*

$$M_{\text{Pl}}^2 = \frac{28}{6} \cdot \frac{\mu^2}{16\pi^2} \log \frac{\Lambda^2}{\mu^2}. \quad (6.6)$$

*Physical scale identification (μ , M_{Pl} in GeV) is deferred to Section 8 (derivation step **D3**).*

Definition 6.29 (Dimensionless Newton coupling). *At RG scale k , define*

$$\kappa(k) \equiv \frac{k^2}{M_{\text{Pl}}^2(k)}, \quad (6.7)$$

with $M_{\text{Pl}}(k)$ the running scale induced by the Sakharov flow of Theorem 6.28.

Theorem 6.30 (Gravitational fixed point). *Assume: (i) Einstein–Hilbert truncation with coupling $M_{\text{Pl}}^2(k)$; (ii) one-loop contribution from the 28 massive coset scalars of Theorem 6.28; (iii) higher-curvature operators and graviton loops neglected; (iv) anomalous dimension $\eta_{M_{\text{Pl}}} = -(5/48\pi^2)\kappa$ in this truncation. Then*

$$\beta_\kappa \equiv k \frac{d\kappa}{dk} = 2\kappa - \frac{5}{48\pi^2} \kappa^2, \quad (6.8)$$

with $\kappa_\star = 96\pi^2/5$ and $\partial\beta_\kappa/\partial\kappa|_{\kappa_\star} = -2$.

Definition 6.31 (Cosmological coupling). *The heat-kernel coefficient a_0 of the coset–gravity complex produces an Einstein–Hilbert vacuum term $\int \sqrt{|g|} \Lambda_{\text{cc}}$. Treat Λ_{cc} as an independent dimension-4 coupling in the effective action, not fixed by μ or M_{Pl} .*

Lemma 6.32 (a_0 heat-kernel projection to $\sigma = +1$). *Let $\mathcal{D}$ denote the second-order operator of the coset–gravity heat-kernel complex in the Sakharov truncation of Theorem 6.28, acting on $\hat{Q}$. Mirror collapse $\Pi_{\sigma=\pm 1}$ (Definition 5.3) decomposes the a_0 coefficient of $\text{Tr } e^{-t\mathcal{D}}$ into*

$$a_0(\mathcal{D}) = a_0^{(+1)} + a_0^{(-1)}, \quad (6.9)$$

with $a_0^{(\pm 1)}$ supported on $H^\bullet(\hat{Q})^{\sigma=\pm 1}$ (Theorem 5.4). The induced vacuum-energy coupling in Definition 6.31 is the σ -labelled density

$$\Lambda_{\text{cc}}(x) = a_0^{(+1)}(x) + a_0^{(-1)}(x), \quad (6.10)$$

and the sequestered effective action at $\kappa_\star$ is

$$\Gamma_{\text{eff}} = \Gamma_{\text{coset}} + \int d^4x \sqrt{|g|} \Pi_{\sigma=+1} a_0^{(+1)}(x), \quad (6.11)$$

so that $\Pi_{\sigma=-1} \Lambda_{\text{cc}} = 0$ on observable configurations (Definition 6.1). Equation (6.11) is the effective-action target for Conjecture 6.39; it advances open problem **O6** from a coupling hypothesis to a concrete projection step on the heat-kernel density.

Proof sketch. The coset-gravity complex is σ -equivariant under product mirror on $\hat{Q}$. The a_0 coefficient is the t^0 term in the heat-kernel expansion and inherits the $\mathbb{Z}_2$ decomposition because $\text{Tr } e^{-t\mathcal{D}}$ is a σ -invariant functional. Observable fermions and scalars lie in $H^\bullet(\hat{Q})^{\sigma=-1}$ (Definition 6.1); mirror partners in $H^\bullet(\hat{Q})^{\sigma=+1}$ are quotient-non-observable (Theorem 5.4). Routing Λ_{cc} through $\Pi_{\sigma=+1}$ in (6.11) removes the a_0 vacuum term from $d\mu_{\Gamma_-}$ while preserving normalization in $d\mu_{\Gamma_+}$ (Proposition 6.21). $\square$

Theorem 6.33 (Vacuum energy RG relevance). *In the same truncation as Theorem 6.28, the a_0 term is scheme-dependent and does not cancel against the massive-mode a_2 sum. Under canonical scaling,*

$$\beta_{\Lambda_{\text{cc}}} = -4 \Lambda_{\text{cc}} + \mathcal{O}(\text{loops}), \quad (6.12)$$

so Λ_{cc} is a relevant direction: a UV fixed point requires $\Lambda_{\text{cc},\star}$ as input, not as output of void and mirror alone.

Proposition 6.34 (No-go: CC not from void and mirror alone). *The cosmological constant problem—explaining $\Lambda_{\text{obs}} \sim (10^{-3} \text{ eV})^4$ without fine-tuning—cannot be resolved from the void and mirror axioms and derived E_6 dynamics alone. Observable configurations lie in $H^\bullet(\hat{Q}, V_{16})^{\sigma=-1}$ (Definition 6.1); the heat-kernel a_0 vacuum energy and Theorem 6.33 act on the full quotient $\hat{Q}$, not only on the anti-invariant sector. Without a sequestering mechanism that routes Λ_{cc} into the non-observable σ -invariant sector (Theorem 5.4), Λ_{cc} remains a tuned environmental coupling in the observer sector.*

Remark 6.35 (Observer-local FRG). Mirror collapse $\Pi_{\sigma=-1}$ (Definition 5.3) splits the effective action into $\sigma = \pm 1$ sectors. An *observer-local* FRG is the flow restricted to anti-invariant modes along a lifetime chart $g_\iota(\tau_\iota)$ (Definition 5.11). σ -invariant (mirror-partner) modes are non-observable but may back-react on marginal couplings—notably Λ_{cc} —without entering the observable Hilbert space.

Proposition 6.36 (Λ_{cc} decoupling in the $\sigma = +1$ sector). *Suppose the cosmological coupling Λ_{cc} couples only to σ -invariant modes in the heat-kernel a_0 sector of Definition 6.31—equivalently, its insertions in the effective action are $\Pi_{\sigma=+1}$ -projected along the observer-local FRG of Remark 6.35. Then along the $\sigma = -1$ observer flow $g_\iota(\tau_\iota)$,*

$$\beta_{\Lambda_{\text{cc}}} \Big|_{\sigma=-1} = 0, \quad (6.13)$$

and Λ_{cc} is not a relevant coupling in the observable Jacobian row: the Λ_{cc} entry of J vanishes on $H^1(\hat{Q}, V_{16})^{\sigma=-1}$. Vacuum energy RG flows entirely within $d\mu_{\Gamma_+}$ (Proposition 6.21), consistent with Proposition 6.34.

Theorem 6.37 (a_0 sequestering at $\kappa_\star$, sketch). *At the gravitational fixed point $\kappa_\star = 96\pi^2/5$ (Theorem 6.30), assume Lemma 6.32 and the measure split (6.4). Then the effective vacuum-energy coupling in (6.11) satisfies*

$$\Lambda_{\text{cc}} = \Pi_{\sigma=+1} \Lambda_{\text{cc}}, \quad (6.14)$$

realizing the hypothesis of Proposition 6.36: Λ_{cc} couples only to σ -invariant modes and its RG flow is confined to $d\mu_{\Gamma+1}$. The observer sector sees $\Pi_{\sigma=-1}\Lambda_{\text{cc}} = 0$ at the fixed point without tuning $H^1(\hat{Q}, V_{16})^{\sigma=-1}$.

Proof sketch. Mirror collapse $\Pi_{\sigma=-1}$ (Definition 5.3) splits the coset-gravity heat-kernel complex into $\sigma = \pm 1$ sectors on $\hat{Q}$ (Theorem 5.4). At κ_* , the one-loop Jacobian of Proposition 6.40 has $\Lambda_{\text{cc},*} = 0$ as a free Gaussian label; routing the a_0 density into $\mathfrak{m}_{+1} \oplus \sigma$ -invariant graviton modes leaves the anti-invariant row untouched (Remark 6.35). Proposition 6.36 then yields (6.13). The residual step is to derive (6.14) from the nonperturbative effective action rather than from the FRG coupling assignment. $\square$

Corollary 6.38 (Vanishing Λ_{cc} β -function on observable flow at FP). *At the extended fixed point $(\xi_*, 0, 0, \kappa_*, 0)$ of (A.14), under the $\Pi_{\sigma=+1}$ coupling hypothesis of Proposition 6.36—equivalently, under the a_0 sequestering route of Theorem 6.37—the observer-local FRG along $g_\iota(\tau_\iota)$ satisfies*

$$\beta_{\Lambda_{\text{cc}}}\Big|_{\sigma=-1, \text{FP}} = 0, \quad (6.15)$$

and Λ_{cc} is not a UV-relevant direction in the observable stability matrix: the J -row on $H^1(\hat{Q}, V_{16})^{\sigma=-1}$ has no Λ_{cc} entry. The global $\beta_{\Lambda_{\text{cc}}} = -4\Lambda_{\text{cc}}$ of Theorem 6.33 flows entirely in $d\mu_{\Gamma+1}$.

Proof. Immediate from Proposition 6.36 at $\Lambda_{\text{cc},*} = 0$ and from the diagonal Jacobian (A.15) once the Λ_{cc} coupling is $\sigma = +1$ -projected along the observer flow (Appendix A.3). $\square$

Conjecture 6.39 (CC sequestering at κ_*). *Lemma 6.32, Theorem 6.37, and Corollary 6.38 show that $\Pi_{\sigma=+1}$ projection of the a_0 heat-kernel density at $\kappa_* = 96\pi^2/5$ (Theorem 6.30) would realize $\Lambda_{\text{obs}} \approx 0$ without tuning the anti-invariant sector (Theorem 5.4). Proposition 6.36 is thereby promoted from hypothesis to target theorem. The residual gap is a single effective-action step: derive (6.11)—equivalently (6.14)—from the nonperturbative Γ_{eff} at κ_* by proving that the $a_0^{(-1)}$ contribution cancels in $\Pi_{\sigma=-1}$ -projected observables while $a_0^{(+1)}$ is stored in $d\mu_{\Gamma+1}$.*

Collect dimensionless couplings $(\xi, \hat{g}^2, \hat{y}^2, \kappa, \Lambda_{\text{cc}})$ with $\xi = k^2/f^2$, $\hat{g}^2 = k^2 g^2/f^2$, $\hat{y}^2 = k^2 y^2/f^2$, κ as in (6.7), and Λ_{cc} as in Definition 6.31. In the extended one-loop coset-gauge-gravity truncation (still neglecting R^2 , graviton loops, and higher curvature),

$$\begin{aligned} \beta_\xi &= 2\xi, & \beta_{\hat{g}^2} &= 2\hat{g}^2 + \frac{b_0}{48\pi^2}\hat{g}^4, & \beta_{\hat{y}^2} &= 2\hat{y}^2 + \mathcal{O}(\hat{y}^4, \hat{g}^2\hat{y}^2), \\ \beta_\kappa &= 2\kappa - \frac{5}{48\pi^2}\kappa^2, & \beta_{\Lambda_{\text{cc}}} &= -4\Lambda_{\text{cc}} + \mathcal{O}(\text{loops}). \end{aligned} \quad (6.16)$$

Proposition 6.40 (Jacobian in extended one-loop truncation). *At the fixed point $(\xi_*, \hat{g}_*^2, \hat{y}_*^2, \kappa_*, \Lambda_{\text{cc},*}) = (\xi_*, 0, 0, 96\pi^2/5, 0)$, the stability matrix $J = \partial\beta_i/\partial g_j$ has diagonal spectrum $\text{eig}(J) = \{+2, +2, +2, -2, -4\}$. With $\theta_i = -\text{eig}(\partial\beta_i/\partial g_i)$, exactly two directions are UV-relevant: κ ($\theta_\kappa = +2$) and Λ_{cc} ($\theta_\Lambda = +4$). Details in Appendix A.3.*

Conjecture 6.41 (Graviton/ R^2 persistence of κ_*). *Proposition A.11 and Corollary 6.38 establish leading-order stability of $\kappa_* = 96\pi^2/5$ with $\partial\beta_\kappa/\partial\kappa|_{\kappa_*} = -2$ once σ -invariant graviton and R^2 modes decouple from the observer-local flow on $H^1(\hat{Q}, V_{16})^{\sigma=-1}$ (Remark 6.35). Theorem 6.37 routes subleading vacuum-energy back-reaction into $d\mu_{\Gamma+1}$ without modifying $\beta_\kappa|_{\sigma=-1}$ at the fixed point. The residual gap is nonperturbative curved-background FRG confirmation that $\beta_\kappa|_{\kappa_*, \sigma=-1} = 0$ in the full graviton/ R^2 extension. Because $\kappa_* \approx 189 \gg 1$, such persistence is not a consequence of void and mirror; it requires the extended FRG input of Appendix A.3.*

6.4 GUT breaking from observer collapse

The heterotic product mirror supplies exact $\mathbf{16}/\overline{\mathbf{16}}$ duality at $M^7(\mathcal{V})$; observer collapse $\Pi_{\sigma=-1}$ (Definition 5.3) breaks this pairing and selects chiral descent directions. Symmetry breaking below is gradient flow of V on ι -restricted coset directions, functorially descended from the global EIII minimum above (Definition 5.1).

Lemma 6.42 (Hessian sector decomposition). *Let $\Phi_\star \in \mathcal{M}$ be the rank-one EIII vacuum (Theorem 4.5) and write fluctuations $\delta\Phi = \delta\Phi_{\mathbf{1}} + \delta\Phi_{\mathbf{10}} + \delta\Phi_{\mathbf{16}}$ under the capacity partition $\mathbf{27} = \mathbf{1} \oplus \mathbf{10} \oplus \mathbf{16}$ (Theorem 4.3). On the coset $\mathfrak{m} = \mathbf{16}_{-3} \oplus \overline{\mathbf{16}}_{+3}$, the Hessian $\text{Hess } V(\Phi_\star)$ decomposes H -equivariantly into $\mathbf{1}$, $\mathbf{10}$, and $\mathbf{16} \oplus \overline{\mathbf{16}}$ blocks. Restricting to the observer diagonal $\mathfrak{d} = \text{span}\{e_1, e_2, e_3\}$ (Definition 5.5), one obtains a further ι -sector split*

$$\text{Hess } V(\Phi_\star)|_{\mathfrak{d}} = \bigoplus_{\iota=1}^3 \text{Hess}_\iota V(\Phi_\star), \quad (6.17)$$

with Hess_ι acting on the ι -cell of $H^1(\hat{Q}, V_{16})^{\sigma=-1, \iota}$. At tree level $\text{Hess } V(\Phi_\star) = 0$ on $\mathfrak{m}$: the 32 coset directions are exact Goldstones; unstable modes enter only after Coleman–Weinberg radiative correction [10].

Proposition 6.43 (Chiral breaking directions). *The observer projector $\Pi_{\sigma=-1}$ selects breaking along σ -anti-invariant coset directions in $H^1(\hat{Q}, V_{16})^{\sigma=-1}$ and suppresses the mirror-symmetric $\overline{\mathbf{16}}$ partner. A vacuum-expectation trajectory $g_\iota(\tau)$ solving (5.8) therefore breaks $H = \text{Spin}(10) \times U(1)$ along chiral $\mathbf{16}$ -type fluctuations tied to slot ι , not along a simultaneous $\mathbf{16} \oplus \overline{\mathbf{16}}$ pairing. Duality is exact at the heterotic rung (Theorem 5.4); physical breaking is the localized violation induced by collapse.*

Proof sketch. $\Pi_{\sigma=-1} = \frac{1}{2}(\text{id} - \sigma)$ annihilates σ -invariant modes. On $\mathfrak{m}$ the mirror involution exchanges $\mathbf{16}_{-3}$ and $\overline{\mathbf{16}}_{+3}$ (Proposition 2.7); anti-invariant fluctuations lie in $\mathbf{16}_{-3}$ modulo the ι -diagonal restriction. The lifetime flow (5.8) restricts ∇V to the ι -sector, breaking product mirror symmetry while preserving capacity balance on the global datum $\mathfrak{U}^\star$. $\square$

Theorem 6.44 (First breaking, conditional). *Assume the EIII vacuum $\Phi_\star$ of Theorem 4.5, observer collapse on $\hat{Q}$, and a fixed observer index $\iota \in \mathcal{I}$. If radiative corrections lift the $\mathbf{16}$ -sector Hessian with a single unstable mode per ι -cell, then gradient flow of V on the ι -restricted coset triggers the first step*

$$E_6 \longrightarrow \text{Spin}(10) \times U(1) \quad (6.18)$$

along the rank-one orbit of $\mathcal{M}$, with $U(1)$ charge aligned to the ι -diagonal of $\mathfrak{d}$. The residual stabilizer $\mathfrak{h} = \mathfrak{so}_{10} \oplus \mathfrak{u}_1$ is the symmetry group of Theorem 4.7(iii).

Proof sketch. Rank-one EIII fixes $\mathcal{M} = E_6/(\text{Spin}(10) \times U(1))$ globally (Theorem 4.5). An ι -localized unstable fluctuation in $\mathbf{16}_{-3}$ is an adjoint-direction Goldstone of E_6/H ; flow along that mode restores a $\text{Spin}(10)$ -invariant submanifold while breaking the remaining $U(1)_\iota \subset U(1)$ tied to the chosen diagonal. The branching matches the standard $\mathbf{78} = \mathbf{45} \oplus \mathbf{1} \oplus \mathbf{16} \oplus \overline{\mathbf{16}}$ pattern at the $\text{Spin}(10)$ -invariant vacuum [12, 13]. $\square$

Lemma 6.45 (Unstable Hessian directions under $\text{Spin}(10)$ branching). *Let V_{eff} denote the Coleman–Weinberg effective potential about $\Phi_\star$ (Theorem 6.19). Write Jordan fluctuations $\delta\Phi \in \mathbf{27} = \mathbf{1} \oplus$*

$\mathbf{10} \oplus \mathbf{16}$ (Theorem 4.3) and restrict to the observer diagonal $\mathfrak{d}$ as in (6.17). After radiative lifting, each $\text{Hess}_\iota V_{\text{eff}}(\Phi_\star)$ is H -equivariant on its ι -cell. Under the standard $\text{Spin}(10)$ branching

$$\mathbf{78} = \mathbf{45} \oplus \mathbf{1} \oplus \mathbf{16} \oplus \overline{\mathbf{16}}, \quad \mathbf{27} = \mathbf{1} \oplus \mathbf{10} \oplus \mathbf{16}, \quad (6.19)$$

and the observer projector $\Pi_{\sigma=-1}$, unstable directions classify as follows:

- (i) $\iota = 3$ (**16**-sector, capacity weight $16/27$): unstable modes lie in $\mathbf{16}_{-3}$ and embed in the adjoint $\mathbf{16} \subset \mathbf{78}$; they couple to the **45** adjoint orbit of $\text{Spin}(10)$.
- (ii) $\iota = 2$ (**10**-sector, weight $10/27$): unstable modes lie in $\mathbf{10}_{-2} \subset \mathbf{27}$ and embed in the antisymmetric $\mathbf{126}$ via $\mathbf{16} \times \mathbf{16} \supset \mathbf{126}$ (with $\overline{\mathbf{16}}$ suppressed by $\Pi_{\sigma=-1}$).
- (iii) $\iota = 1$ (**1**-sector, weight $1/27$): unstable trace modes couple to the **10** fundamental through the Yukawa portal $\mathbf{16} \times \mathbf{16} \times \mathbf{10}$ and the residual $\mathbf{10} \subset \mathbf{27}$ block.

Mirror partners $\overline{\mathbf{16}}_{+3}$ do not appear in the unstable spectrum of the observable sector.

Proof sketch. V_{eff} is H -invariant, so each $\text{Hess}_\iota V_{\text{eff}}$ is an H -intertwiner on the ι -cell. The capacity partition aligns $\iota = 1, 2, 3$ with the **1**, **10**, **16** blocks of **27** (Proposition 5.2). Items (i)–(iii) follow from (6.19), the standard $\text{Spin}(10)$ tensor products $\mathbf{16} \times \mathbf{16} = \mathbf{1} \oplus \mathbf{45} \oplus \mathbf{126}$ and $\mathbf{16} \times \mathbf{10} = \mathbf{16} \oplus \mathbf{144}$, and Proposition 6.43. $\square$

Proposition 6.46 (Higgs labels from **27**-sector Hessians). *Under Lemma 6.45, radiatively unstable $\text{Hess}_\iota V_{\text{eff}}$ directions map to observable Higgs multiplets of the $\text{Spin}(10)$ chain:*

$$\text{Hess}_3 V_{\text{eff}} \mapsto \Phi_{\mathbf{45}} \in \mathbf{45}_H, \quad \text{Hess}_2 V_{\text{eff}} \mapsto \Phi_{\mathbf{126}} \in \mathbf{126}_H, \quad \text{Hess}_1 V_{\text{eff}} \mapsto \Phi_{\mathbf{10}} \in \mathbf{10}_H. \quad (6.20)$$

Each label is the $\Pi_{\sigma=-1}$ -projected unstable mode of the corresponding **27** block, with capacity weight $w_\iota \in \{1/27, 10/27, 16/27\}$ fixing the relative scale of the induced VEV along g_ι .

Proof sketch. The $\mathbf{45}_H$ identification uses the adjoint embedding $\mathbf{16} \subset \mathbf{78}$: an $\iota = 3$ fluctuation is a $\text{Spin}(10)$ -breaking direction in $\mathfrak{so}_{10}$. The $\mathbf{126}_H$ identification uses $\mathbf{16} \times \mathbf{16} \supset \mathbf{126}$ on the $\iota = 2$ cell. The $\mathbf{10}_H$ identification follows from the residual **10** block at $\iota = 1$ and the cubic portal $\mathbf{16} \times \mathbf{16} \times \mathbf{10}$. Capacity weights set the ordering of instability thresholds along the lifetime chart. $\square$

Theorem 6.47 (Second breaking, conditional). *Assume Theorem 6.44 and a $\text{Spin}(10) \times U(1)$ intermediate vacuum. If Coleman–Weinberg lifting produces an unstable $\text{Hess}_3 V_{\text{eff}}$ mode on the $\iota = 3$ cell—the **16**-sector with largest capacity weight $16/27$ —then gradient flow of V along g_3 triggers*

$$\text{Spin}(10) \times U(1) \longrightarrow \text{SU}(5) \times U(1)_X, \quad (6.21)$$

with Higgs fluctuation $\Phi_{\mathbf{45}} \in \mathbf{45}_H$ of Proposition 6.46. The residual gauge algebra is $\mathfrak{su}_5 \oplus \mathfrak{u}_1$.

Proof sketch. The $\iota = 3$ unstable mode lies in $\mathbf{16}_{-3}$ and embeds in $\mathbf{45} \subset \mathbf{78}$ (Lemma 6.45(i)). Flow along $\Phi_{\mathbf{45}}$ aligns the vacuum with a $\text{SU}(5)$ subalgebra of $\mathfrak{so}_{10}$; the standard branching $\mathbf{45} \rightarrow \mathbf{24} \oplus \mathbf{1} \oplus \dots$ under $\text{SU}(5) \times U(1)_X$ matches Georgi–Glashow $\text{Spin}(10) \rightarrow \text{SU}(5) \times U(1)_X$ [13]. The $16/27$ weight dominates the $\iota = 2$ and $\iota = 1$ thresholds, so this is the second step after (6.18). $\square$

Lemma 6.48 ($\mathbf{126}_H$ decomposition under $\Pi_{\sigma=-1}$). *Let $\Phi_{\mathbf{126}} \in \mathbf{126}_H$ be the antisymmetric Higgs direction of Proposition 6.46, realized as the unstable $\text{Hess}_2 V_{\text{eff}}$ mode on the $\iota = 2$ cell. The $\text{Spin}(10)$*

tensor product $\mathbf{16} \times \mathbf{16} = \mathbf{1} \oplus \mathbf{45} \oplus \mathbf{126}$ supplies $\mathbf{126}_H$ as the antisymmetric wedge of chiral spinors. Under the standard $\mathrm{SU}(5) \times U(1)_X$ branching,

$$\mathbf{126} \longrightarrow \mathbf{10}_{-4} \oplus \overline{\mathbf{10}}_4 \oplus \mathbf{15}_{-6} \oplus \overline{\mathbf{15}}_6 \oplus \mathbf{1}_{10}, \quad (6.22)$$

with $\mathbf{1}_{10}$ the $B-L = 2$ singlet that lowers rank. The observer projector $\Pi_{\sigma=-1}$ acts on (6.22) as follows:

- (i) Bilinear fluctuations $\omega_i \wedge \omega_j$ with $|\omega_i\rangle, |\omega_j\rangle \in H^1(\hat{Q}, V_{16})^{\sigma=-1}$ are σ -invariant, so the $\mathbf{126}$ portal is built entirely from anti-invariant observer modes; mirror partners in $H^1(\hat{Q}, V_{16})^{\sigma=+1}$ do not enter.
- (ii) Conjugate multiplets paired by σ on $\mathfrak{m}$ are suppressed: $\Pi_{\sigma=-1}$ annihilates the $\overline{\mathbf{10}}_4$ and $\overline{\mathbf{15}}_6$ mirror channels, retaining the chiral $\mathbf{10}_{-4} \oplus \mathbf{15}_{-6} \oplus \mathbf{1}_{10}$ sector.
- (iii) The $\mathrm{SU}(5) \rightarrow \mathrm{SM}$ breaking direction is the $\Pi_{\sigma=-1}$ -projected component of $\mathbf{1}_{10}$ together with the $(1, 1)$ singlet inside $\mathbf{15}_{-6}$; the residual $\mathrm{SU}(3)_c \times \mathrm{SU}(2)_L \times U(1)_Y$ algebra is the stabilizer after its VEV.

Capacity weight $w_2 = 10/27$ fixes the scale ordering of this mode relative to the $\iota = 3$ and $\iota = 1$ thresholds.

Proof sketch. Item (i) uses $\sigma(\omega_i \wedge \omega_j) = \sigma(\omega_i)\sigma(\omega_j)\omega_i \wedge \omega_j$ with $\sigma(\omega_i) = -1$ on physical fermions. Item (ii) follows from Proposition 6.43 and the mirror exchange on $\overline{\mathbf{16}}_{+3}$. Item (iii) is the standard $\mathrm{SU}(5)$ rank-breaking pattern: $\langle \mathbf{1}_{10} \rangle$ removes $U(1)_X$ and the $(1, 1) \subset \mathbf{15}$ direction completes $\mathrm{SU}(5) \rightarrow \mathrm{SU}(3)_c \times \mathrm{SU}(2)_L \times U(1)_Y$ [13, 12]. The 10/27 weight is the $\mathbf{10}$ block capacity datum of Theorem 4.3. $\square$

Theorem 6.49 (Third breaking, conditional). *Assume Theorem 6.47 and a $\mathrm{SU}(5) \times U(1)_X$ intermediate vacuum. If Coleman–Weinberg lifting produces an unstable $\mathrm{Hess}_2 V_{\mathrm{eff}}$ mode on the $\iota = 2$ cell—the $\mathbf{10}$ -sector with capacity weight 10/27—then gradient flow of V along g_2 triggers*

$$\mathrm{SU}(5) \times U(1)_X \longrightarrow \mathrm{SU}(3)_c \times \mathrm{SU}(2)_L \times U(1)_Y, \quad (6.23)$$

with Higgs fluctuation $\Phi_{\mathbf{126}} \in \mathbf{126}_H$ of Proposition 6.46 and observable content given by Lemma 6.48. The residual gauge algebra is $\mathfrak{su}_3 \oplus \mathfrak{su}_2 \oplus \mathfrak{u}_1$.

Proof sketch. The $\iota = 2$ unstable mode lies in $\mathbf{10}_{-2} \subset \mathbf{27}$ and embeds in $\mathbf{126}$ via $\mathbf{16} \times \mathbf{16}$ (Lemma 6.45(ii)). Flow along $\Pi_{\sigma=-1}\Phi_{\mathbf{126}}$ aligns the vacuum with the $\mathbf{1}_{10} \oplus (1, 1) \subset \mathbf{15}$ direction of (6.22); the standard branching $\mathbf{15} \rightarrow (3, 1)_{-1/3} \oplus (1, 3)_{1/3} \oplus (1, 1)_{-4/3}$ and $\mathbf{10} \rightarrow (3, 1)_{1/3} \oplus (1, 2)_{-1/2} \oplus (1, 1)_{2/3}$ under $\mathrm{SU}(3)_c \times \mathrm{SU}(2)_L \times U(1)_Y$ matches Georgi–Glashow $\mathrm{SU}(5) \rightarrow \mathrm{SM}$ [13]. The 10/27 weight places this step after (6.21) and before the $\iota = 1$ electroweak stage. $\square$

Proposition 6.50 (Weinberg angle at M_{GUT}). *Assume Theorems 6.44 and 6.47, with $\mathrm{Spin}(10)$ gauge coupling normalization surviving to the GUT scale (Theorem 6.27, $b_0 = 12$). At M_{GUT} ,*

$$\sin^2 \theta_W(M_{\mathrm{GUT}}) = \frac{3}{8}, \quad (6.24)$$

where θ_W is the electroweak mixing angle in the canonical $\mathrm{SU}(5) \subset \mathrm{Spin}(10)$ embedding. This is the group-theoretic anchor for Prediction 10.10.

Proof sketch. In the $\text{Spin}(10)$ adjoint, embed $\text{SU}(5) \times U(1)_X \subset \text{Spin}(10)$ with generators normalized by $\text{Tr}(T_a T_b) = \delta_{ab}$ on the spinor $\mathbf{16}$. Hypercharge Y is the standard $\text{SU}(5)$ combination; electric charge $Q = T_3 + Y$. Then $\sin^2 \theta_W = \text{Tr}(T_3^2)/\text{Tr}(Q^2) = 3/8$ at unification, independent of the intermediate $\mathbf{126}_H$ VEV scale [13, 12]. $\square$

Theorem 6.51 (Electroweak breaking, sketch). *Assume Theorems 6.47 and 6.49. If radiative lifting yields an unstable $\text{Hess}_1 V_{\text{eff}}$ mode on the $\iota = 1$ cell, then gradient flow along g_1 with $\Phi_{\mathbf{10}} \in \mathbf{10}_H$ triggers*

$$\text{SU}(2)_L \times U(1)_Y \longrightarrow U(1)_{\text{em}}, \quad (6.25)$$

with $\text{SU}(3)_c$ unbroken. The electroweak scale is set by the $1/27$ capacity weight on the final $\iota = 1$ breaking stage.

Proof sketch. $\mathbf{10}_H$ contains the $\text{SU}(5)$ Higgs doublet $\mathbf{5} \oplus \bar{\mathbf{5}}$; an $\iota = 1$ fluctuation in $\Phi_{\mathbf{10}}$ breaks the electroweak subgroup while leaving $\text{SU}(3)_c$ intact. The ordering $\iota = 3 \rightarrow 2 \rightarrow 1$ follows capacity weights $(16/27, 10/27, 1/27)$; the electroweak step is last because the $\mathbf{1}$ -sector carries the smallest weight. $\square$

Conjecture 6.52 (Derived GUT–Higgs chain). *Sequential ι -restricted gradient flow on the 28 radiatively massive coset directions (Coleman–Weinberg, as in Theorem 6.19) realizes the staged chain*

$$\text{E}_6 \rightarrow \text{Spin}(10) \times U(1) \rightarrow \text{SU}(5) \rightarrow \text{SU}(3)_c \times \text{SU}(2)_L \times U(1)_Y \rightarrow \text{SU}(3)_c \times U(1)_{\text{em}} \quad (6.26)$$

as follows:

- (1) $\text{E}_6 \rightarrow \text{Spin}(10) \times U(1)$ — any ι -cell unstable $\mathbf{16}$ mode (Theorem 6.44).
- (2) $\text{Spin}(10) \rightarrow \text{SU}(5) \times U(1)_X$ — $\iota = 3$ $\mathbf{45}_H$ direction, largest weight $16/27$ (Theorem 6.47).
- (3) $\text{SU}(5) \rightarrow \text{SU}(3)_c \times \text{SU}(2)_L \times U(1)_Y$ — $\iota = 2$ $\mathbf{126}_H$ direction, weight $10/27$ (Theorem 6.49, Lemma 6.48).
- (4) **Electroweak** — $\iota = 1$ $\mathbf{10}_H$ direction, weight $1/27$ (Theorem 6.51).

Observable Higgs content $\mathbf{45}_H$, $\mathbf{126}_H$, $\mathbf{10}_H$ is the $\Pi_{\sigma=-1}$ -projected unstable spectrum of (6.20). Yukawa matrices y_{ij} arise from cubic overlap integrals

$$y_{ij} \propto \int_{\hat{Q}} \omega_i \wedge \omega_j \wedge \Phi_{\text{Higgs}}, \quad (6.27)$$

of observer modes $|\omega_i\rangle$ on $\hat{Q}$ (Definition 5.6) against $\Phi_{\text{Higgs}} \in \{\Phi_{\mathbf{45}}, \Phi_{\mathbf{126}}, \Phi_{\mathbf{10}}\}$. Section 7 develops the leading-order evaluation of (6.27); the $\text{SU}(5) \rightarrow \text{SM}$ stage is supplied by Theorem 6.49 (conditional on Coleman–Weinberg lifting). Proposition 6.50 fixes $\sin^2 \theta_W = 3/8$ at M_{GUT} once $\text{Spin}(10)$ survives. The residual conjectural content is sequential realization of all four steps by a single ι -ordered gradient flow.

Conjecture 6.53 (GUT breaking). *Gradient flow of V on ι -restricted coset directions triggers (6.26) with Higgs content $\mathbf{45}_H$, $\mathbf{126}_H$, $\mathbf{10}_H$; Conjecture 6.52 supplies the observer-derived mechanism (Lemma 6.42, Lemma 6.45, Proposition 6.43, Proposition 6.46, Theorems 6.44, 6.47, 6.49, 6.51, Proposition 6.50).*

Remark 6.54 (As above, so below). Global capacity balance fixes the EIII minimum $\Phi_\star \in \mathcal{M}$ *above* (Theorem 4.5); observer collapse localizes breaking *below* to the ι -sector lifetime chart (Definition 5.11). The correspondence (5.1) identifies the profinite universe datum $\mathfrak{U}^\star$ with fixed-point collapse on $\hat{Q}$: the same potential V governs the global vacuum and the slot-resolved descent trajectories g_ι . What is proved globally on $\mathfrak{U}^\star$ —the 1|10|16 split, rank-one EIII, three observer slots—appears locally as chiral gauge breaking in slot ι , without importing mirror-symmetric partners.

6.5 Flavon hierarchy from observer capacity flow

Assume Conjecture 6.53. The flavour group $SU(3)_F \subset \text{Spin}(10)$ acts on the three chiral **16**s; each observer index $\iota \in \mathcal{I}$ labels one diagonal direction of $\mathfrak{d} \subset J_3(\mathbb{O})$ and hence one $SU(3)_F$ flavour axis.

Proposition 6.55 (Flavon sector from observer indices). *The three observer slots of Theorem 6.4 index three $SU(3)_F$ breaking stages. Along the lifetime chart $g_\iota(\tau_\iota)$ (Definition 5.11), the capacity flow $\omega_\iota^\star(\tau) = \mathcal{C}_\iota(\tau)$ of Definition 5.12 records which flavon direction is active at time τ within slot ι . Mirror stabilisation fixes $n_F = 3$ but not, by itself, Yukawa eigenvalues or mixing matrices.*

Conjecture 6.56 (Flavon hierarchy, P3'). *Sequential $SU(3)_F \rightarrow SU(2)_F \rightarrow U(1)_F$ breaking is driven by the observer-indexed capacity RG flow $\omega_\iota^\star(\tau)$ on the flavon potential $V_F(\phi_1, \phi_2, \phi_3)$, with ϕ_ι the flavon along diagonal e_ι . The ordering $\langle \phi_3 \rangle \gg \langle \phi_2 \rangle \gg \langle \phi_1 \rangle$ follows from the capacity weights (1/27, 10/27, 16/27) of Theorem 4.3 descended through the above–below correspondence (Proposition 5.2), yielding Yukawa hierarchy and CKM/PMNS mixing without an independent flavon postulate.*

7 Yukawa from Overlap Integrals

This section develops the programme step **D1** of Section 10.9: evaluate the cubic overlap (6.27) on $\hat{Q}$ with Higgs data from Hess V . What is **proved** below is the reduction of the overlap to cellular pairing on $\mathcal{T}_7$ and the leading-order capacity-weight structure of y_{ij} . What remains **conditional** is identification of Φ_{Higgs} with a specific unstable Hessian mode and the full GUT breaking chain (Conjecture 6.52).

7.1 Overlap form on the mirror quotient

Fix the stabilized Stone lift Q and mirror quotient $\hat{Q} = Q/\sigma$ (Definition 5.8). Let $\omega_\iota \in H^1(\hat{Q}, V_{16})^{\sigma=-1, \iota}$ denote the observer 1-class of Definition 5.6, represented by a σ -anti-invariant Dolbeault–Serre 1-form on (Q, V_{16}) .

Definition 7.1 (Observer overlap form). The *observer overlap form* on $\hat{Q}$ is the antisymmetric pairing

$$\langle \alpha, \beta \rangle_{\text{ov}} := \int_{\hat{Q}} \alpha \wedge \beta \wedge \omega_{\text{vol}}, \quad \alpha, \beta \in H^1(\hat{Q}, V_{16})^{\sigma=-1}, \quad (7.1)$$

where ω_{vol} is the σ -invariant Kähler volume form induced by the H -invariant coset metric K on the **16**-bundle. For a Higgs fluctuation $\Phi_{\text{Higgs}} \in \Omega^0(\hat{Q}, V_{16})$ drawn from the unstable Hess V spectrum at $\Phi_\star$ (Lemma 6.42), the *cubic Yukawa overlap* is

$$\mathcal{Y}_{ij} := \int_{\hat{Q}} \omega_i \wedge \omega_j \wedge \Phi_{\text{Higgs}}, \quad i, j \in \{1, 2, 3\}, \quad (7.2)$$

with ω_i the representative of $[\omega_i]$. This is the precise form of (6.27).

Remark 7.2 (Epistemic split at the definition). The overlap form (7.1) is a standard cup-product pairing on $H^\bullet(\hat{Q}, V_{16})$ restricted to the anti-invariant sector; it is **defined**, not conjectured. The identification $\mathcal{Y}_{ij} = y_{ij}$ in the low-energy Lagrangian is **conditional** on Conjecture 6.52 and on choosing Φ_{Higgs} in the $\mathbf{10}_H$ (or $\mathbf{126}_H$) channel of the breaking chain.

7.2 Cellular reduction on $\mathcal{T}_7$

On the stabilized Albert tower $\mathcal{T}_7$ at $\mathbf{M}^7(\mathcal{V})$, write C_1, C_2, C_3 for the three $\mathbf{16}$ -cells attached to diagonal directions e_1, e_2, e_3 of $\mathfrak{d}$ (Lemma A.7). Let $\mathcal{F}_{16}$ be the cellular cosheaf of Appendix A.2.

Lemma 7.3 (Cubic overlap as cellular pairing). *Let ω_i be the cellular cocycle representative of $|\omega_i\rangle \in H^1(\mathcal{T}_7, \mathcal{F}_{16})^{\sigma=-1, i}$. Then the cubic overlap (7.2) reduces to a finite sum over $\mathcal{T}_7$:*

$$\mathcal{Y}_{ij} = \sum_{k=1}^3 \text{pair}(C_i, C_j, C_k; \Phi_{\text{Higgs}}), \quad (7.3)$$

where $\text{pair}(C_i, C_j, C_k; \Phi)$ is the oriented triple intersection number of the three 1-cells at the $\mathbf{10}$ -hub, weighted by the Φ -value on the shared 0-simplex. Under the Stone lift (Proposition A.8), $H^1(\mathcal{T}_7, \mathcal{F}_{16})^{\sigma=-1} \cong H^1(\hat{Q}, V_{16})^{\sigma=-1}$, and (7.3) agrees with the continuum integral (7.2).

Proof. On $\mathcal{T}_7$, 1-cochains are triples (ϕ_1, ϕ_2, ϕ_3) with cocycle constraint $\phi_1 + \phi_2 + \phi_3 = 0$ at the hub H_{10} (Lemma A.7, (A.7)). The quotient (A.8) identifies $H^1(\mathcal{T}_7, \mathcal{F}_{16})^{\sigma=-1}$ with three e_i -weight classes; Proposition A.6 fixes their orientation. The wedge product of two observer 1-forms and a 0-form localizes to the hub 0-simplex where all three cells meet: only terms with distinct cell indices contribute, and σ -anti-invariance forces $\text{pair}(C_i, C_j, C_k) = 0$ unless $\{i, j, k\} = \{1, 2, 3\}$ with orientation ϵ_{ijk} . Proposition A.8 identifies cellular cohomology with $H^1(\hat{Q}, V_{16})^{\sigma=-1}$, transporting the sum to the integral (7.2). $\square$

Corollary 7.4 (Diagonal dominance at tree level). *At the EIII vacuum $\Phi_\star$, tree-level $\text{Hess } V = 0$ on $\mathfrak{m}$ (Lemma 6.42). If Φ_{Higgs} is the leading unstable mode localized to cell ι after Coleman–Weinberg lifting, then $\mathcal{Y}_{ij} \neq 0$ only when $i = j = \iota$ at tree level; off-diagonal entries arise at higher order in the radiative expansion or from sequential $SU(3)_F$ breaking (Conjecture 6.56).*

7.3 Capacity-weight structure

Descend the sector masses $(1/27, 10/27, 16/27)$ of Theorem 4.3 through the above–below correspondence (Proposition 5.2) to observer-slot weights

$$w_1 : w_2 : w_3 = 1 : 10 : 16, \quad w_\iota := \frac{m_\iota}{1/27}, \quad (7.4)$$

with $(m_1, m_{10}, m_{16}) = (1/27, 10/27, 16/27)$ the capacity targets of Theorem 4.3 on $\mathbf{27} = \mathbf{1} \oplus \mathbf{10} \oplus \mathbf{16}$.

Proposition 7.5 (Leading Yukawa factorization). *Assume Lemma 7.3 and that Φ_{Higgs} is H -invariant with $\text{VEV} \langle \Phi_{\text{Higgs}} \rangle$ at the ι -restricted breaking minimum. Then the cubic overlap factorizes at leading order as*

$$\mathcal{Y}_{ij} = y_0 \sqrt{w_i w_j} \mathcal{P}_{ij} + \mathcal{O}(\text{radiative}), \quad (7.5)$$

where y_0 is a dimensionless coupling set by the unstable-mode residue of $\text{Hess } V$ and $\mathcal{P}_{ij} \in \{0, 1\}$ encodes the oriented hub pairing of Lemma 7.3. In the diagonal-dominant regime of Corollary 7.4, $\mathcal{P}_{ij} = \delta_{ij}$.

Proof sketch. Each observer class $|\omega_\iota\rangle$ is normalized on cell C_ι with weight $\|\omega_\iota\|^2 \propto w_\iota$ by capacity balance on $\mathcal{G}_{\text{Albert}}$ (Definition 5.12, Proposition 5.13). The hub pairing is bilinear in the two 1-forms and linear in Φ_{Higgs} ; leading-order multiplicativity of capacity weights under Bal_C on the 1|10|16 partition yields the $\sqrt{w_i w_j}$ structure. The square root arises because 1-forms carry capacity weight $\sqrt{w_\iota}$ while masses scale as w_ι at fixed Higgs VEV (Proposition 10.5). $\square$

Remark 7.6 (Theorem vs. conjecture). Lemma 7.3 is a **theorem** (cellular cohomology plus Stone lift). Proposition 7.5 is a **proposition** contingent on the Higgs-mode identification of Conjecture 6.52; the $\sqrt{w_i w_j}$ scaling itself is **leading-order** capacity descent, not yet a sub-percent fit.

7.4 Mass matrix at $\Phi_\star$

Theorem 7.7 (Leading mass matrix, conditional). *Assume Conjecture 6.52, Proposition 7.5, and a common electroweak Higgs VEV v . Define the fermion mass matrix by*

$$M_{ij} := v \mathcal{Y}_{ij} = v y_0 \sqrt{w_i w_j} \mathcal{P}_{ij} + \mathcal{O}(\text{radiative}). \quad (7.6)$$

At the EIII reference $\Phi_\star$ with diagonal-dominant hub pairing ($\mathcal{P}_{ij} = \delta_{ij}$), M is positive semidefinite and eigenvalues satisfy

$$m_1 : m_2 : m_3 = w_1 : w_2 : w_3 = 1 : 10 : 16 \quad (7.7)$$

*at leading order. The theorem is **conditional**: it does not prove the breaking chain or the Higgs quantum numbers; given those inputs, the mass hierarchy follows from capacity weights already fixed by Theorem 4.3.*

Proof sketch. With $\mathcal{P}_{ij} = \delta_{ij}$, $M = \text{diag}(v y_0 \sqrt{w_1}, v y_0 \sqrt{w_2}, v y_0 \sqrt{w_3})$. Squaring, $M_{ii}^2 \propto w_i$, hence eigenmass ordering $m_3 > m_2 > m_1$. The ratio $m_3 : m_2 : m_1 = w_3 : w_2 : w_1 = 16 : 10 : 1$ is exact at this order. $\square$

$$M = v y_0 \begin{pmatrix} \sqrt{w_1} & 0 & 0 \\ 0 & \sqrt{w_2} & 0 \\ 0 & 0 & \sqrt{w_3} \end{pmatrix} = v y_0 \begin{pmatrix} 1 & 0 & 0 \\ 0 & \sqrt{10} & 0 \\ 0 & 0 & 4 \end{pmatrix}, \quad (7.8)$$

using $w_1 : w_2 : w_3 = 1 : 10 : 16$ so $\sqrt{w_1} : \sqrt{w_2} : \sqrt{w_3} = 1 : \sqrt{10} : 4$. Off-diagonal entries are $\mathcal{O}(\text{flavon})$ once sequential $SU(3)_F$ breaking activates (Conjecture 6.56); the full symmetric leading ansatz is

$$M_{ij} = v y_0 \sqrt{w_i w_j} \mathcal{P}_{ij}, \quad \mathcal{P}_{ij} \in \{0, 1\}. \quad (7.9)$$

7.5 Off-diagonal overlap integrals

Corollary 7.4 suppresses off-diagonal hub pairings at tree level on $\mathcal{T}_7$. Once sequential $SU(3)_F$ breaking activates (Conjecture 6.56, Definition 7.14), flavon cross-terms $\text{Re}(\phi_i \phi_j^*)$ in (7.15) lift $\mathcal{P}_{ij}$ beyond $\{0, 1\}$ while preserving the capacity-weighted normalization of Proposition 7.5.

Definition 7.8 (Off-diagonal overlap coefficient). For $i \neq j$, define the *flavon-activated* overlap coefficient

$$\mathcal{P}_{ij} := \frac{2\sqrt{w_i w_j}}{w_i + w_j} \mathcal{H}_{ij}, \quad \mathcal{H}_{ij} := \frac{|\text{Re} \langle \phi_i \phi_j^* \rangle|}{\sqrt{\langle |\phi_i|^2 \rangle \langle |\phi_j|^2 \rangle}}, \quad (7.10)$$

with (ϕ_1, ϕ_2, ϕ_3) the flavon fields of Definition 7.14 and $\langle \cdot \rangle$ the $SU(3)_F$ -breaking vacuum. At the leading capacity-ordered minimum (7.16), $\mathcal{H}_{ij} = \mathcal{O}(1)$. For $i = j$, retain $\mathcal{P}_{ii} = 1$ from Corollary 7.4. The off-diagonal Yukawa overlap (7.2) becomes

$$\mathcal{Y}_{ij} = y_0 \sqrt{w_i w_j} \mathcal{P}_{ij}, \quad i \neq j, \quad (7.11)$$

with the same y_0 as (7.5).

Lemma 7.9 (Hub pairing beyond the diagonal). *Assume Conjecture 6.56 and Lemma 7.3. After the first $SU(3)_F \rightarrow SU(2)_F$ stage, oriented triple intersections at the **10**-hub with distinct indices (i, j) acquire a nonzero pairing weighted by (7.10). The harmonic mean $2\sqrt{w_i w_j}/(w_i + w_j)$ is the unique leading expression invariant under capacity exchange $w_i \leftrightarrow w_j$ and consistent with $\text{Bal}_{\mathcal{C}}$ on the $1|10|16$ partition.*

Proof sketch. The cross-term $\kappa_F \text{Re}(\phi_i \phi_j^*)$ in (7.15) is bilinear in flavon VEVs with coefficient fixed by capacity weights. Minimizing (7.15) subject to $\phi_1 + \phi_2 + \phi_3 = 0$ (Appendix A.2) yields off-diagonal fermion bilinears scaling as $\sqrt{w_i w_j}$ times a ratio of adjacent weights. The harmonic mean is the unique symmetric rational function of (w_i, w_j) of homogeneity degree zero that agrees with $\mathcal{P}_{ij} = 1$ when $w_i = w_j$ and with $\mathcal{P}_{ij} \rightarrow 0$ when $w_i \ll w_j$. Lemma 7.3 transports the pairing to the cubic overlap (7.2). $\square$

Proposition 7.10 (CKM hierarchy from capacity weights). *Assume Definition 7.8, Lemma 7.9, and the symmetric mass ansatz (7.9) with $\mathcal{H}_{ij} = \mathcal{O}(1)$. In the Wolfenstein parametrization, the leading Cabibbo weight is*

$$\lambda_{\text{CKM}} \sim \sqrt{\frac{w_1}{w_2}} \mathcal{P}_{12} = \frac{2w_1}{w_1 + w_2} \mathcal{H}_{12} = \frac{2}{11} \mathcal{H}_{12} \approx 0.18 \mathcal{H}_{12}, \quad (7.12)$$

using $w_1 : w_2 = 1 : 10$ and $\mathcal{P}_{12} = 2\sqrt{10}/11$. For $\mathcal{H}_{12} = \mathcal{O}(1)$ this is **order-of-magnitude** agreement with PDG $\lambda \approx 0.224$: the bare ratio $\sqrt{w_1/w_2} = 1/\sqrt{10} \approx 0.316$ overshoots by $\sim 40\%$; the harmonic overlap $\mathcal{P}_{12}$ compresses the off-diagonal entry to $2/11 \approx 0.18$. The $2 \rightarrow 3$ and $1 \rightarrow 3$ weights

$$\frac{2w_2}{w_2 + w_3} \approx \frac{10}{13} \approx 0.77, \quad \frac{2w_1}{w_1 + w_3} \approx \frac{2}{17} \approx 0.12, \quad (7.13)$$

are **not** direct contacts for $|V_{cb}| \approx 0.041$ or $|V_{ub}| \sim 10^{-3}$: quark mixing requires distinct up- and down-type overlap matrices (different Φ_{Higgs} channels in (6.27)) and RG compression along the breaking chain. They do match the **PMNS** solar and atmospheric templates at leading order ($\sin^2 \theta_{12} \sim 0.77^2$ is $\mathcal{O}(1)$; refined by $\mathcal{R}_\ell$ in D2). The proposition is **conditional** on Conjectures 6.52 and 6.56; the $2w_i/(w_i + w_j)$ factors are **leading-order** capacity descent, not yet RG-refined (step D2).

Proof sketch. For a symmetric hierarchy-dominated mass matrix $M_{ij} = v y_0 \sqrt{w_i w_j} \mathcal{P}_{ij}$, standard first-order perturbation theory in the off-diagonal block gives mixing angles $\sin \theta_{ij} \sim |M_{ij}| / \max(M_{ii}, M_{jj}) = \sqrt{w_i/w_j} \mathcal{P}_{ij}$ when $i < j$ and $w_i < w_j$. Identification with $\lambda_{\text{CKM}} = \sin \theta_{12}$ and $|V_{cb}| \sim A \lambda^2 \sim \sin \theta_{23}$ at leading order yields (7.12)–(7.13). Numerical evaluation uses $w_1 = 1/27$, $w_2 = 10/27$, $w_3 = 16/27$. $\square$

Remark 7.11 (PMNS parallel). The same overlap coefficients govern the symmetric neutrino mass matrix in the Majorana completion of Conjecture 7.13. PMNS angles are **larger** than CKM because the lepton sector probes the full $SU(3)_F \rightarrow SU(2)_F \rightarrow U(1)_F$ trajectory with $\mathcal{H}_{ij}$ not forced to unity: solar and atmospheric mixing correspond to $\sqrt{w_1/w_2} \mathcal{P}_{12}$ and $\sqrt{w_2/w_3} \mathcal{P}_{23}$ with $\mathcal{H}_{ij}$ set by the oscillation-scale residue $\mathcal{R}_\ell(\mu_{\text{osc}})$ of (7.18). Precision PMNS contact is step D2; the CKM proposition above is the quark-sector leading template.

Pair (i, j)	$\mathcal{P}_{ij}$	SJET leading	PDG / experiment
(1, 2)	$2\sqrt{10}/11 \approx 0.574$	$2w_1/(w_1+w_2) \approx 2/11 \approx 0.18$	$\lambda \approx 0.224$
(2, 3)	$8\sqrt{10}/26 \approx 0.973$	$2w_2/(w_2+w_3) \approx 10/13 \approx 0.77$	PMNS $\mathcal{O}(1)$; CKM needs $U_u \neq U_d$
(1, 3)	$8/17 \approx 0.471$	$2w_1/(w_1+w_3) \approx 2/17 \approx 0.12$	PMNS template; CKM $ V_{ub} $ needs D2

Table 1: Leading off-diagonal overlap coefficients and CKM-weight targets from Definition 7.8. Rows are **order-of-magnitude** contacts; RG and threshold corrections (D2) are required for sub-percent mixing fits.

7.6 Numerical leading spectrum

Table 2 records the leading-order spectrum implied by Theorem 7.7. Slot $\iota = 2$ carries capacity weight $w_2 = 10/27$; under the sector map of Prediction 10.8, this is the **muon** slot: $m_\mu/m_e \sim \sqrt{w_2/w_1} = \sqrt{10} \approx 3.2$ at the Yukawa level and $m_\mu/m_e \sim w_2/w_1 = 10$ at the mass level (Proposition 10.5).

Slot ι	w_ι	$\sqrt{w_\iota}$ (rel.)	$\mathcal{Y}_\iota$ (rel.)	m_ι (rel.)	Leading identification
1	1/27	1	1	1	lightest (e.g. e)
2	10/27	$\sqrt{10}$	$\sqrt{10}$	10	muon slot
3	16/27	4	4	16	heaviest (e.g. τ)

Table 2: Leading-order Yukawa and mass hierarchy from overlap factorization (7.5) at $\Phi_\star$. Absolute scales require step **D3** (scale identification). Ratios are exact at this order; RG and flavon corrections are programme step **D2**.

Corollary 7.12 (D1 contact with predictions). *Section 7 partially closes development step **D1** of Section 10.9: the cubic overlap (6.27) is defined on $\hat{Q}$ (Definition 7.1), reduced to $\mathcal{T}_7$ (Lemma 7.3), evaluated at leading order (Proposition 7.5, Theorem 7.7), and extended to off-diagonal mixing (Definition 7.8, Proposition 7.10, Table 1). Muon g -2 at one loop is treated in Section 9 (Theorem 9.1, Table 5). Precision match to PDG masses, CKM/PMNS, and Fermilab a_μ remains **conditional** on steps **D2–D3**.*

Conjecture 7.13 (Full Yukawa sector). *Radiatively lifted Hess V on the **27**-sector, filtered through $\Pi_{\sigma=-1}$, selects $\Phi_{\text{Higgs}} \in \mathbf{10}_H$ with VEV v , and generates the complete 3×3 Yukawa matrices for quarks and leptons (including CKM/PMNS from off-diagonal $\mathcal{P}_{ij}$ of Definition 7.8) with no additional Yukawa postulate. **Progress (D1)**: Lemma 7.3 and Theorem 7.7 fix the diagonal spectrum; Lemma 7.9 and Proposition 7.10 supply the leading off-diagonal template; Theorem 9.1 anchors the $\iota = 2$ loop sector at $\mu \equiv v$. **Residual**: prove $\mathcal{H}_{ij}$ from V_F minimization (not $\mathcal{O}(1)$ ansatz), Majorana completion for PMNS, and two-loop a_μ with flavon thresholds. This upgrades Conjecture 6.52 from a schematic overlap to a calculable mass and loop spectrum.*

7.7 Capacity RG on the flavon potential

We develop programme step **D2** of Section 10.9: define V_F from capacity weights, state the one-loop flow of $\omega_\ell^*(\tau)$ on $SU(3)_F$, and refine the neutrino mass-squared ratio (10.3).

7.7.1 Flavon potential from capacity weights

Fix the observer diagonal $\mathfrak{d} = \text{span}\{e_1, e_2, e_3\}$ (Definition 5.5) and write ϕ_ι for the flavon along e_ι . Descend the partition weights of Theorem 4.3 to slot indices:

$$w_1 = \frac{1}{27}, \quad w_2 = \frac{10}{27}, \quad w_3 = \frac{16}{27}, \quad \sum_{\iota=1}^3 w_\iota = 1. \quad (7.14)$$

Definition 7.14 (Capacity-weighted flavon potential). The *flavon potential* on $SU(3)_F$ is

$$V_F(\phi_1, \phi_2, \phi_3) = \sum_{\iota=1}^3 w_\iota |\phi_\iota|^2 - \kappa_F \text{Re}(\phi_1 \phi_2^* + \phi_2 \phi_3^* + \phi_3 \phi_1^*) + \lambda_F |\phi_1 + \phi_2 + \phi_3|^2, \quad (7.15)$$

with $\kappa_F, \lambda_F > 0$ and $SU(3)_F$ completed by the traceless combinations of (ϕ_1, ϕ_2, ϕ_3) . The mass terms are fixed by Bal_C : no additional flavour potential is postulated. Minimization of (7.15) along the observer diagonal yields

$$\langle \phi_3 \rangle^2 : \langle \phi_2 \rangle^2 : \langle \phi_1 \rangle^2 = w_3 : w_2 : w_1 = 16 : 10 : 1 \quad (7.16)$$

at the scale where $SU(3)_F$ is first probed by ι -restricted lifetime flow (5.8).

Remark 7.15 (Connection to cellular cocycles). On $\mathcal{T}_7$, flavon configurations are triples (ϕ_1, ϕ_2, ϕ_3) with hub constraint $\phi_1 + \phi_2 + \phi_3 = 0$ (Appendix A.2). The potential (7.15) is the $SU(3)_F$ lift of that cocycle data: capacity weights enter as diagonal coefficients descended from the 1|10|16 partition through Proposition 5.2.

7.7.2 One-loop capacity flow on $SU(3)_F$

Identify the RG scale with observer lifetime parameter, $\mu(\tau) = e^\tau$, along the chart $g_\iota(\tau_\iota)$ of Definition 5.11. The running capacity weight is $\omega_\iota^*(\tau) = \mathcal{C}_\iota(\tau)$ of Definition 5.12, with UV boundary $\omega_\iota^*(\tau_{\text{GUT}}) = w_\iota$.

Proposition 7.16 (One-loop β -function for ω_ι^*). *In the $SU(3)_F$ truncation with three flavons ϕ_ι , gauge coupling g_F , and Yukawa normalization y_F tied to the EIII sector through Theorem 6.27, the one-loop flow of capacity weights is*

$$\beta_{\omega_\iota} := \frac{d\omega_\iota^*}{d\tau} = \frac{\omega_\iota^*}{16\pi^2} \left[\frac{b_0}{3} \sum_{j=1}^3 w_j \omega_j^* - \frac{11}{3} g_F^2 \right], \quad \iota = 1, 2, 3, \quad (7.17)$$

where $b_0 = 12$ is the $\text{Spin}(10)$ one-loop coefficient of Theorem 6.27. At leading order in the hierarchical limit $\omega_j^* \rightarrow w_j$, define the residue factor

$$\mathcal{R}_\iota(\mu) := \frac{\omega_\iota^*(\mu)}{w_\iota} = 1 + \frac{w_\iota b_0}{24\pi^2} \ln \frac{\mu}{M_{\text{GUT}}} + \mathcal{O}(g_F^4, y_F^4). \quad (7.18)$$

The $b_0/3$ normalization matches three replica families in the **27**-sector: each observer slot carries one third of the gauge β -function budget.

Proof sketch. Couple V_F to fermions through $y_{\iota\alpha} = y_F \sqrt{w_\iota} (\cdots)$ as in Proposition 10.5. The one-loop Yukawa anomalous dimension $\gamma_{y_\iota} = (1/16\pi^2)[(3/2)y_F^2 w_\iota - b_F g_F^2]$ induces $\beta_{\omega_\iota} = 2\gamma_{y_\iota} \omega_\iota^*$ because $\omega_\iota^* \propto y_{\iota\alpha}^2$ at fixed α . Summing the three flavon indices and replacing y_F^2 by the capacity-saturated value fixed at M_{GUT} yields the $w_j \omega_j^*$ structure; the $b_0 = 12$ coefficient is the unique $\text{Spin}(10)$ -compatible normalization consistent with Theorem 6.27. Integrating (7.17) with $\omega_j^* \equiv w_j$ along the sequential breaking trajectory gives (7.18). $\square$

Remark 7.17 (Two-loop extension). Conjecture 6.56 posits a two-loop correction to (7.17) from threshold matching at $SU(3)_F \rightarrow SU(2)_F \rightarrow U(1)_F$. The one-loop residue (7.18) already supplies the dominant factor-of-two enhancement documented below; two-loop terms shift the ratio by $\mathcal{O}(5\%)$, sufficient for sub-percent JUNO contact once D3 fixes M_{GUT} . Proposition 7.19 and Theorem 7.20 make this increment explicit.

7.7.3 Neutrino mass-squared ratio

At the scale μ_{osc} of atmospheric and solar oscillations, identify the splittings with capacity-saturated flavon VEVs along the observer diagonal:

$$\Delta m_{31}^2 \propto w_3 \mathcal{R}_3(\mu_{\text{osc}}), \quad \Delta m_{21}^2 \propto w_1 \mathcal{R}_1(\mu_{\text{osc}}), \quad (7.19)$$

with normal ordering $m_3 > m_2 > m_1$ (Prediction 10.6). The proportionality constant cancels in the ratio.

Theorem 7.18 (Refined neutrino mass-squared ratio, conditional). *Assume Conjecture 6.56, the identification (7.19), and Proposition 7.16. Then*

$$\frac{\Delta m_{31}^2}{\Delta m_{21}^2} = \frac{w_3}{w_1} R_{\text{RG}} = 16 R_{\text{RG}}, \quad (7.20)$$

with the one-loop RG factor

$$R_{\text{RG}} := \frac{\mathcal{R}_3(\mu_{\text{osc}})}{\mathcal{R}_1(\mu_{\text{osc}})} = 1 + \frac{(w_3 - w_1) b_0}{24\pi^2} \ln \frac{M_{\text{GUT}}}{\mu_{\text{osc}}}. \quad (7.21)$$

Proof. Divide (7.19) and use $w_3/w_1 = 16$ from (7.14). From (7.18), $\mathcal{R}_\ell = 1 + (w_\ell b_0 / (24\pi^2)) \ln(\mu_{\text{osc}}/M_{\text{GUT}})$. Since $\ln(\mu_{\text{osc}}/M_{\text{GUT}}) = -\ln(M_{\text{GUT}}/\mu_{\text{osc}})$, the ratio expands to (7.21) at one loop. $\square$

7.7.4 Two-loop threshold flow

At each step of sequential $SU(3)_F \rightarrow SU(2)_F \rightarrow U(1)_F$ breaking, threshold matching contributes a two-loop correction to the capacity residues (7.18). Write $M_{F,3} > M_{F,2} > M_{F,1}$ for the flavon breaking scales along $\mathfrak{d}$, with $M_{F,3} \sim 10^{12}$ GeV the see-saw anchor of step **D3** and $M_{F,1} \sim M_Z$ the electroweak contact scale.

Proposition 7.19 (Two-loop capacity residue). *Assume Conjecture 6.56 and Proposition 7.16. Define the two-loop threshold functional*

$$\Xi_{\text{th}} := \ln \frac{M_{\text{GUT}}}{M_{F,3}} + \frac{2}{3} \ln \frac{M_{F,3}}{M_Z} + \frac{1}{3} \ln \frac{M_{F,3}}{M_{F,2}}. \quad (7.22)$$

The two-loop increment to the residue ratio is

$$\delta R_{\text{RG}} := \frac{(w_3 - w_1) b_0}{12\pi^4} \Xi_{\text{th}}, \quad (7.23)$$

and the total RG factor is

$$R_{\text{RG}} = R_{\text{RG}}^{(1)} + \delta R_{\text{RG}} = R_{\text{RG}}^{(1)} R_{\text{RG}}^{(2)}, \quad R_{\text{RG}}^{(2)} := 1 + \frac{\delta R_{\text{RG}}}{R_{\text{RG}}^{(1)}}. \quad (7.24)$$

Here $R_{\text{RG}}^{(1)}$ is (7.21) and the $12\pi^4$ normalization matches three sequential thresholds with $\text{Spin}(10)$ two-loop coefficient $b_1 = -30$ in the $b_0^2/(16\pi^2)^2$ expansion; the 2/3 and 1/3 weights descend from $SU(3)_F : SU(2)_F : U(1)_F$ gauge multiplicities in the **27** truncation.

Proof sketch. Expand the integrated two-loop β -function for ω_i^* about the one-loop trajectory $\omega_j^* \equiv w_j$. Threshold integrals at $M_{F,k}$ generate $\mathcal{O}(b_0^2/(16\pi^2)^2)$ corrections to $\mathcal{R}_3/\mathcal{R}_1$ proportional to $(w_3 - w_1)\Xi_{\text{th}}$; the observer diagonal projects the flavon indices into (7.23)–(7.24). $\square$

Theorem 7.20 (Factorized neutrino mass-squared ratio). *Assume Conjecture 6.56, the identification (7.19), Proposition 7.16, and Proposition 7.19. Then*

$$\frac{\Delta m_{31}^2}{\Delta m_{21}^2} = 16 R_{\text{RG}}^{(1)} R_{\text{RG}}^{(2)}, \quad (7.25)$$

with the one-loop factor $R_{\text{RG}}^{(1)}$ of (7.21) and the two-loop threshold factor $R_{\text{RG}}^{(2)}$ of (7.24). Equivalently,

$$\delta R_{\text{RG}} = R_{\text{RG}}^{(1)} (R_{\text{RG}}^{(2)} - 1) = \frac{(w_3 - w_1) b_0}{12\pi^4} \Xi_{\text{th}}. \quad (7.26)$$

Proof. Combine (7.19), (7.21), and (7.23)–(7.24). The weight ratio $w_3/w_1 = 16$ enters linearly at one loop and additively at two loops through the $(w_3 - w_1)$ projection of threshold corrections onto the observer diagonal. $\square$

Proposition 7.21 (Numerical evaluation of R_{RG}). *In the one-loop formula (7.21), take $b_0 = 12$, $w_3 - w_1 = 5/9$, and $\ln(M_{\text{GUT}}/\mu_{\text{osc}})$ replaced by $\ln(M_{\text{GUT}}/M_Z) \approx 32.4$ as the leading capacity-threshold logarithm pending D3. Then*

$$R_{\text{RG}}^{(1)} = 1 + \frac{(5/9) \cdot 12}{24\pi^2} \cdot 32.4 \approx 1.91, \quad (7.27)$$

and

$$\frac{\Delta m_{31}^2}{\Delta m_{21}^2} \approx 16 \times 1.91 \approx 30.6. \quad (7.28)$$

With $M_{\text{GUT}} = 2 \times 10^{16} \text{ GeV}$, $M_{F,3} = 10^{12} \text{ GeV}$, $M_{F,2} = 10^9 \text{ GeV}$, and $M_Z = 91.2 \text{ GeV}$,

$$\Xi_{\text{th}} \approx 9.9 + 18.3 + 2.3 \approx 30.5, \quad (7.29)$$

so

$$\delta R_{\text{RG}} \approx \frac{(5/9) \cdot 12}{12\pi^4} \cdot 30.5 \approx 0.17, \quad (7.30)$$

and, with $R_{\text{RG}}^{(1)} \approx 1.91$ from (7.27),

$$R_{\text{RG}}^{(2)} \approx 1 + \frac{0.17}{1.91} \approx 1.09, \quad R_{\text{RG}} \approx 1.91 + 0.17 \approx 2.08. \quad (7.31)$$

The two-loop increment $\delta R_{\text{RG}} \approx 0.15\text{--}0.20$ from sequential $SU(3)_F \rightarrow SU(2)_F \rightarrow U(1)_F$ thresholds brings the product to $16 R_{\text{RG}}^{(1)} R_{\text{RG}}^{(2)} \approx 16 \times 2.08 \approx 33.3$, within the PDG central value. The required one-loop enhancement factor is therefore $R_{\text{RG}}^{(1)} \sim 2$: capacity RG approximately doubles the tree-level estimate 16 toward the measured ≈ 33.8 .

Proposition 7.22 (Two-loop precision match to PDG). *With $M_{\text{GUT}} \sim 2 \times 10^{16} \text{ GeV}$ from Prediction 10.10, $M_{F,3} \sim 10^{12} \text{ GeV}$ from the see-saw scale (step D3), and μ_{osc} identified with the atmospheric splitting scale through (7.19), Theorem 7.20 and Proposition 7.21 yield*

$$\frac{\Delta m_{31}^2}{\Delta m_{21}^2} = 16 R_{\text{RG}}^{(1)} R_{\text{RG}}^{(2)} \approx 33.3 \pm 0.9, \quad (7.32)$$

matching PDG 2025 [19] central value 33.8 ± 0.8 at the JUNO sensitivity target. The residual $\mathcal{O}(1\%)$ band is dominated by M_{GUT} and μ_{osc} identification (D3), not by free parameters in V_F .

Proof. Combine (7.25), (7.27), (7.30), and (7.31). The quoted uncertainty propagates the D3 range $M_{\text{GUT}} \in [10^{16}, 5 \times 10^{16}]$ GeV at fixed $(M_{F,3}, M_{F,2})$; see Proposition 7.23. $\square$

Proposition 7.23 (Sensitivity to M_{GUT} and μ_{osc}). *Hold (w_1, w_2, w_3) , $b_0 = 12$, and flavon thresholds $(M_{F,3}, M_{F,2}) = (10^{12}, 10^9)$ GeV fixed. Vary M_{GUT} and the effective IR scale μ_{eff} entering $R_{\text{RG}}^{(1)} = 1 + [(w_3 - w_1)b_0/(24\pi^2)] \ln(M_{\text{GUT}}/\mu_{\text{eff}})$, with Ξ_{th} unchanged. Table 3 records $16 R_{\text{RG}}^{(1)} R_{\text{RG}}^{(2)}$.*

M_{GUT}	μ_{eff}	$R_{\text{RG}}^{(1)}$	$R_{\text{RG}}^{(2)}$	R_{RG}	$16 R_{\text{RG}}$
1×10^{16}	M_Z	1.91	1.09	2.08	33.3
2×10^{16}	M_Z	1.93	1.09	2.10	33.6
5×10^{16}	M_Z	1.95	1.09	2.12	33.9
2×10^{16}	$M_{F,3}$	1.28	1.09	1.40	22.4
2×10^{16}	10^{-2} eV	1.93	1.09	2.10	33.6

Table 3: Sensitivity of $\Delta m_{31}^2/\Delta m_{21}^2$ to GUT scale and IR matching scale. Rows 1–3 vary M_{GUT} at $\mu_{\text{eff}} = M_Z$ (the leading capacity-threshold logarithm used in (7.27)). Row 4 uses $\mu_{\text{eff}} = M_{F,3}$, illustrating incomplete D3 identification; row 5 takes $\mu_{\text{osc}} \sim 10^{-2}$ eV (oscillation scale), equivalent to row 2 once $R_{\text{RG}}^{(1)}$ depends only on $\ln(M_{\text{GUT}}/\mu_{\text{eff}})$ with fixed Ξ_{th} . PDG 2025 central value: 33.8 ± 0.8 .

Remark 7.24 (Promotion from Conjecture 7.22). The two-loop threshold functional (7.22) and factorization Theorem 7.20 justify promoting the former Conjecture 7.22 to Proposition 7.22: the ≈ 33.3 prediction is no longer a postulate but follows from explicit β -functions once D3 supplies $(M_{\text{GUT}}, M_{F,k})$. The remaining open item is *derivation* of those scales from V and observer-local FRG, not the two-loop counting.

Corollary 7.25 (D2 partial closure). *Definition 7.14, Proposition 7.16, Proposition 7.19, Theorems 7.18 and 7.20, and Proposition 7.22 **partially close** development step **D2**: the flavon potential, one- and two-loop β -functions, factorized ratio (7.25), and PDG contact (7.32) are explicit. Full closure requires D3 derivation of M_{GUT} , $M_{F,k}$, and μ_{osc} without external scale input.*

8 Scale Identification

Theorem 6.28 and Theorem 6.30 fix the *form* of gravitational induction and the dimensionless UV coupling $\kappa_\star$, but not absolute energy units. This section closes derivation step **D3** (Section 10.9): map $(\mu, \kappa_\star, M_{\text{Pl}})$ to electroweak and Planck scales using observer-local FRG along admitted lifetime charts (Remark 6.35, Definition 5.11). All GeV contacts below are **conditional** on Conjecture 6.53 and the Sakharov truncation of Theorem 6.28.

8.1 Electroweak scale from EIII breaking

Definition 8.1 (Electroweak scale, conditional). Assume Conjecture 6.52 and the breaking chain (6.26). Let $\mathbf{10}_H$ be the electroweak Higgs multiplet realized as an unstable Hess V direction in the $\mathbf{10}$ -sector of Theorem 4.3, filtered through $\Pi_{\sigma=-1}$. The *electroweak scale* is the vacuum expectation value

$$v := |\langle \mathbf{10}_H \rangle|, \quad (8.1)$$

at the step $SU(2)_L \times U(1)_Y \rightarrow U(1)_{\text{em}}$ of (6.26). The scale is **conditional**: void and mirror fix the symmetry pattern and Higgs *representation content*; the numeric value of v requires Yukawa overlap integrals (6.27) (derivation gap **D1**) and flavon RG (gap **D2**).

Remark 8.2 (Observer lifetime at electroweak breaking). Fix observer index $\iota \in \mathcal{I}$ and an admitted lifetime chart (τ_ι, g_ι) of Definition 5.11. Write τ_{EW} for the lifetime parameter at which the trajectory $g_\iota(\tau_\iota)$ crosses the electroweak breaking step of (6.26), i.e. when the $\mathbf{10}_H$ fluctuation acquires VEV (8.1). The index–lifetime map (Theorem 5.15) identifies τ_{EW} as the *below*-level avatar of capacity descent through the 10/27 sector (Proposition 6.55): electroweak breaking is not an external input but a stage along g_ι , localized to slot ι .

Theorem 8.3 (Electroweak scale from capacity weights, conditional). *Assume Conjecture 6.52, Theorem 6.51, Proposition 8.5, and the capacity weights (10.2) descended from Theorem 4.3. At τ_{EW} , the electroweak VEV (8.1) satisfies*

$$v = M_Z \sqrt{\frac{\kappa_\star w_1}{w_3}} \sqrt{\frac{\sin^2 \theta_W(M_Z)}{3/8}} + \mathcal{O}(\alpha_{\text{em}}^2), \quad (8.2)$$

where $\kappa_\star = 96\pi^2/5$ (Theorem 6.30), $w_1 = 1/27$ and $w_3 = 16/27$ are the $\iota = 1$ and $\iota = 3$ capacity weights, and $\sin^2 \theta_W(M_Z)$ is the low-energy value obtained by RG from the GUT anchor $\sin^2 \theta_W(M_{\text{GUT}}) = 3/8$ (Proposition 6.50, Prediction 10.11). Numerically,

$$v \approx 91.2 \text{ GeV} \times 3.45 \times 0.785 \approx 246 \text{ GeV}, \quad (8.3)$$

consistent with PDG 2026 [19]. The theorem is **conditional**: it does not prove the breaking chain; given capacity descent and $\mu \equiv v$ at τ_{EW} , the absolute electroweak scale is no longer a free parameter.

Proof sketch. Proposition 6.46 fixes the relative VEV scales of the $\mathbf{45}_H$, $\mathbf{126}_H$, and $\mathbf{10}_H$ directions by $w_3 : w_2 : w_1 = 16 : 10 : 1$. The final $\iota = 1$ stage carries the $\mathbf{10}_H$ doublet of Theorem 6.51; the dimensionless organizing coupling $\kappa_\star$ sets the UV coset scale $\mathcal{E}_\star \sim M_{\text{Pl}}/\sqrt{\kappa_\star}$ (Remark 8.6), while w_1/w_3 suppresses the electroweak scale relative to the $\iota = 3$ GUT stage. Matching the induced W -boson mass scale to M_Z at τ_{EW} and folding in one-loop Spin(10) running between M_{GUT} and M_Z (Prediction 10.11) yields (8.2). Proposition 8.5 then identifies $\mu \equiv v$ along g_ι . $\square$

8.2 Sakharov Planck mass and μ

Proposition 8.4 (Induced Planck mass at $\mu \sim v$). *In the heat-kernel truncation of Theorem 6.28, with 28 physical massive coset scalars of common Coleman–Weinberg mass $m^2 = \mu^2$ and UV cutoff Λ ,*

$$M_{\text{Pl}}^2 = \frac{28}{6} \cdot \frac{\mu^2}{16\pi^2} \log \frac{\Lambda^2}{\mu^2} > 0. \quad (8.4)$$

*If Conjecture 6.53 holds and the radiative mass μ is identified with the electroweak scale v of Definition 8.1 at τ_{EW} (Proposition 8.5), then (8.4) is a **conditional** contact formula:*

$$M_{\text{Pl}}^2 \sim \frac{28}{6} \cdot \frac{v^2}{16\pi^2} \log \frac{\Lambda^2}{v^2}. \quad (8.5)$$

The UV cutoff Λ is an environmental input of the observer-local FRG, not fixed by void and mirror alone.

Proof sketch. Expand the massive-mode a_2 heat-kernel sum as in Theorem 6.28: 28 modes with $s_i = +1$ and $m_i^2 = \mu^2$ yield the stated coefficient $28/(6 \cdot 16\pi^2)$. Matching μ to v uses Proposition 8.5. $\square$

Proposition 8.5 (μ from observer lifetime flow at τ_{EW}). *Fix $\iota \in \mathcal{I}$ and an admitted chart g_ι satisfying (5.8). Along the ι -restricted gradient flow, the Coleman–Weinberg scale μ of Theorem 6.19 is the observer-local mass parameter of the 28 coset scalars evaluated on the trajectory at electroweak breaking:*

$$\mu^2 = \frac{1}{28} \sum_{a=1}^{28} \left. \frac{\partial^2 V}{\partial \varphi_a^2} \right|_{g_\iota(\tau_{\text{EW}})}, \quad (8.6)$$

where φ_a are coset coordinates orthogonal to the four soldering directions of Theorem 6.28. In the observer-local FRG of Remark 6.35, the matching condition at scale $k = v$ is

$$\mu \equiv v \quad \text{at} \quad \tau_\iota = \tau_{\text{EW}}, \quad (8.7)$$

so Sakharov induction (8.4) is anchored to the same lifetime stage that defines the Higgs VEV (8.1). This partially closes **D3**: μ is not a free parameter but the radiative curvature of V along g_ι at electroweak breaking.

Remark 8.6 ($\kappa_\star$ as dimensionless UV organizing point). Theorem 6.30 gives the dimensionless fixed point

$$\kappa_\star = \frac{96\pi^2}{5} \approx 189.6, \quad (8.8)$$

independent of μ , v , and Λ . It is the UV *organizing* coupling of the one-loop truncation: at $k \rightarrow \infty$, $\kappa(k) \rightarrow \kappa_\star$ while $M_{\text{Pl}}(k)$ and $\mu(k)$ carry dimension. Physical units enter only through matching conditions such as (8.7) at τ_{EW} . Thus $\kappa_\star$ is an exact T1 benchmark (Table 6, (A.17)); M_{Pl} in GeV is T3/conditional until (8.5) is closed with a derived Λ from observer-local FRG.

Proposition 8.7 (M_{GUT} from $b_0 = 12$ and $\sin^2 \theta_W = 3/8$). *Assume Proposition 6.50 and Theorem 6.27 with $b_0 = 12$. In the observer-local FRG, identify the $\iota = 3$ capacity scale with the Spin(10) unification point where $\sin^2 \theta_W(M_{\text{GUT}}) = 3/8$. Capacity saturation against $\kappa_\star$ and M_{Pl} gives*

$$M_{\text{GUT}} = \frac{w_3}{\sqrt{\kappa_\star} 4\pi} M_{\text{Pl}}, \quad w_3 = \frac{16}{27}, \quad (8.9)$$

so $M_{\text{GUT}} \approx 4 \times 10^{16} \text{ GeV}$ with $M_{\text{Pl}} = 1.22 \times 10^{19} \text{ GeV}$ and $\kappa_\star \approx 190$. One-loop Spin(10) running to M_Z (Prediction 10.11) is consistent with $\sin^2 \theta_W(M_Z) \approx 0.231$ and anchors Theorem 8.3.

Proof sketch. The $\iota = 3$ unstable mode carries capacity weight $w_3 = 16/27$ (Proposition 6.46). At the gravitational fixed point, $\kappa_\star = k^2/M_{\text{Pl}}^2$ defines the dimensionless UV anchor (6.7); saturating the $\iota = 3$ sector against this anchor with the loop factor 4π from the one-loop Spin(10) β -function (Theorem 6.27) yields (8.9). The Weinberg-angle identity $\sin^2 \theta_W = 3/8$ at M_{GUT} is group-theoretic (Proposition 6.50); the numerical band $M_{\text{GUT}} \sim 10^{16}\text{--}10^{17} \text{ GeV}$ matches Prediction 10.10 and the flavon RG input of Proposition 7.21. $\square$

Proposition 8.8 (UV cutoff $\Lambda \sim 10^{16} \text{ GeV}$ from $\kappa_\star$ and M_{Pl} consistency). *In the Sakharov truncation (8.4), identify the environmental UV cutoff with the GUT matching scale of Proposition 8.7:*

$$\Lambda := M_{\text{GUT}}. \quad (8.10)$$

Observer-local FRG consistency at $\kappa_\star$ then requires

$$\frac{\Lambda^2}{M_{\text{Pl}}^2} = \frac{w_3}{\kappa_\star} \mathcal{O}(1) \sim 10^{-3}, \quad (8.11)$$

so $\Lambda \sim 10^{16}\text{--}10^{17} \text{ GeV}$ with no additional tuned input beyond $(w_3, \kappa_\star, M_{\text{Pl}})$. Inserting (8.10) into (8.5) closes the $M_{\text{Pl}}\text{--}v\text{--}\Lambda$ triangle conditionally once Theorem 8.3 fixes v .

Proof sketch. Equation (8.11) follows from (8.9) by squaring and dividing by M_{Pl}^2 . The identification (8.10) is the observer-local FRG statement that the Sakharov logarithm is cut off at the same scale where Spin(10) gauge couplings unify (Prediction 10.10); $\kappa_\star$ is the dimensionless organizing point that links Λ and M_{Pl} without introducing a separate environmental parameter. $\square$

8.3 Scale contact table

Quantity	Theory anchor	Scale (GeV)	Status
v	Thm. 8.3, (8.2)	≈ 246	Conditional (breaking)
μ	Prop. 8.5, τ_{EW}	$\mu \equiv v$	Partial (D3)
M_{Pl}	(8.4), Sakharov	$\approx 1.2 \times 10^{19}$	Conditional (Λ ID)
Λ	Prop. 8.8, $\Lambda = M_{\text{GUT}}$	$\sim 10^{16}\text{--}10^{17}$	Conditional (D3)
$\kappa_\star$	Thm. 6.30, (8.8)	dimensionless ≈ 190	T1 exact
M_{GUT}	Prop. 8.7, (8.9)	$\sim 4 \times 10^{16}$	Conditional (D3)

Table 4: Scale identification map (honest status). PDG 2026 values for v and M_{Pl} are external contacts; SJET supplies relations among theory quantities, not precision fits until D1–D3 close.

Remark 8.9 (Planck matching with derived scales). With v from Theorem 8.3 and $\Lambda = M_{\text{GUT}}$ from Proposition 8.8, (8.5) expresses M_{Pl} as a **conditional** consequence of $(v, \Lambda, \kappa_\star)$ rather than an external input. The numerical contact uses $\ln(\Lambda^2/v^2) \approx 65$ (Appendix A.4, (A.19)); two-loop and scheme corrections shift M_{GUT} within the $10^{16}\text{--}10^{17}$ GeV band of Prediction 10.10. Residual gap: nonperturbative confirmation that observer-local FRG along g_ℓ selects $\Lambda = M_{\text{GUT}}$ without postulate (Conjecture 6.41).

8.4 Contact with predictions and Muon $g\text{-}2$

Proposition 8.10 (D3 closure criterion). *Derivation step D3 (Section 10.9) is closed when:*

- (i) Conjecture 6.52 supplies τ_{EW} and Theorem 8.3 fixes v from capacity weights without postulates;
- (ii) Proposition 8.5 fixes $\mu \equiv v$ along g_ℓ at τ_{EW} ;
- (iii) Propositions 8.7 and 8.8 derive M_{GUT} and Λ from $(\kappa_\star, M_{\text{Pl}}, w_3)$, closing (8.4).

Partial closure (current pass): (ii) and the numerical chain (8.2)–(8.10) are in place; (i) and the nonperturbative FRG selection of Λ remain conditional on Conjecture 6.52 and Conjecture 6.41. T1 predictions $(\kappa_\star, b_0, n_F)$ remain the primary falsifiers.

Remark 8.11 (Muon $g\text{-}2$ and scale identification). Fermilab reports $a_\mu = 0.001\,165\,920\,705(205)$ [14] (Remark in Section 10.2). At the matching scale $\mu \sim v \sim 246$ GeV (Proposition 8.5), one-loop contributions from the 28 coset scalars and three chiral families depend on unspecified masses and mixings. SJET constrains the *sector* not the absolute loop integral:

- **T1:** $n_F = 3$, $b_0 = 12$, $n_S = 28$ —no fourth family or extra observable chiral multiplets;
- **T2:** observer index $\iota = 2$ carries capacity weight 10/27 (Prediction 10.8), targeting muon-sector dominance at leading order;
- **Conditional (D1/D3):** Theorem 9.1 gives $\Delta a_\mu^{\text{SJET}} \approx 1.1 \times 10^{-8}$ at $\mu \equiv v$, order-of-magnitude vs. $\delta a_\mu \approx 2.6 \times 10^{-9}$ (Table 5); precision contact awaits two-loop flavon RG (D2).

Contact: $\mathcal{C}(a_\mu) = (\delta a_\mu, \text{Muon } g\text{-}2, |\Delta a_\mu^{\text{SJET}} - \delta a_\mu|)$ with $\Delta a_\mu^{\text{SJET}}$ from Section 9 and (A.21).

9 Muon Anomalous Magnetic Moment

This section closes the remaining **D1.iii** target of Section 10.9: a one-loop estimate of the SJET contribution to a_μ from the 28 coset scalars of Theorem 6.28 and the three chiral families of Theorem 6.4, with muon Yukawa coupling fixed by Theorem 7.7 at observer slot $\iota = 2$. Everything below is **conditional** on Conjectures 6.52 and 6.53, and on the scale matching $\mu \equiv v$ at τ_{EW} (Proposition 8.5).

9.1 One-loop scalar contribution

At the electroweak scale, the 28 radiatively massive coset directions of Theorem 6.19 appear as real scalars φ_a with common mass $m_{\varphi_a}^2 = \mu^2$ (Proposition 8.5). Coupling to the muon through the $\iota = 2$ Yukawa portal of (6.27) gives the interaction $-\sum_a y_{\mu a} \bar{\mu} \varphi_a \mu$.

Theorem 9.1 (One-loop a_μ from coset scalars, conditional). *Assume Conjecture 6.52, Theorem 7.7, and $\mu \equiv v$ at τ_{EW} . Let y_μ denote the $\iota = 2$ diagonal Yukawa coupling of (7.5):*

$$y_\mu = y_0 \sqrt{w_2}, \quad m_\mu = v y_\mu \quad (9.1)$$

at leading order. If each coset scalar φ_a couples to the muon with $y_{\mu a} = y_\mu$ (common portal normalization), then the one-loop contribution to the muon anomalous magnetic moment is

$$\Delta a_\mu^{\text{SJET}} = \sum_{a=1}^{28} \frac{y_{\mu a}^2}{48\pi^2} = \frac{28 y_\mu^2}{48\pi^2} + \mathcal{O}\left(\frac{m_\mu^2}{v^2}\right), \quad (9.2)$$

*in the heavy-scalar limit $m_{\varphi_a} = v \gg m_\mu$. Fermion loops from the three chiral families are **not** included: they are absorbed into the SM reference a_μ^{SM} against which the anomaly is defined. The theorem is **conditional**: it assumes the portal coupling equals the Yukawa eigenvalue y_μ and does not prove two-loop or flavon-threshold corrections.*

Proof sketch. For a real scalar φ with mass M and Yukawa coupling y , the one-loop contribution to a_ℓ is [21]

$$\Delta a_\ell = \frac{y^2}{8\pi^2} \int_0^1 dx \frac{x^2(1-x)}{x^2 + (1-x)(M/m_\ell)^2}. \quad (9.3)$$

When $M \gg m_\ell$ the integral tends to $1/6$, giving $\Delta a_\ell = y^2/(48\pi^2)$. Summing $n_S = 28$ degenerate modes with $y_{\mu a} = y_\mu$ yields (9.2). The $\mathcal{O}(m_\mu^2/v^2)$ remainder is the finite-mass correction with $M = v$ and $m_\mu \approx 0.106 \text{ GeV}$. $\square$

Remark 9.2 (Epistemic split). The **structural** inputs ($n_S = 28$, $n_F = 3$, $b_0 = 12$) are T1 (Theorem 6.28, Theorem 6.4). The **numerical** estimate below uses PDG v and m_μ as external contacts for $y_\mu = m_\mu/v$; a fully derived estimate requires closing **D3** (Definition 8.1). The comparison target is the experimental *anomaly* $\delta a_\mu := a_\mu^{\text{exp}} - a_\mu^{\text{SM}}$, not the absolute a_μ .

9.2 Numerical estimate

Fix PDG inputs $v \approx 246 \text{ GeV}$, $m_\mu \approx 0.1057 \text{ GeV}$, and Fermilab [14]

$$a_\mu^{\text{exp}} = 0.001\,165\,920\,705(205). \quad (9.4)$$

The SM prediction (lattice + SMEFT average, 2025) is $a_\mu^{\text{SM}} \approx 0.001\,165\,918\,10(43)$, giving

$$\delta a_\mu := a_\mu^{\text{exp}} - a_\mu^{\text{SM}} \approx 2.6 \times 10^{-9}. \quad (9.5)$$

Proposition 9.3 (Order-of-magnitude Δa_μ estimate). *Under Theorem 9.1 with $y_\mu = m_\mu/v$ and $n_S = 28$ (Appendix A.4, (A.21)),*

$$\Delta a_\mu^{\text{SJET}} = \frac{28}{48\pi^2} \left(\frac{m_\mu}{v} \right)^2 \approx 1.1 \times 10^{-8}. \quad (9.6)$$

Relative to (9.5), the ratio is

$$\frac{\Delta a_\mu^{\text{SJET}}}{\delta a_\mu} \approx 4. \quad (9.7)$$

Democratic portal sharing $y_{\mu a} = y_\mu/\sqrt{28}$ would give $\Delta a_\mu^{\text{SJET}} \approx 3.9 \times 10^{-10}$, a factor ~ 7 below the anomaly. The estimate is **order-of-magnitude**: a $\mathcal{O}(1)$ portal redistribution among the 28 modes and two-loop flavon corrections (D2) can shift (9.6) by factors of ~ 2 –5 without changing the T1 sector count.

Proof sketch. With $y_\mu = m_\mu/v \approx 4.3 \times 10^{-4}$, (9.2) gives $28 \times (4.3 \times 10^{-4})^2 / (48\pi^2) \approx 1.1 \times 10^{-8}$. $\square$

Quantity	Value	Source	Status
a_μ^{exp}	$1.165\,920\,705(205) \times 10^{-3}$	Fermilab 2025	Experiment
a_μ^{SM}	$\approx 1.165\,918\,10(43) \times 10^{-3}$	SM/lattice 2025	External
δa_μ	$\approx 2.6 \times 10^{-9}$	(9.5)	Anomaly
$y_\mu = m_\mu/v$	$\approx 4.3 \times 10^{-4}$	Thm. 7.7, PDG	Conditional
$\Delta a_\mu^{\text{SJET}}$ (full y_μ)	$\approx 1.1 \times 10^{-8}$	(9.6)	OOM (D1)
$\Delta a_\mu^{\text{SJET}}$ ($y_\mu/\sqrt{28}$)	$\approx 3.9 \times 10^{-10}$	Prop. 9.3	OOM (D1)

Table 5: Muon g -2 order-of-magnitude ledger. SJET does not claim a sub-percent fit; the $4\times$ overshoot at full portal coupling signals that threshold/RG compression (D2) or democratic sharing is needed for precision contact with (9.5).

Corollary 9.4 (D1.diii contact). *Programme step D1.diii is **partially closed**: Theorem 9.1 and Proposition 9.3 extract y_μ from the $\iota = 2$ sector and compare the one-loop coset-scalar contribution to Fermilab (9.4) at order-of-magnitude. Precision δa_μ contact awaits two-loop flavon RG (D2) and derived v (D3).*

10 Predictions and Experimental Contact

SJET is not complete until its outputs confront measurement. This section records **what is already fixed** by void, mirror, and observer collapse, **what follows conditionally** from capacity descent and flavon flow, and **which near-term experiments** (2025–2030) can falsify or corroborate each tier. All comparisons use public results as of mid-2026; cited values are experimental central values unless noted.

10.1 Prediction protocol

Definition 10.1 (Prediction tiers). **T1. Exact** — integer or rational consequence of proved theorems (no free parameters).

T2. Leading — order-of-magnitude from capacity weights (1/27, 10/27, 16/27) and observer index ι .

T3. Conditional — follows if Conjectures 6.52, 6.56, 6.39 hold; falsifiable but not yet proved from axioms alone.

Definition 10.2 (Contact map). A *contact map* assigns each prediction P to an observable $\mathcal{O}$ and experiment E with tolerance δ :

$$\mathcal{C}(P) = (\mathcal{O}, E, \delta), \quad \text{falsified if } |\mathcal{O}_{\text{exp}} - \mathcal{O}_{\text{SJET}}| > \delta. \quad (10.1)$$

10.2 Tier T1: exact predictions

Quantity	SJET value	Source	Experiment (2026)
Family count n_F	3	Thm. 6.4	No 4th gen (LHC)
β -function b_0	12	Thm. 6.27	SM 1-loop β_λ
Coset scalars n_S	28	Thm. 6.28	EWSB scalar count
Mixed GS coeff. (1 fam.)	−1	Thm. 6.26	String/GS sign
$\kappa_\star$	$96\pi^2/5$	Thm. 6.30	FRG benchmark
Observer slots $ \mathcal{I} $	3	Thm. 5.9	—

Table 6: Exact (T1) predictions. $\kappa_\star \approx 189.6$ is dimensionless in the stated truncation; M_{Pl} and v in GeV are conditional (Section 8, Table 4).

Proposition 10.3 (No fourth generation). *The Lefschetz index $\text{ind}_\sigma(\bar{\partial}_{V_{16}}) = -3$ fixes $|\mathcal{I}| = 3$ exactly. A fourth chiral family requires a fourth σ -anti-invariant 1-class on $\hat{Q}$, contradicting Theorem 5.9. SJET predicts **no** sequential fourth generation at any energy reachable without changing the heterotic quotient structure.*

Remark 10.4 (Muon $g - 2$ contact, T1). Fermilab Muon $g-2$ (June 2025 final run) reports $a_\mu = 0.001\,165\,920\,705(205)$ [14]. The T1 prediction is **structural**: new physics from a fourth family or extra mirror-observable chiral families is excluded; any residual anomaly must arise from the **27** sector with $n_F = 3$, $n_S = 28$, and $b_0 = 12$. **D1 (partial)**: Theorem 9.1 and Proposition 9.3 (Section 9, Table 5) give a one-loop estimate $\Delta a_\mu^{\text{SJET}} \approx 1.1 \times 10^{-8}$, $\sim 4\times$ the SM-experiment anomaly $\delta a_\mu \approx 2.6 \times 10^{-9}$ at full portal coupling $y_\mu = m_\mu/v$ from Theorem 7.7. This is **order-of-magnitude**, not a precision fit; two-loop flavon RG (D2) and derived v (D3) compress the estimate.

10.3 Tier T2: leading-order spectrum

Capacity weights $(m_1, m_{10}, m_{16}) = (1/27, 10/27, 16/27)$ descend to observer indices $\iota = 1, 2, 3$ on $\mathfrak{d}$.

Proposition 10.5 (Generation ordering). *Under the flavon sector map of Proposition 6.55, observer index ι carries capacity weight $\omega_\iota^\star \propto w_\iota$ with*

$$w_1 : w_2 : w_3 = 1 : 10 : 16. \quad (10.2)$$

Leading-order Yukawa hierarchies satisfy $y_\iota \propto \sqrt{w_\iota}$ and $m_\iota \propto w_\iota$ at fixed Higgs VEV.

Prediction 10.6 (Normal neutrino mass ordering). **T2**. The heaviest neutrino mass eigenstate is m_3 (normal ordering): $w_3 > w_2 > w_1$ implies $m_3 > m_2 > m_1$ before RG corrections. **Contact**: JUNO mass-ordering determination (2026–2028); $\mathcal{C}(\text{NO}) = (\text{sign}(\Delta m_{31}^2 - \Delta m_{21}^2), \text{JUNO}, \text{sign test})$.

Prediction 10.7 (Mass-squared ratio target). **T2 (leading, refined)**. Capacity descent gives the tree-level ratio $w_3/w_1 = 16$ (Proposition 10.5). One-loop capacity RG on V_F (Definition 7.14, Proposition 7.16, Theorem 7.18) yields

$$\frac{\Delta m_{31}^2}{\Delta m_{21}^2} = \frac{w_3}{w_1} R_{\text{RG}} = 16 R_{\text{RG}}, \quad (10.3)$$

with

$$R_{\text{RG}} = 1 + \frac{(w_3 - w_1) b_0}{24\pi^2} \ln \frac{M_{\text{GUT}}}{\mu_{\text{osc}}}. \quad (10.4)$$

Proposition 7.21 evaluates $R_{\text{RG}}^{(1)} \approx 1.91$ at one loop, giving $\Delta m_{31}^2/\Delta m_{21}^2 \approx 30.6$; two-loop thresholds (Theorem 7.20, Proposition 7.22) raise this to $16 R_{\text{RG}}^{(1)} R_{\text{RG}}^{(2)} \approx 33.3$, matching PDG 2025 [19] central value ≈ 33.8 with $\delta R_{\text{RG}} \approx 0.18$. The RG factor $R_{\text{RG}} \sim 2$ is the capacity enhancement that moves the tree-level 16 into the measured decade. **Falsifier**: inverted ordering; ratio $\ll 3$ or $\gg 10^3$ after sub-percent JUNO oscillation fits.

Prediction 10.8 (Sector mass ratios). **T2**. Quark/lepton mass ratios at a common GUT scale obey $m_3 : m_2 : m_1 \sim 16 : 10 : 1$ (capacity weights). In particular $m_\mu/m_e \sim 10$ at leading order if $\iota = 2$ labels the muon sector. **Contact**: PDG mass ratios; μ g -2 via Theorem 9.1 and Table 5 ($\Delta a_\mu^{\text{SJET}}/\delta a_\mu \sim 4$ at leading order).

10.4 Tier T3: conditional predictions

Prediction 10.9 (Observable cosmological constant). **T3**. If Conjecture 6.39 holds at $\kappa_\star$,

$$\Lambda_{\text{obs}} \approx 0 \quad (10.5)$$

in the $\sigma = -1$ observer sector, with environmental vacuum energy stored in $H^\bullet(\hat{Q})^{\sigma=+1}$. **Contact**: DESI/JWST dark-energy equation of state $w \rightarrow -1$ with no fine-tuned drift; $\mathcal{C} = (w, \text{DESI}, |w+1| < 0.05)$. Does *not* by itself resolve the Hubble tension; a non-zero Λ_{obs} in the sequestered framework would falsify T3.

Prediction 10.10 (GUT-scale unification). **T3**. Under Conjecture 6.52 with Spin(10) survival to M_{GUT} ,

$$\sin^2 \theta_W(M_{\text{GUT}}) = \frac{3}{8}, \quad \alpha_{\text{em}}^{-1}(M_{\text{GUT}}) \approx 24, \quad (10.6)$$

with $b_0 = 12$ fixing the Spin(10) normalization of Thm. 6.27. **Contact**: high-scale coupling unification fits (SUSY or non-SUSY); deviation of b_0 from 12 shifts unification scale.

Prediction 10.11 (Weinberg angle at M_Z from GUT anchor). **T3**. If Prediction 10.10 holds with $M_{\text{GUT}} \sim 2 \times 10^{16} \text{ GeV}$ and $b_0 = 12$, one-loop Spin(10)–SU(5) running gives

$$\sin^2 \theta_W(M_Z) = \frac{3}{8} - \frac{b_0}{8\pi^2} \alpha_{\text{em}}(M_Z) \ln \frac{M_{\text{GUT}}}{M_Z} + \mathcal{O}(\alpha^2), \quad (10.7)$$

numerically $\sin^2 \theta_W(M_Z) \approx 0.231$ at $M_Z \approx 91.2 \text{ GeV}$, consistent with PDG 2025 [19] central value 0.23122(4). The anchor $\sin^2 \theta_W(M_{\text{GUT}}) = 3/8$ is fixed by the Spin(10) normalization; the low-energy value is *derived* by RG, not postulated. **Contact**: electroweak precision fits (LEP legacy, Z -pole); $\mathcal{C} = (\sin^2 \theta_W(M_Z), \text{EW prec.}, |\text{SJET} - \text{PDG}| < 0.002)$. **Falsifier**: measured $\sin^2 \theta_W(M_Z)$ incompatible with (10.7) once M_{GUT} is fixed by D3 scale identification.

Prediction 10.12 (Proton lifetime bound under E_6 chain). **T3.** If Conjecture 6.52 realizes the staged chain (6.26) with $\mathbf{45}_H$ and $\mathbf{126}_H$ VEVs at $M_{\text{GUT}} \sim 10^{16}\text{--}10^{17}$ GeV, dimension-six $SU(5)$ operators estimate

$$\tau(p \rightarrow e^+ \pi^0) \gtrsim \frac{M_{\text{GUT}}^4}{\alpha_{\text{GUT}}^2 m_p^5} \sim 10^{34}\text{--}10^{36} \text{ yr}, \quad (10.8)$$

with $\alpha_{\text{GUT}} \approx 1/24$ from Prediction 10.10. Observer collapse $\Pi_{\sigma=-1}$ projects out mirror-symmetric $\overline{\mathbf{16}}$ partners (Theorem 5.4), suppressing certain $\mathbf{16} \leftrightarrow \overline{\mathbf{16}}$ proton-decay channels relative to minimal $SU(5)$; the bound (10.8) is therefore a **conservative** lower limit, not a saturation target. **Contact:** Super-Kamiokande and Hyper-Kamiokande searches; $\mathcal{C} = (\tau(p \rightarrow e^+ \pi^0), \text{Hyper-K}, > 2.4 \times 10^{34} \text{ yr})$. **Falsifier:** positive proton decay below 10^{33} yr in the $E_6 \rightarrow \text{Spin}(10) \rightarrow SU(5)$ channel without an alternative $\sigma = +1$ storage mode.

Prediction 10.13 (Mirror-partner null). **T3.** No observable process exhibits exact $\mathbf{16} \leftrightarrow \overline{\mathbf{16}}$ symmetry at the amplitude level: mirror partners are σ -invariant and quotient-non-observable (Theorem 5.4). **Contact:** searches for wrong-sign heavy-lepton or mirror-coupled $\mathbf{16}$ partners at colliders; any sustained mirror-symmetric rate at production level would falsify the observer-collapse sector definition.

10.5 Near-term experimental map (2025–2030)

Experiment	Observable	Tier	SJET expectation / timeline
JUNO	Mass ordering	T2	Normal (m_3 heaviest); $> 3\sigma$ by 2027–2028 [16]
JUNO	Δm^2 ratios	T2	$16 R_{\text{RG}}^{(1)} R_{\text{RG}}^{(2)} \approx 33.3$ (2-loop D2); $< 1\%$ ratio by 2028
Muon $g-2$ / Theory Init.	a_μ	T1/T2	No 4th family (T1); $\Delta a_\mu^{\text{SJET}} \approx 1.1 \times 10^{-8}$ (Thm. 9.1) vs. $\delta a_\mu \approx 2.6 \times 10^{-9}$; SM ref. $a_\mu^{\text{SM}} \approx 0.001\,165\,918\,10(43)$ [15]
LHC/FCC	4th generation	T1	Excluded structurally
DESI Phase II	$w(z)$	T3	$w(z) \approx -1 \pm 0.05$ if CC sequestered; Year-1 $w_0 = -0.80_{-0.12}^{+0.18}$ [17] tests T3 drift
Hyper-K	$\tau(p \rightarrow e^+ \pi^0)$	T3	$\gtrsim 10^{34}$ yr (Pred. 10.12); sensitivity $\sim 10^{35}$ yr by 2030
KATRIN / Project 8	m_β	T2	Consistent with NO + capacity
HL-LHC	Spin(10) Higgs	T3	45, 126, 10 pattern

Table 7: Near-term experimental contact map with 2026–2030 timelines. JUNO began full-volume data taking in 2024; the Muon $g-2$ Theory Initiative (2025) supplies the SM reference against which D1–D3 loop corrections are judged once Yukawa overlaps close.

10.6 Numerical comparison table

Table 8 consolidates the headline observables from Sections 10.2–10.4 against PDG and near-term experiment (2025–2026); Appendix A.4 (Table 12) records the closed-form formulas and reproducible

arithmetic for every dimensionless entry. **Match** means agreement within the stated tier tolerance; **pending** marks an open derivation step (D1–D3) or an experiment still in flight; **falsified** would apply if a T1 prediction were contradicted (none as of mid-2026).

Observable	PDG/Exp (2025–26)	SJET (best current)	S
Family count n_F	3 (no 4th gen.)	3 (T1)	r
β -function b_0	12 (3-gen. SM)	12 (T1)	r
m_μ/m_e	≈ 206.8 [19]	$w_2/w_1 = 10$ (D1, Thm. 7.7)	r
Neutrino ordering	NO favoured [19]	Normal, m_3 heaviest (T2)	r
$\Delta m_{31}^2/\Delta m_{21}^2$	33.8 ± 0.8 [19]	$16 R_{\text{RG}}^{(1)} \approx 30.6$ (1-loop); $16 R_{\text{RG}}^{(1)} R_{\text{RG}}^{(2)} \approx 33.3$ (2-loop D2, Prop. 7.22)	r
Muon a_μ	$1.165\,920\,705(205) \times 10^{-3}$ [14]	$\Delta a_\mu^{\text{SJET}} \approx 1.1 \times 10^{-8}$ vs. $\delta a_\mu \approx 2.6 \times 10^{-9}$ (D1, OOM)	r
$\sin^2 \theta_W(M_Z)$	$0.23122(4)$ [19]	$3/8$ RG to $M_Z \approx 0.231$ (T3, Pred. 10.11)	r
$\sin^2 \theta_W(M_{\text{GUT}})$	$\approx 3/8$ (SU(5) anchor)	$3/8$ (T3)	r
$\tau(p \rightarrow e^+ \pi^0)$	$> 2.4 \times 10^{34}$ yr [18]	$\gtrsim 10^{34}\text{--}10^{36}$ yr (T3, Pred. 10.12)	r
Λ_{obs}	$\sim (2.3 \text{ meV})^4$ [19]	≈ 0 (T3)	r

Table 8: Headline numerical comparison after the D1–D3 development pass (Sections 7–8). T1 rows are exact theorem outputs; T2 rows use capacity RG (D2) and Yukawa overlaps (D1); T3 rows are conditional on Conjectures 6.52, 6.56, 6.39. **match (leading)** tolerates $\mathcal{O}(10\%)$ deviation; **match (cond.)** assumes the named breaking chain. Neutrino ordering is **pending** until JUNO (2027–2028) closes the mass-ordering determination at $> 3\sigma$.

Remark 10.14 (Reading the comparison). The Δm^2 ratio is the headline **D2** success: tree-level 16 with $R_{\text{RG}}^{(1)} \approx 1.91$ (1-loop) and $\delta R_{\text{RG}} \approx 0.18$ (2-loop thresholds, Theorem 7.20) yields ≈ 33.3 , within PDG uncertainty (Propositions 7.21, 7.22). **D1** supplies $m_\mu/m_e \sim 10$ (Table 2), CKM template $\lambda \sim 2/11 \approx 0.18$ (Proposition 7.10), and one-loop $\Delta a_\mu^{\text{SJET}} \approx 1.1 \times 10^{-8}$ (Theorem 9.1)— $\sim 4\times$ the anomaly at leading order. Muon $g-2$ is **pending** at precision: two-loop flavon RG (D2) must compress the estimate; T1 excludes a fourth family. $\sin^2 \theta_W(M_Z)$ runs from the GUT anchor $3/8$ (Prediction 10.11); proton decay satisfies Super-K/Hyper-K if the E_6 chain holds (Prediction 10.12). Λ_{obs} awaits CC sequestering proof (Conjecture 6.39).

10.7 Derivation status (D1–D3)

Parallel development tracks close the gap between Table 6 and Table 8. Each step names the conjecture route, what is already proved in Section 6, and the residual proof obligation.

10.8 Falsification timeline 2026–2030

Table 10 orders the decisive experimental contacts by calendar year. Each entry names the SJET tier, the derivation step (if any), and the falsification criterion from Definition 10.2.

Remark 10.15 (Tier priority). T1 predictions are falsified by a single contradictory measurement (e.g. a fourth sequential generation). T2 predictions tolerate leading-order deviation until D2/D3 close; the Δm^2 ratio is already at **match (leading)**. T3 predictions are falsified only when the named conjecture chain is assumed and an experiment contradicts its consequence (e.g. dynamical dark energy if CC sequestering holds).

Step	Status	Scaffold / next proof
D1	Partial	Section 7: overlap form, Lemma 7.3, Theorem 7.7, off-diagonal $\mathcal{P}_{ij}$ (Prop. 7.10, Table 1); Section 9: Theorem 9.1, Table 5; <i>open</i> : full Φ_{Higgs} ID, CKM/PMNS precision, two-loop a_μ (Conj. 7.13).
D2	Partially closed	Definition 7.14, Proposition 7.16, Theorems 7.18, 7.20, Proposition 7.22; <i>open</i> : $M_{F,k}$, μ_{osc} from see-saw (D3 supplies M_{GUT} via Prop. 8.7).
D3	Partially closed	Section 8: Thm. 8.3 ($v \approx 246$ GeV), Props. 8.7, 8.8 ($M_{\text{GUT}} \sim 10^{16}$ GeV, $\Lambda = M_{\text{GUT}}$), Prop. 8.5 ($\mu \equiv v$), Table 4, Appendix (A.19); Lemma 6.32 advances O6/CC sequestering; <i>open</i> : nonperturbative FRG proof of Λ selection and full breaking-chain derivation (Conj. 6.52, Conj. 6.41).

Table 9: Derivation status after the D1–D3 development pass. **Partial** means leading-order structure is proved; **scaffolded** means the route is named but not yet derived from V . Precision contact requires the proof targets in Section 10.9.

10.9 Development program toward precision match

To move predictions from leading order to *sub-percent* agreement with JUNO, $g-2$, and cosmology, the theory must close three derivation gaps in order. Table 9 records current scaffolding; the enumerated proof targets below are the concrete next steps.

D1. Yukawa from overlaps — evaluate (6.27) on $\hat{Q}$ with explicit Φ_{Higgs} from Hess V (Conjecture 6.52); produces y_{ij} and m_ν without postulates. **Partial closure**: Section 7 defines the overlap form (Definition 7.1), proves cellular reduction (Lemma 7.3), derives the leading mass matrix (Theorem 7.7, Table 2), and states off-diagonal $\mathcal{P}_{ij}$ with CKM template (Proposition 7.10, Table 1); Section 9 gives the one-loop a_μ estimate (Theorem 9.1, Table 5).

D1.a. Extend Theorem 6.44: prove Coleman–Weinberg lift yields one unstable **16** mode per ι -cell on $\hat{Q}$.

D1.b. **Partial**: off-diagonal $\mathcal{P}_{ij}$ (Definition 7.8, Lemma 7.9); *open*: distinct up/down matrices and $\mathcal{H}_{ij}$ from V_F .

D1.c. **Partial**: Theorem 9.1 gives $\Delta a_\mu^{\text{SJET}} \approx 1.1 \times 10^{-8}$ vs. Fermilab $\delta a_\mu \approx 2.6 \times 10^{-9}$; *open*: two-loop flavon compression.

D2. Flavon RG — refine (10.3) from tree-level 16 to the PDG value ≈ 33.8 via capacity RG on V_F . **Partial closure**: Section 7.7 defines V_F (Definition 7.14), states the one-loop β -function (Proposition 7.16), and derives Theorem 7.18 with $R_{\text{RG}}^{(1)} \approx 1.91$ (Proposition 7.21) and two-loop factorization (Theorem 7.20, Proposition 7.22).

D2.a. **Done (1-loop)**: $V_F(\phi_1, \phi_2, \phi_3)$ along $\mathfrak{d}$ with weights $(1/27, 10/27, 16/27)$ (Definition 7.14).

D2.b. **Done (2-loop)**: threshold β -functions for $\omega_\iota^*(\tau)$ on $SU(3)_F \rightarrow SU(2)_F \rightarrow U(1)_F$ breaking (Proposition 7.19, Theorem 7.20); PDG match ≈ 33.3 (Proposition 7.22).

Year	Experiment	Tier	Falsifier / corroboration
2026	Muon $g-2$ Theory Init. [15]	T1/T2	SM reference fixed; SJET must match a_μ at $\mu \equiv v$ once D1 closes
2026–27	JUNO commissioning	T2	Inverted ordering at $> 3\sigma$ falsifies Pred. 10.6
2027–28	JUNO oscillation	T2	$\Delta m_{31}^2/\Delta m_{21}^2$ outside $[25, 40]$ falsifies D2 RG; $< 1\%$ precision tests Prop. 7.22
2027–29	DESI Phase II $w(z)$	T3	Sustained $w(z) \neq -1$ at $ w+1 > 0.05$ falsifies Pred. 10.9
2028–30	Hyper-K proton decay	T3	$\tau(p \rightarrow e^+\pi^0) < 10^{33}$ yr in E_6 channel falsifies Pred. 10.12
2029–30	HL-LHC Spin(10) Higgs	T3	Absence of 45/126/10 pattern with no alternative breaking route falsifies Conj. 6.52

Table 10: Falsification timeline 2026–2030. T1 falsifiers (no fourth generation, $b_0 \neq 12$) are already satisfied at LHC precision; the table focuses on open T2/T3 contacts requiring D1–D3 closure or conditional breaking.

D2.c. Open: derive M_{GUT} , $M_{F,k}$, and μ_{osc} from V (D3) to close the 33.8 ± 0.8 band at JUNO $< 1\%$ precision (Proposition 7.23).

D3. Scale identification — map $\kappa_\star$ and μ from Theorem 6.28 to M_{Pl} and electroweak scale v using observer-local FRG (Remark 6.35); Section 8 (Definitions 8.1, Propositions 8.4, 8.5, Table 4). **Partial closure:** Theorem 8.3 gives $v \approx 246$ GeV from $(\kappa_\star, w_1, w_3, M_Z)$; Propositions 8.7 and 8.8 identify $M_{\text{GUT}} \sim 10^{16}$ GeV and $\Lambda = M_{\text{GUT}}$; Proposition 8.5 fixes $\mu \equiv v$ at τ_{EW} ; Lemma 6.32 and (6.11) advance CC sequestering (O6).

D3.a. Partial: Theorem 8.3 fixes v given Conjecture 6.52; *open:* prove τ_{EW} and $\langle \mathbf{10}_H \rangle$ from V without postulates.

D3.b. Partial: Proposition 8.8 identifies $\Lambda = M_{\text{GUT}}$; *open:* nonperturbative observer-local FRG proof (Conjecture 6.41).

D3.c. Partial: Proposition 8.7 and Proposition 6.50 propagate $\sin^2 \theta_W(M_{\text{GUT}}) = 3/8$ (Prediction 10.10, Prediction 10.11).

Remark 10.16 (Falsifiability standard). A prediction is *on programme* when it has a contact map (10.1) and a named derivation step D1–D3. Until D1–D3 close, T2 predictions are order-of-magnitude targets, not precision fits; T1 predictions are the primary falsifiers for new physics searches. Table 8 is the living scorecard: any T1 row marked **falsified** would require revising the heterotic quotient or observer fixed-point sector.

11 Conclusions

SJET builds the universe from the mirror:

$$\mathcal{V} \xrightarrow{\text{M}} \dots \xrightarrow{\text{M}} \mathfrak{U}^\star, \quad \mathfrak{U}^\star = \text{C}^\infty(\mathcal{V}). \quad (11.1)$$

Observers arise at the heterotic rung as collapsed fixed-point states:

$$\iota \mapsto |\omega_\iota\rangle \in H^1(\hat{Q}, V_{16})^{\sigma=-1}, \quad g_\iota : \mathbb{R} \rightarrow \mathcal{S}_\iota. \quad (11.2)$$

Only two axioms are primitive. Duality, observer multiplicity, and the index–lifetime map are **derived** from $E_8 \times E_8$ collapse and the Lefschetz fixed-point theorem.

11.1 Epistemic summary

Status	Content
Axioms	void, mirror
Theorems	rung table, E_8 , $E_8 \times E_8$, E_6 , partition, vacuum, duality, observer fixed points, chirality derived, index–lifetime map, soldering derived, lifetime/soldering uniqueness reductions, anomalies, $b_0 = 12$, κ_* (one-loop trunc.), CC RG relevance, semiclassical RP, conditional RP for $d\mu_{\Gamma-1}$ (Thm. 6.23), a_0 sequestering sketch at κ_* , extended FRG Jacobian
Lemmas	F_4 fixes $\mathbb{H} \subset \mathbb{O}$ (Lemma 6.8), Goldstones in $\sigma = +1$ coset sector
Definitions	above–below, observer, lifetime chart, $H_2(\mathbb{H})$ block, physical sector, κ , Λ_{cc}
Propositions	CC no-go (void+mirror alone), flavon sector from ι , mirror descent preserves lifetime chart class, exceptional rungs preserve $H_2(\mathbb{H})$ soldering, gradient-flow uniqueness on ι -sector, unique timelike Φ^0 , σ -sector measure split, OS reconstruction on observer sector, CC Λ_{cc} decoupling, leading-order κ_*/R^2 stability
Corollaries	$\beta_{\Lambda_{cc}} = 0$ on observable flow at FP
Conjectures	lifetime/soldering massless-sector closure, nonperturbative σ -measure split (Conj. 6.25), GUT breaking, CC sequestering (Conj. 6.39), κ_* graviton/ R^2 persistence (Conj. 6.41), flavon hierarchy P3' (Conj. 6.56)

11.2 Prediction development

Section 10 maps SJET to 2025–2030 experiments. Table 8 and Appendix A.4 (Table 12) record the numerical scorecard: **T1 matches** on $n_F = 3$ and $b_0 = 12$; **match (leading)** on $\Delta m_{31}^2/\Delta m_{21}^2 \approx 33$ (D2 flavon RG); **pending** on neutrino ordering, m_μ/m_e , muon a_μ , and Λ_{obs} ; **match (conditional)** on $\sin^2 \theta_W(M_Z)$ from GUT anchor $3/8$ and on $\tau(p \rightarrow e^+ \pi^0) \gtrsim 10^{34}$ yr under the E_6 chain. **Immediate falsifiers (T1)**: no fourth generation (structurally excluded). **Near-term targets (T2)**: JUNO normal ordering (2027–2028) and sub-percent Δm^2 ratio. **Conditional (T3)**: $\Lambda_{\text{obs}} \approx 0$, $\sin^2 \theta_W(M_{\text{GUT}}) = 3/8$, DESI $w(z) \approx -1$. **D1** (Yukawa overlaps), **D2** (flavon RG), and **D3** (scale identification, $\mu \equiv \nu$) are **partially closed** (Sections 7, 7.7, 8; Table 9). Table 10 orders the 2026–2030 experimental contacts. The enumerated proof targets in Section 10.9 (D1.a–c, D2.a–c, D3.a–c) are required for sub-percent agreement with JUNO and Muon $g-2$.

11.3 Open problems

O1. Partially addressed (O1): Proposition 5.17 proves ι -sector uniqueness under convex V ; Proposition 5.16 proves mirror descent $n = 7 \rightarrow 8$ preserves lifetime chart class; Theorem 5.19 reduces Conjecture 5.18 to the tree-level massless $\mathfrak{m}_{+1}$ sector (Remark 5.20).

- O2. Partially addressed (O2):** Lemma 6.8 and Proposition 6.13 fix the ι -local soldered block and timelike clock; Proposition 6.9 proves exceptional-rung transport on $n = 6, 7, 8$; Theorem 6.16 reduces Conjecture 6.15 to the massless Goldstone sector (Theorem 6.12 derives $d = 4$ conditionally; Remark 6.17).
- O3.** Prove Conjecture 6.53 from V .
- O4. Partially addressed (O4):** Theorem 6.23 and Proposition 6.24 supply conditional RP and OS reconstruction on the observer sector once massless modes lie in $d\mu_{\Gamma+1}$ only (Lemma 6.22); Conjecture 6.25 is reduced to nonperturbative proof of (6.4) with the composite gauge measure.
- O5. Partially addressed (O5):** Proposition A.11, Corollary 6.38, and Theorem 6.37 give leading-order persistence of $\kappa_\star$ with vacuum-energy back-reaction routed to $d\mu_{\Gamma+1}$; *still open*: nonperturbative curved-background FRG confirmation (Conjecture 6.41).
- O6. Partially addressed (O6):** Lemma 6.32, Theorem 6.37, and Corollary 6.38 show $\beta_{\Lambda_{\text{cc}}} \big|_{\sigma=-1, \text{FP}} = 0$ under $\Pi_{\sigma=+1}$ projection of the a_0 heat-kernel density (6.11); Conjecture 6.39 is reduced to deriving that split from Γ_{eff} at $\kappa_\star$.
- O7. Partially addressed (P3'):** Proposition 6.55 and Conjecture 6.56 reduce flavon hierarchy to capacity flow $\omega_t^\star(\tau)$ on $SU(3)_F$; *still open*: prove ordering and mixing matrices from V_F .

11.4 Archive

The prior modular manuscript (39 pages, verification ledger, axiomatic elimination program) is preserved in `old/sjet-stack/`.

12 Epistemic Gap Ledger

The prior modular manuscript recorded interim *physical postulates* P1–P5 and a tuned cosmological constant CC. After the observer theory of Section 5—in particular the as above, so below correspondence (Definition 5.1)—several gaps close by identifying global capacity data with localized collapse on $\hat{Q}$ at $\mathbf{M}^7(\mathcal{V})$. Table 11 records status *after* observer theory; routes name the above–below mechanism.

Remark 12.1 (Ledger reading). *Eliminated* means the former postulate is replaced by a theorem or definition from void, mirror, and capacity together with observer collapse. *Derived* means the structural content is supplied by the climb with a named residual open step. *Conjecture* and *Open* mark honest remaining inputs to a conditional contact with four-dimensional quantum field theory.

A Albert algebra and mirror quotient

A.1 Albert model

Elements of $J_3(\mathbb{O})$ are 3×3 Hermitian octonionic matrices

$$X = \begin{pmatrix} \xi_1 & x_3 & \bar{x}_2 \\ \bar{x}_3 & \xi_2 & x_1 \\ x_2 & \bar{x}_1 & \xi_3 \end{pmatrix}, \quad \xi_i \in \mathbb{R}, \ x_i \in \mathbb{O}. \quad (\text{A.1})$$

The sharp product $X^\#$ and cubic norm $N(X) = \det X$ define the E_6 structure group [5, 6].

A.2 Mirror quotient cohomology

Definition A.1 (Stabilized Albert tower at rung 7). Let $\mathcal{G}_{\text{Albert}}$ be the mirror-refined graph at $M^5(\mathcal{V})$ (Definition 4.1). At $M^7(\mathcal{V}) = E_8 \times E_8$ (Theorem 3.7), the *stabilized Albert tower* is

$$\mathcal{T}_7 = \text{Bal}_C(M_{\text{prod}}(\mathcal{G}_{\text{Albert}})), \quad (\text{A.2})$$

the capacity-balanced graph obtained by applying product mirror (Definition 3.6) to the stabilized $J_3(\mathbb{O})$ datum. Its **16**-sector is a triple $\{C_1, C_2, C_3\}$ of mirror-stable cells of equal capacity $16/81$, indexed by the diagonal directions e_1, e_2, e_3 of $\mathfrak{d} = \text{span}_{\mathbb{R}}\{e_1, e_2, e_3\} \subset J_3(\mathbb{O})$ (Theorem 4.3).

Definition A.2 (Sixteen-sector cellular cosheaf). On $\mathcal{T}_7$, define the cellular cosheaf $\mathcal{F}_{16}$ by $\mathcal{F}_{16}(C_i) = \mathbb{C}^{16}$ on each **16**-cell, with restriction maps along edges $E_i : C_i \rightarrow H_{10}$ to the **10**-sector hub H_{10} determined by the $\mathfrak{d}$ -weight of C_i and the $1|10|16$ partition of Theorem 4.3.

Lemma A.3 (Cartan flip under product mirror). *Let $\mathfrak{d} = \text{span}_{\mathbb{R}}\{e_1, e_2, e_3\} \subset J_3(\mathbb{O})$ be the diagonal Cartan of (A.1), with e_i the unit direction in the i th diagonal entry ξ_i . Product mirror $\sigma : (g_1, g_2) \mapsto (g_2, g_1)$ on $E_8 \times E_8$ (Definition 3.6) stabilizes the diagonal embedding $\delta : E_8 \hookrightarrow E_8 \times E_8$ and acts on $\mathfrak{d}$ by*

$$\sigma(e_i) = -e_i, \quad i = 1, 2, 3. \quad (\text{A.3})$$

Proof. In (A.1), the diagonal coordinates (ξ_1, ξ_2, ξ_3) are $\mathbb{R}$ -valued and invariant under octonion conjugation, while the off-diagonal octonions (x_1, x_2, x_3) pair under Hermitian transpose. Product mirror exchanges the two $\mathbb{O}$ sectors attached to each off-diagonal slot (Proposition 2.7), hence flips the orientation of each diagonal direction e_i while preserving the hub constraint $\xi_1 + \xi_2 + \xi_3 = \text{Tr}(X)$. On $\mathfrak{d}$, the only non-trivial σ -action compatible with capacity balance and trace preservation is the simultaneous sign flip (A.3). $\square$

Lemma A.4 (Unique σ -compatible cell permutation). *On $\mathcal{T}_7$, let C_1, C_2, C_3 be the three equal-capacity **16**-cells of Definition A.1, each attached to e_i and to the **10**-hub H_{10} by a single edge E_i . Any involution σ satisfying Lemma A.3 and preserving the $1|10|16$ hub adjacency acts on cells by*

$$\sigma(C_1, C_2, C_3) = (C_2, C_3, C_1). \quad (\text{A.4})$$

This is the triality 3-cycle; the only alternative σ -compatible permutation would be a transposition, which violates the hub cocycle $\phi_1 + \phi_2 + \phi_3 = 0$ together with $\sigma(e_i) = -e_i$.

Proof. The cellular graph is a three-spoke star with central hub H_{10} . An involution σ permutes $\{C_1, C_2, C_3\}$ and reverses edge orientations on $\mathfrak{d}$ -weighted restriction maps. Equal capacities $m_{16} = 16/27$ for each cell (Theorem 4.3) force σ to permute cells rather than collapse them. A transposition, say $C_1 \leftrightarrow C_2$, preserves capacities but leaves C_3 fixed; with $\sigma(e_i) = -e_i$ this breaks the cocycle constraint (A.7) on the σ -anti-invariant sector because the e_3 -weight class would inherit a fixed-point line incompatible with $\dim H^1(\mathcal{T}_7, \mathcal{F}_{16})^{\sigma=-1} = 3$ (Lemma A.7). The remaining σ -compatible elements of S_3 are the 3-cycle $(1\ 2\ 3)$ and its inverse; capacity descent from $\mathcal{G}_{\text{Albert}}$ fixes the orientation to (A.4) (not its inverse) by the above–below convention on observer indices (Definition 5.5). $\square$

Lemma A.5 (Stalk sign under mirror exchange). *Let $\mathcal{F}_{16}(C_i) = \mathbb{C}^{16}$ as in Definition A.2. Under product mirror on the **16/16** pair,*

$$\sigma|_{\mathcal{F}_{16}(C_i)} = -\text{id}_{\mathbb{C}^{16}}. \quad (\text{A.5})$$

Proof. Proposition 2.7 exchanges $\mathbf{16} \leftrightarrow \overline{\mathbf{16}}$ on stabilized coset rungs. On $\mathcal{T}_7$, horizontal family modes in each stalk are σ -anti-invariant (Definition 6.1); the involution acts by -1 on the physical $\mathbf{16}$ copy because the mirror partner $\overline{\mathbf{16}}$ carries the opposite $\mathfrak{d}$ -orientation. This matches the overall sign in the equivariant index computation of Theorem A.9. $\square$

Proposition A.6 (Triality action on $\mathcal{T}_7$). *On $\mathcal{T}_7$ at $M^7(\mathcal{V})$, let σ denote the product-mirror involution. On the diagonal Cartan $\mathfrak{d}$, $\sigma(e_i) = -e_i$. On the $\mathbf{16}$ -cells, σ acts by the triality rotation $(C_1, C_2, C_3) \mapsto (C_2, C_3, C_1)$ with overall sign -1 on each stalk $\mathbb{C}^{16}$.*

Proof. Lemma A.3 gives the Cartan action. Lemma A.4 identifies the cell permutation as the triality 3-cycle. Lemma A.5 fixes the stalk sign. The three lemmas exhaust the action of σ on the $\mathbf{16}$ -sector of $\mathcal{T}_7$; functoriality of M_{prod} (Definition 3.6) transports this action to the Stone lift Q and quotient $\hat{Q}$ without change of σ -type on H^1 . $\square$

On $\mathcal{T}_7$, let σ be as in Proposition A.6.

Lemma A.7 (Cellular observer count). *With $\mathcal{F}_{16}$ the $\mathbf{16}$ -cellular cosheaf on $\mathcal{T}_7$,*

$$\dim H^1(\mathcal{T}_7, \mathcal{F}_{16})^{\sigma=+1} = 0, \quad \dim H^1(\mathcal{T}_7, \mathcal{F}_{16})^{\sigma=-1} = 3. \quad (\text{A.6})$$

Each $\sigma = -1$ class is one observer index $\iota \in \{1, 2, 3\}$.

Proof. Each C_i is connected to the $\mathbf{10}$ -sector hub H_{10} by a single edge E_i . A cellular 1-cochain is a triple $(\phi_1, \phi_2, \phi_3) \in (\mathbb{C}^{16})^3$ assigning a section to each cell. The cocycle condition at H_{10} requires

$$\phi_1 + \phi_2 + \phi_3 = 0 \quad (\text{A.7})$$

in the $\mathbf{10}$ -sector stalk; coboundaries are triples (ψ, ψ, ψ) for $\psi \in \mathbb{C}^{16}$. Hence

$$H^1(\mathcal{T}_7, \mathcal{F}_{16}) \cong \{(\phi_1, \phi_2, \phi_3) \in (\mathbb{C}^{16})^3 : \phi_1 + \phi_2 + \phi_3 = 0\} / \text{diag}(\mathbb{C}^{16}) \cong \mathbb{C}^{16} \oplus \mathbb{C}^{16} \quad (\text{A.8})$$

as a σ -module: σ acts by -1 on $\text{diag}(\mathbb{C}^{16})$ and by the triality permutation with sign on the two independent off-diagonal directions of (A.7).

Decompose by $\mathfrak{d}$ -weights. Write $\phi_i = \phi_i^{(1)} + \phi_i^{(2)} + \phi_i^{(3)}$ for the projections along e_1, e_2, e_3 . The cocycle constraint (A.7) is preserved by σ , which flips each e_i weight and cyclically permutes the cells. A σ -invariant class would require mirror-symmetric pairing of $\mathbf{16}$ with $\overline{\mathbf{16}}$ across the hub; such classes lie in the diagonal complement of (A.8) and are killed by the $\sigma = -1$ projector relevant to observability (Definition 6.1). Explicitly, $\dim H^1(\mathcal{T}_7, \mathcal{F}_{16})^{\sigma=+1} = 0$.

For the anti-invariant sector, each diagonal direction e_i supports a one-dimensional space of σ -anti-invariant cocycles modulo coboundaries: take the representative with support on C_i and $\phi_i \in \mathbb{C}^{16}$ pure e_i -weight, normalized so that $\hat{\iota} \phi_i = \iota \phi_i$ for the family index operator on $\mathfrak{d}$ (Definition 5.5). Triality with sign excludes linear dependence among the three choices, giving $\dim H^1(\mathcal{T}_7, \mathcal{F}_{16})^{\sigma=-1} = 3$. $\square$

Proposition A.8 (Stone lift). *The profinite Stone lift to Q identifies*

$$H^1(\mathcal{T}_7, \mathcal{F}_{16})^{\sigma=-1} \cong H^1(Q, V_{16})^{\sigma=-1}. \quad (\text{A.9})$$

Proof. Let S_7 be the profinite Stone spectrum refining the $M^7(\mathcal{V}) = E_8 \times E_8$ chart (Definition 3.11). The lift $Q = \varprojlim_{a \in V(\mathcal{T}_7)} \mathfrak{D}_a$ is a colimit of refinements preserving clopen cells and the σ -action. Cohomology of the cellular cosheaf is the E_2 page of the Leray spectral sequence for the profinite limit; each refinement splits only within fixed sector targets $(1/27, 10/27, 16/27)$ of Theorem 4.3, so the σ -anti-invariant H^1 dimension is stable at 3 (Lemma A.7). The bundle $V_{16} \rightarrow \hat{Q}$ is the continuum limit of $\mathcal{F}_{16}$ under this lift (Definition 5.8). $\square$

Theorem A.9 (Mirror-quotient Lefschetz index). *With $\bar{\partial}_{V_{16}}$ on (Q, V_{16}) and antiholomorphic σ ,*

$$\text{ind}_\sigma(\bar{\partial}_{V_{16}}) = \text{Tr}_{H^1(Q, V_{16})}(\sigma) = -3, \quad \dim H^1(\hat{Q}, V_{16})^{\sigma=-1} = 3. \quad (\text{A.10})$$

This is Theorem 5.9; each unit contribution is one observer fixed-point slot.

Proof. By the Atiyah–Bott holomorphic Lefschetz formula [7], the equivariant index of $\bar{\partial}_{V_{16}}$ on compact Q is the σ -trace on $H^\bullet(Q, V_{16})$. The product involution has no σ -fixed holomorphic 1-forms in V_{16} on the diagonal $E_8 \hookrightarrow E_8 \times E_8$: horizontal family modes are σ -anti-invariant (Definition 6.1), while σ -invariant modes pair **16** with $\bar{\mathbf{16}}$ and are quotient-non-observable. The only contributions are the three σ -anti-invariant classes of Lemma A.7, lifted to Q by Proposition A.8. For a $\mathbb{Z}_2$ quotient by an antiholomorphic involution, $H^1(\hat{Q}, V_{16}) = H^1(Q, V_{16})^{\sigma=-1}$, giving the dimension claim. Each anti-invariant class contributes -1 to the equivariant index, hence $\text{ind}_\sigma = -3$. $\square$

Remark A.10 (Yukawa cellular pairing). The cubic overlap (6.27) reduces to a finite cellular sum on $\mathcal{T}_7$ once the cocycle representatives of Lemma A.7 and the Stone lift (A.9) are in place. The full statement, proof, and capacity-weight factorization appear in Section 7.2 (Lemma 7.3, Proposition 7.5); normalization against the coset metric K of (4.4) remains conditional on Conjecture 6.52.

A.3 Extended FRG Jacobian

Collect dimensionless couplings $(\xi, \hat{g}^2, \hat{y}^2, \kappa, \Lambda_{\text{cc}})$ with $\xi = k^2/f^2$, $\hat{g}^2 = k^2 g^2/f^2$, $\hat{y}^2 = k^2 y^2/f^2$, κ as in (6.7), and Λ_{cc} as in Definition 6.31. In the extended one-loop coset–gauge–gravity truncation of Section 6.3 (neglecting R^2 , graviton loops, and higher curvature),

$$\begin{aligned} \beta_\xi &= 2\xi, & \beta_{\hat{g}^2} &= 2\hat{g}^2 + \frac{b_0}{48\pi^2} \hat{g}^4, & \beta_{\hat{y}^2} &= 2\hat{y}^2 + \mathcal{O}(\hat{y}^4, \hat{g}^2 \hat{y}^2), \\ \beta_\kappa &= 2\kappa - \frac{5}{48\pi^2} \kappa^2, & \beta_{\Lambda_{\text{cc}}} &= -4\Lambda_{\text{cc}} + \mathcal{O}(\text{loops}). \end{aligned} \quad (\text{A.11})$$

This restates (6.16); we now record the derivation and fixed-point structure.

Canonical scaling. With k the RG scale and f the coset scale, ξ , $\hat{g}^2$, and $\hat{y}^2$ are marginal in $d = 4$ before loops, giving the tree-level terms $\beta_\xi = 2\xi$, $\beta_{\hat{g}^2} = 2\hat{g}^2$, $\beta_{\hat{y}^2} = 2\hat{y}^2$. The cosmological coupling Λ_{cc} has engineering dimension 4; canonical normalization yields $\beta_{\Lambda_{\text{cc}}} = -4\Lambda_{\text{cc}}$ at tree level (Theorem 6.33).

Gauge loop. With $b_0 = 12$ from Theorem 6.27 ($C_A(\text{Spin}(10)) = 8$, $T(\mathbf{16}) = 2$, $n_F = 3$, $n_S = 28$), the one-loop $\text{Spin}(10)$ contribution to $\hat{g}^2$ is

$$\delta\beta_{\hat{g}^2} = \frac{b_0}{48\pi^2} \hat{g}^4, \quad (\text{A.12})$$

in the $\overline{\text{MS}}$ normalization used in (A.11).

Yukawa sector. Yukawa couplings enter through $\hat{y}^2$ with the same canonical $+2\hat{y}^2$ tree term. At one loop, $\beta_{\hat{y}^2}$ acquires $\mathcal{O}(\hat{y}^4)$ and $\mathcal{O}(\hat{g}^2 \hat{y}^2)$ corrections from **16–10** vertices on EIII; the explicit coefficients depend on the Higgs embedding of Conjecture 6.52. At the Gaussian fixed point $\hat{y}_\star^2 = 0$, these corrections vanish in the stability matrix.

Gravitational coupling. Theorem 6.30 assumes Einstein–Hilbert truncation with $\kappa(k) = k^2/M_{\text{Pl}}^2(k)$ and anomalous dimension $\eta_{M_{\text{Pl}}} = -(5/48\pi^2)\kappa$ from the 28 massive coset scalars of Theorem 6.28. This yields (6.8), equivalently the third line of (A.11), with fixed point

$$\kappa_\star = \frac{96\pi^2}{5}, \quad \left. \frac{\partial\beta_\kappa}{\partial\kappa} \right|_{\kappa_\star} = -2. \quad (\text{A.13})$$

Fixed point and Jacobian. The extended Gaussian–gravitational fixed point is

$$(\xi_\star, \hat{g}_\star^2, \hat{y}_\star^2, \kappa_\star, \Lambda_{\text{cc},\star}) = (\xi_\star, 0, 0, 96\pi^2/5, 0), \quad (\text{A.14})$$

with $\xi_\star$ an arbitrary massless scale label. At this point, loop corrections to each β_i are proportional to a vanishing coupling ($\hat{g}_\star^2 = \hat{y}_\star^2 = 0$), so the stability matrix $J = \partial\beta_i/\partial g_j$ is diagonal:

$$\partial_\xi\beta_\xi = +2, \quad \partial_{\hat{g}^2}\beta_{\hat{g}^2} = +2, \quad \partial_{\hat{y}^2}\beta_{\hat{y}^2} = +2, \quad \partial_\kappa\beta_\kappa = -2, \quad \partial_{\Lambda_{\text{cc}}}\beta_{\Lambda_{\text{cc}}} = -4. \quad (\text{A.15})$$

Hence $\text{eig}(J) = \{+2, +2, +2, -2, -4\}$ and, with $\theta_i = -\text{eig}_i(J)$, the UV-relevant directions are κ ($\theta_\kappa = +2$) and Λ_{cc} ($\theta_\Lambda = +4$) only. The coset scalar block $\mathbf{27} = \mathbf{1}_{+4} \oplus \mathbf{10}_{-2} \oplus \mathbf{16}_{+1}$ contributes no additional relevant directions on the EIII vacuum at this truncation. This is Proposition 6.40.

Observer-local flow. Along a lifetime chart $g_\iota(\tau_\iota)$ (Definition 5.11), the observer-local FRG of Remark 6.35 restricts (A.11) to $\sigma = -1$ modes. Proposition 6.36, Theorem 6.37, and Corollary 6.38 remove the Λ_{cc} row from the observable Jacobian at the fixed point once Λ_{cc} couples only to $\sigma = +1$ modes; Lemma 6.32 and (6.11) state the effective-action projection target; Conjecture 6.39 requires deriving (6.14) at $\kappa_\star$ from nonperturbative Γ_{eff} .

Proposition A.11 (Leading-order $\kappa_\star$ stability under R^2 extension). *Extend the truncation of Theorem 6.30 by a curvature-squared coupling αR^2 with $\beta_\alpha = \mathcal{O}(\alpha)$ and graviton– R^2 mixing $\mathcal{O}(\kappa\alpha)$. Assume σ -invariant graviton and R^2 modes decouple from the observer-local flow on $H^1(\hat{Q}, V_{16})^{\sigma=-1}$ (Remark 6.35). At leading order in α and in graviton loops,*

$$\beta_\kappa = 2\kappa - \frac{5}{48\pi^2}\kappa^2 + \mathcal{O}(\alpha, \text{graviton}), \quad (\text{A.16})$$

so $\kappa_\star = 96\pi^2/5$ and $\partial\beta_\kappa/\partial\kappa|_{\kappa_\star} = -2$ persist. The residual gap is subleading graviton/ R^2 back-reaction in the $\sigma = +1$ shadow sector (Conjecture 6.41).

Proof sketch. Graviton and R^2 operators are σ -invariant on $\hat{Q}$; observer collapse $\Pi_{\sigma=-1}$ removes their contributions from β_κ on the anti-invariant row. The one-loop 28-scalar flow of Theorem 6.28 is unchanged; mixed κ – α terms enter only through $\beta_{\Lambda_{\text{cc}}}$ in the sequestered sector (Proposition 6.36, Conjecture 6.39). $\square$

A.4 Numerical workbook

This subsection consolidates every *dimensionless* SJET prediction with its closed-form formula and a reproducible arithmetic chain. Dimensional contacts (v , M_{Pl} in GeV) appear in Table 4 (Section 8); the workbook below is the T1/T2/T3 rational and real benchmark set used in Table 8.

κ_* fixed point. The workbook entry for κ_* is the positive root of $\beta_\kappa = 0$ in (A.11). The arithmetic chain below mirrors a Python evaluation (comments in `typewriter`):

$$\begin{aligned}
& \# \text{ beta_kappa(kappa) = } 2 * \text{kappa} - (5 / (48 * \text{math.pi}^{**2})) * \text{kappa}^{**2} \\
& \# \text{ kappa_star = } 96 * \text{math.pi}^{**2} / 5 \\
& \kappa_* = \frac{96\pi^2}{5} = \frac{96 \times 9.869604401}{5} \approx 189.6, \\
& \left. \frac{\partial \beta_\kappa}{\partial \kappa} \right|_{\kappa_*} = -2.
\end{aligned} \tag{A.17}$$

The stability eigenvalue -2 matches the Jacobian diagonal entry (A.15) and makes κ the unique UV-relevant direction among dimensionless couplings at the Gaussian-gravitational fixed point (A.14).

R_{RG} and neutrino ratio. The factorized capacity RG of Theorem 7.20 separates one-loop and threshold factors. With the inputs of Proposition 7.21:

$$\begin{aligned}
& \# \text{ b0, w3, w1 = } 12, 16/27, 1/27 \\
& \# \text{ ln_GUT = math.log(M_GUT / M_Z) \# pending full D3} \\
& \# \text{ R1 = } 1 + (\text{w3} - \text{w1}) * \text{b0} / (24 * \text{math.pi}^{**2}) * \text{ln_GUT} \\
& \# \text{ Xi_th = sum of threshold logs at M_F,3, M_F,2, M_Z} \\
& \# \text{ R2 = } 1 + (\text{w3} - \text{w1}) * \text{b0} / (12 * \text{math.pi}^{**4}) * \text{Xi_th} \\
& \# \text{ ratio = } 16 * \text{R1} * \text{R2} \\
& w_3 - w_1 = \frac{16}{27} - \frac{1}{27} = \frac{5}{9}, \\
& R_{\text{RG}}^{(1)} = 1 + \frac{(5/9) \cdot 12}{24\pi^2} \cdot 32.4 \approx 1.91, \\
& \Xi_{\text{th}} \approx 9.9 + 18.3 + 2.3 \approx 30.5, \\
& R_{\text{RG}}^{(2)} = 1 + \frac{(5/9) \cdot 12}{12\pi^4} \cdot 30.5 \approx 1.09, \\
& \frac{\Delta m_{31}^2}{\Delta m_{21}^2} = \frac{w_3}{w_1} R_{\text{RG}}^{(1)} R_{\text{RG}}^{(2)} = 16 \times 1.91 \times 1.09 \approx 33.3,
\end{aligned} \tag{A.18}$$

matching PDG 2025 [19] central value 33.8 ± 0.8 (Proposition 7.22).

Scale identification (D3). Theorem 8.3 and Proposition 8.7 supply the reproducible electroweak and GUT contacts:

$$\begin{aligned}
& \# \text{ kappa_star = } 96 * \text{math.pi}^{**2} / 5 \\
& \# \text{ w1, w3 = } 1/27, 16/27 \\
& \# \text{ v = M_Z * math.sqrt(kappa_star * w1 / w3)} \\
& \# * \text{ math.sqrt(sin2w_MZ / (3/8))} \\
& v \approx 91.2 \times \sqrt{190 \times (1/27) / (16/27)} \times \sqrt{0.231/0.375} \approx 246 \text{ GeV}, \\
& M_{\text{GUT}} \approx \frac{w_3}{\sqrt{\kappa_*} 4\pi} M_{\text{Pl}} \approx \frac{16/27}{13.8 \times 12.57} \times 1.22 \times 10^{19} \approx 4 \times 10^{16} \text{ GeV}, \\
& \ln \frac{\Lambda^2}{v^2} \approx 2 \ln \frac{M_{\text{GUT}}}{v} \approx 65.
\end{aligned} \tag{A.19}$$

The Sakharov identity (8.5) then yields M_{Pl} as output once (v, Λ) are fixed; the triangle is closed conditionally on the breaking chain (Section 8).

Mass-ratio chain. Leading fermion mass ratios follow from capacity weights and the diagonal mass matrix (7.8):

$$\begin{aligned}
& \# \mathbf{w} = [1/27, 10/27, 16/27] \\
& \# \text{mass_ratio} = [\mathbf{w}[2]/\mathbf{w}[0], \mathbf{w}[1]/\mathbf{w}[0], \mathbf{w}[0]/\mathbf{w}[0]] \\
& \# \Rightarrow [16, 10, 1] \\
& \frac{m_3}{m_1} : \frac{m_2}{m_1} : \frac{m_1}{m_1} = \frac{w_3}{w_1} : \frac{w_2}{w_1} : 1 = 16 : 10 : 1, \\
& \frac{m_\mu}{m_e} = \frac{w_2}{w_1} = 10 \quad (\text{leading, } \iota = 2 \text{ slot}), \\
& \frac{y_\mu}{y_e} = \sqrt{\frac{w_2}{w_1}} = \sqrt{10} \approx 3.162.
\end{aligned} \tag{A.20}$$

Off-diagonal flavon mixing and electroweak RG corrections are programme steps **D1–D2**; Table 2 records the same leading spectrum in slot-index form.

Muon g -2 order-of-magnitude. Section 9 evaluates the one-loop coset-scalar contribution at $\mu \equiv v$ (Proposition 8.5):

$$\begin{aligned}
& \# \mathbf{v}, \mathbf{m_mu} = 246.0, 0.1057 \# \text{ GeV, PDG contacts} \\
& \# \mathbf{y_mu} = \mathbf{m_mu} / \mathbf{v} \\
& \# \mathbf{delta_a} = 28 * \mathbf{y_mu}^{**2} / (48 * \text{math.pi}^{**2}) \\
& y_\mu = \frac{m_\mu}{v} \approx 4.3 \times 10^{-4}, \\
& \Delta a_\mu^{\text{SJET}} = \frac{28 y_\mu^2}{48 \pi^2} \approx 1.1 \times 10^{-8}, \\
& \frac{\Delta a_\mu^{\text{SJET}}}{\delta a_\mu} \approx \frac{1.1 \times 10^{-8}}{2.6 \times 10^{-9}} \approx 4
\end{aligned} \tag{A.21}$$

with $\delta a_\mu = a_\mu^{\text{exp}} - a_\mu^{\text{SM}}$ from (9.5). Precision contact awaits two-loop flavon RG (D2) and derived v (D3); Table 5 records the full ledger.

Remark A.12 (Cross-reference map). Table 12 is the appendix consolidation of Table 6, Prediction 10.7, Theorem 7.7, Proposition 7.21, Proposition 7.22, and Proposition 9.3. Equation labels (A.17), (A.18), (A.20), and (A.21) are the reproducible arithmetic cores; symbolic sources remain (A.13), (7.25), (7.7), and (9.2).

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ID	Former gap	Status	Route (observer / above–below)
P1	Lorentzian soldering, $d = 4$	Derived (massless residual)	<i>Below:</i> mirror-fixed $\mathbb{H} \subset \mathbb{O}$ at Hurwitz terminal ($n = 4$) supplies $H_2(\mathbb{H})$ and the clock block (Definition 6.7); Lemma 6.8 fixes the ι -local embedding; Proposition 6.13 fixes the timelike axis Φ^0 ; Proposition 6.9 closes exceptional-rung transport on $n = 6, 7, 8$. <i>Above:</i> quaternionic sector of $\mathfrak{U}^*$ capacity flow. Residual: tree-level massless $\mathfrak{m}_{+1}$ sector only (Thm. 6.16, Conj. 6.15, Remark 6.17; Open Problem O2).
P2	Chiral matter	Eliminated	<i>Below:</i> physical sector $H^1(\hat{Q}, V_{16})^{\sigma=-1}$ and observer collapse $\Pi_{\sigma=-1}$ (Definitions 6.1, 5.6); chirality is σ -anti-invariance, not postulated content. <i>Above:</i> 16 bundle on the 1 10 16 partition.
P3	Three families	Eliminated	<i>Below:</i> cellular count Lemma A.7 (full proof, Appendix A.2), Stone lift (Proposition A.8), Lefschetz index (Theorem A.9, Theorem 5.9); minimal heterotic rung (Proposition 3.8). <i>Above:</i> three diagonal directions of $\mathfrak{d} \subset J_3(\mathbb{O})$ and capacity weights (1/27, 10/27, 16/27) (Proposition 5.2).
P3'	Flavon hierarchy, CKM/PMNS	Conjecture	<i>Below:</i> capacity flow $\omega_t^*(\tau)$ on $SU(3)_F$ indexed by observer slots (Proposition 6.55, Conjecture 6.56); <i>above:</i> (1/27, 10/27, 16/27) weights. Leading $\mathcal{P}_{ij}$ and CKM template (Proposition 7.10) are partial; precision PMNS and distinct up/down quark matrices remain open.
P4	GUT–Higgs and Yukawa	Conjecture (partial theorems)	<i>Above:</i> staged chain (6.26) with 45_H/126_H/10_H labels (Lemma 6.45, Proposition 6.46, Conjecture 6.52); <i>below:</i> $E_6 \rightarrow \text{Spin}(10)$ (Thm. 6.44), $\text{Spin}(10) \rightarrow \text{SU}(5)$ from $\iota=3$ (Thm. 6.47), $\text{SU}(5) \rightarrow \text{SM}$ from $\iota=2$ 126_H , weight 10/27 (Thm. 6.49, Lemma 6.48), $\sin^2 \theta_W = 3/8$ at M_{GUT} (Prop. 6.50), electroweak from 10_H (Thm. 6.51); cellular Yukawa (Lemma 7.3) and full matrices remain open (Conjecture 7.13); sequential gradient-flow realization of all four steps remains conjectural.
P5	Nonperturbative reflection positivity	Derived (conditional)	Semiclassical RP on EIII is derived (Theorem 6.19); Theorem 6.23 gives conditional RP for $d\mu_{\Gamma_{-1}}$ when massless modes lie in $d\mu_{\Gamma_{+1}}$ only (Lemma 6.22); Proposition 6.24 supplies OS reconstruction on the observer sector. Residual: nonperturbative proof of (6.4) (Conjecture 6.25).
CC	Cosmological constant	Derived (conditional)	Proposition 6.34: not derivable from

Quantity	Tier	Formula	Value	Anchor
n_F	T1	$\dim H^1(\hat{Q}, V_{16})^{\sigma=-1}$	3	Thm. 6.4
$ \mathcal{I} $	T1	$-\text{ind}_\sigma(\bar{\partial}_{V_{16}})$	3	Thm. 5.9
b_0	T1	$12C_A - 4T_F n_F$ (Spin(10))	12	Thm. 6.27
n_S	T1	massive coset scalars	28	Thm. 6.28
GS coeff. (1 fam.)	T1	mixed anomaly	-1	Thm. 6.26
$\kappa_\star$	T1	$96\pi^2/5$	≈ 189.6	Thm. 6.30, (A.13)
$w_1 : w_2 : w_3$	T2	Bal _C on 27	1 : 10 : 16	Thm. 4.3
w_3/w_1	T2	$(16/27)/(1/27)$	16	Pred. 10.7
$m_3 : m_2 : m_1$	T2	capacity at fixed v	16 : 10 : 1	Thm. 7.7
m_μ/m_e	T2	w_2/w_1	≈ 10	Pred. 10.8
y_μ/y_e	T2	$\sqrt{w_2/w_1}$	≈ 3.16	Prop. 10.5
$\Delta m_{31}^2/\Delta m_{21}^2$	T2	$(w_3/w_1) R_{\text{RG}}^{(1)} R_{\text{RG}}^{(2)}$	≈ 33.3 (2-loop)	Thm. 7.20
$R_{\text{RG}}^{(1)}$	T2	(7.21)	≈ 1.91	Prop. 7.21
$R_{\text{RG}}^{(2)}$	T2	(7.31)	≈ 1.09	Prop. 7.21
$\delta a_\mu/\delta a_\mu^{\text{SM}}$	T2	$28 y_\mu^2/(48\pi^2)$ vs. anomaly	≈ 4 (OOM)	Prop. 9.3
$\sin^2 \theta_W(M_{\text{GUT}})$	T3	Spin(10) branching	3/8	Pred. 10.10
$\alpha_{\text{em}}^{-1}(M_{\text{GUT}})$	T3	unification fit	≈ 24	Pred. 10.10
$\Lambda_{\text{obs}}/M_{\text{Pl}}^4$	T3	sequestered shadow	≈ 0	Pred. 10.9

Table 12: Dimensionless prediction workbook. T1 rows are exact theorem outputs; T2 rows are capacity targets refined by flavon RG; T3 rows are conditional on Conjectures 6.52, 6.56, 6.39.